GAUL's Elerrína

Final Design Report

Competing Project 16

GAUL



LAUNCH CANADA CHALLENGE

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Abstract

This report presents the final design of GAUL Project 16's two-stage high-altitude rocket developed for the Launch Canada 2026 challenge. The project focuses on the development of a reliable, minimum-diameter launch vehicle optimized for mass efficiency, aerodynamic performance, and structural robustness while remaining manufacturable and compliant with competition safety requirements. The vehicle integrates several key subsystems, including commercially available solid rocket motors, lightweight composite structures, a reliable staging mechanism, redundant recovery systems, and a hybrid COTS/SRAD avionics architecture. The current design builds upon the team's previous experience with two-stage flights and the proven 4-inch rocket developed and flown in earlier competitions, allowing the reuse of validated concepts while refining and optimizing key subsystems. Since the Preliminary Design Review, several modifications have been implemented, including a change in the sustainer motor due to supply constraints, the development of an in-house line cutter system, and significant improvements to the avionics bay following lessons learned from hardware damage sustained at IREC 2026. The final vehicle is predicted to reach an apogee of approximately 97,000 ft while maintaining adequate stability margins and structural safety factors throughout all phases of flight. Flight performance, aerodynamic characteristics, and recovery operations have been validated through multiple simulation tools and in-house analyses. Conservative mass budgeting, redundant recovery architectures, and progressive subsystem testing have been implemented to mitigate technical risks and increase overall mission reliability.

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Abbreviations

AGL Above ground level
CAD Computer-aided design
COTS Commercial off the shelf
GAUL Groupe Aérospatial de l'Université Laval
PCB Printed Circuit Board
SRAD Student Researched and developed

Nomenclature

m/s^2 Acceleration
CO_2 Carbon Dioxide
SRAD Student Researched and Developed
Isp Engine Specific Impulse
ft Feet
g Gravitational Constant of 9.81 m/s^2
kg Kilograms
m Meters
N Newtons
s Seconds
Ns Total Impulse
Cv Valve Flow Coefficient
m/s Velocity (also denoted by ft/s)

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1 Introduction

This project focuses on the design and construction of a high-performance two-stage rocket developed for the Launch Canada 2026 competition. Created by GAUL (Groupe Aérospatial de l'Université Laval) Project 16, the vehicle uses a minimum-diameter configuration to maximize mass efficiency while maintaining strong aerodynamic performance and structural integrity. The design integrates several key systems, including high-performance propulsion, lightweight composite structures, a reliable staging mechanism, redundant recovery systems, and a hybrid COTS/SRAD avionics architecture to achieve both high altitude performance and flight reliability.

The main design objectives are to maximize the thrust-to-weight ratio at liftoff, maintain sufficient stability throughout the entire flight trajectory, and ensure reliable stage separation and recovery in realistic flight conditions. To achieve these goals, the team combines previous design experience with iterative simulations, detailed mass optimization, and progressive subsystem testing. This approach allows the development of a vehicle capable of reaching near-space altitudes while remaining feasible to manufacture, test, and operate safely within the requirements of the competition.

1.1 Updates and Lessons Learned

Since the previous report, we changed the sustainer motor due to a supply issue, as Cesaroni did not have the motor we originally wanted. We also changed the line cutter system, going from the Halfcat Rocketry CatClaw to one that we designed ourselves. The avionics bay underwent significant modifications since we lost a considerable amount of hardware at IREC 2026 due to damage caused by strong winds on the ground after the flight and before recovery. For instance, we upgraded the structure to maximize its robustness and its ability to resist impacts.

For the rest, the system architecture builds largely on the team's previous experience with two-stage flights and on the proven 4-inch rocket platform developed and flown last year, allowing us to reuse validated concepts while refining and optimizing key subsystems.

1.2 Team Organization

The team is split into 3 main subteams for the construction of the rocket: Aerostructure, Avionics and Payload.

The aerostructure division is responsible for the design, analysis, and manufacturing of the rocket's primary structure and aerodynamic surfaces. This includes the development of the composite airframe, fins, bulkheads, and structural interfaces between stages, with a strong focus on minimizing mass while ensuring sufficient strength and

stiffness under aerodynamic and inertial loads. The team also leads aerodynamic optimization and validates structural integrity through analytical and numerical methods, while coordinating the mechanical integration of the recovery systems and parachute compartments.

The avionics division oversees the development and integration of all onboard electronics and embedded software. Their responsibilities include flight computers, sensor fusion, telemetry, GPS tracking, staging and recovery logic, as well as ground communication systems. They ensure reliable real-time data acquisition and robust event sequencing through redundant architectures and extensive testing. In addition, this division supports the electrical and software interfaces of the scientific payload to guarantee seamless operation throughout the mission.

Finally, the payload division is responsible for the design, integration, and validation of the onboard payload carried by the rocket. Their work focuses on defining the payload objectives, developing the associated instrumentation and electronics, and ensuring reliable operation throughout the flight while respecting strict constraints on mass, volume, and power consumption. The team also oversees the mechanical and electrical integration of the payload within the vehicle.

We also have a business and marketing team explained in section 1.4

1.2.1 Team Development & Succession Planning

Most senior team members have graduated this year, making the transition to the next cohort a significant challenge. To ensure continuity and effective knowledge transfer, we have proactively trained and certified around ten new members with CAR and CQF (Level 1) qualifications, all of whom are expected to remain with the team for at least the next two years. In parallel, we have actively mentored these newer members and progressively entrusted them with leadership responsibilities throughout the year so they will be fully prepared to take on key roles next year.

1.3 Outreach

To engage the community and generate interest in aerospace and rockets, GAUL conducts a range of outreach activities. We can categorize these activities into 3 main efforts.

The first focus point is recruitment. Activities linked to this goal are necessary to keep the association alive through the years and ensure enough manpower to complete ambitious projects. As an example, twice a year, Université Laval organizes an open house for future students. During this event, GAUL has a kiosk to present the association to visitors. The open house is not the only event GAUL is present: we also take part in the

student project presentation day organized by the faculty of Science and Engineering.

The second focus point is to spread interest. In the last year, the members of GAUL presented their projects at different science events, such as the Débrouillard event in Quebec city, Astrofest in (Montréal, Québec) and Lévisium (Lévis, Québec). The main goal of these activities is to exchange our knowledge with the public and the people interested in the field of aerospace but also to promote interest with those from other fields.

Lastly, GAUL has a presence on social media. This year we appointed a member of the GAUL as social media manager and formed a new communication committee to increase our visibility across our platforms, including Facebook, Instagram but also on our Web page. These platforms allow us to share with those interested in our projects, but also various aerospace-related information. The posts play a vital role in actively involving the community in our projects.

1.4 Business & Marketing

Rocket development requires significant financial resources in addition to technical expertise. For the 2025–2026 campaign, GAUL implemented a structured sponsorship plan to fund propulsion, materials, avionics, manufacturing, testing, and competition travel. Budget management is conservative, with cost tracking and margins included early to mitigate risks from design changes and unexpected expenses.

To reduce costs, the team actively seeks both financial support and contributions such as materials, components, and technical mentorship. Sponsorship is organized into four contribution tiers offering increasing levels of visibility, including logos on the rocket and merch, mentions in publications and presentations, and exposure at provincial, national, and international events.

Beyond visibility, GAUL emphasizes long-term partnerships through networking events at Université Laval, company visits, and direct interaction with students for recruitment opportunities. Major competitions provide large-scale exposure, reaching thousands of attendees on site and many more online each year.

While securing funding alongside technical work is challenging, dedicated outreach and strong partner relationships have enabled sustained support and renewed collaborations, providing the financial and professional ecosystem necessary to achieve the project's ambitious goals.

Fundraising activities have focused on strengthening relationships with existing partners while expanding the sponsor base. Notable successes include renewed multi-year

collaborations and increased engagement through dedicated social media features and public outreach events. The main difficulty encountered is the time investment required to secure partnerships while maintaining progress on technical development; this is addressed by dedicating specific team members to business development and sponsor communications.

Overall, the sponsorship plan provides both financial stability and strategic industry connections, enabling GAUL to pursue ambitious technical goals while offering meaningful visibility and recruitment opportunities to its partner.

2 Concept of Operations (CONOPS)

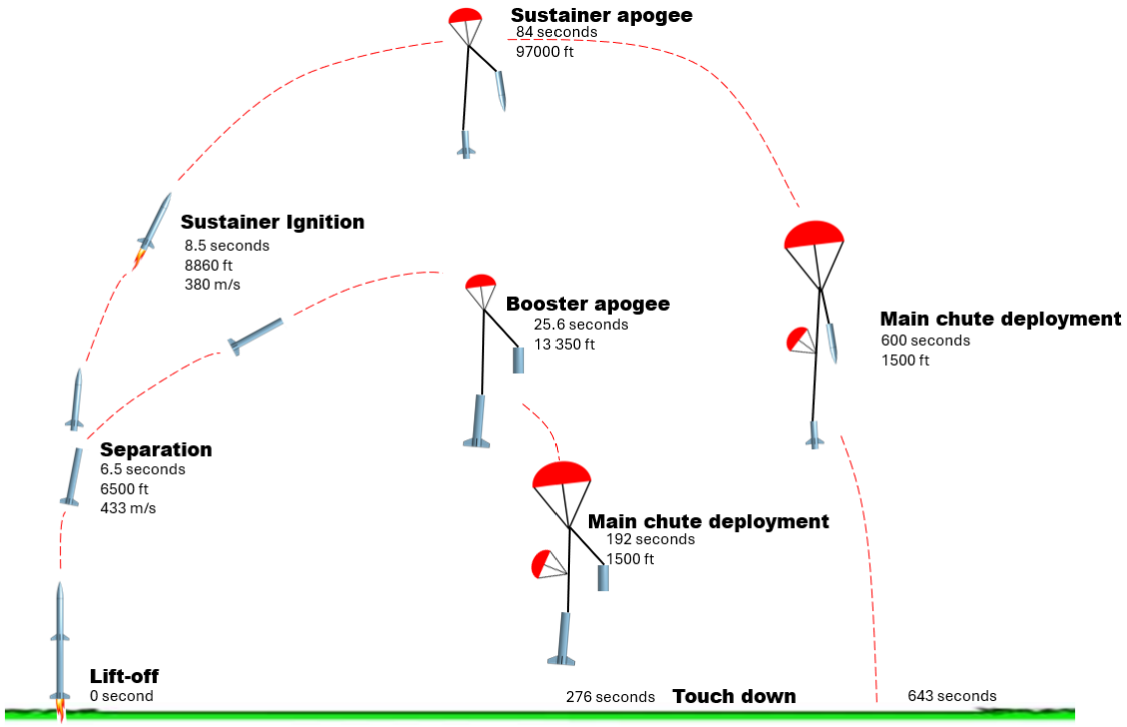


Figure 1: **CONOPS**

Ground Handling

Start: The rocket has left the main assembly area and is in transit to the launch-pad. The push pins are out of the rocket.
End: The launch rail has been raised and secured and the rocket is ready for launch.

Once all components are assembled and the rocket is transported to the launchpad, the sustainer igniter is prepared and tested with power outside of the motor. The rocket is then mounted on the rail, and the avionics system is powered on by inserting push pins one at a time, following the predefined order outlined in the checklist.

Abort Conditions: If any of the following event occurs during the rocket's preparation or waiting time on the rail, the flight shall be postponed until the problem is resolved. The procedures of an abort consist mainly in the removal of the arming system of the rocket and the rocket of the rail.

1. Flight computer malfunction

2. GPS malfunction

3. Empty battery

Booster Ignition

Start: The ignition signal has been sent through the igniter.
End: The rocket begins to accelerate.

Within the propulsion subsystem, the propellant will ignite resulting in the rocket accelerating up the launch rail.

Liftoff

Start: The rocket has begun to accelerate.
End: Only one rail button is in contact with the rail.

The thrust produced by the booster motor lifts the rocket off the ground and accelerates it to 31 m/s before leaving the rail.

Powered Ascent

Start: The rocket has left the rail
End: Burnout of the first motor

The rocket ascends freely, under the thrust of the motor and stabilized only by the fins. The motor burns out in 6.14 seconds and the maximum thrust encountered is 4700 N.

Stage Separation

Start: Burnout of the first motor
End: Lost of contact between the stages

When the thrust of the motor ends, the rocket is only submitted to the aerodynamic loads and the sustainer slides out of the interstage in 0.15 second. The pyrotechnic backup charge fires 0.5 seconds after motor burnout.

First Coast Phase

Start: Lost of contact between the stages
End: Ignition of sustainer motor and Apogee of booster

After being separated, both stages continue their ascent completely freely. The sus-

tainer fins keep it's trajectory straight and the booster is highly slowed with the forward end now open.

Sustainer Ignition

Start: The ignition signal is sent through the igniter
End: The sustainer begins to accelerate

The sustainer motor is ignited with the featherweight blue raven altimeter two seconds after burnout of the booster motor.

Abort Conditions: For the ignition signal to happen, the altimeter must respect the condition of trajectory orientation measured as the tilt angle directly or the altitude planned at that precise moment.

Sustainer Powered Ascent

Start: The sustainer motor produces thrust
End: Burnout of sustainer motor

The upper stage is accelerated again during 13.4 seconds and endures a max thrust of 2120 N.

Second Coast Phase

Start: Burnout of sustainer motor
End: Apogee of sustainer

After the burnout of the sustainer motor, the rocket continues its ascent stabilized by the fins and decelerated by drag forces and gravity.

Booster Apogee and Parachute Deployment

Start: The booster has reached its apogee
End: The Parachute has deployed and inflated

When the air pressure stops decreasing, meaning the apogee of 13 350 ft has been reached, the first altimeter fires a pyrotechnic charge, separating the rocket in two parts and freeing the parachute and recovery harness. The cords extend and the parachute begins inflating. The redundant pyrotechnic charge controlled by the second altimeter fires one second after the first.

Booster Descent Under Drogue Parachute

Start: Cords and parachute are fully extended
End: Ignition of main parachute deployment system

The drogue parachute of the booster slows it down to a velocity of 26 to 20 m/s during the descent until it reaches 1500 ft.

Sustainer Apogee and Drogue Deployment

Start: The sustainer has reached apogee
End: The drogue parachute has deployed and inflated

When the air pressure stops decreasing, meaning the apogee of 97 000 ft has been reached, the first altimeter fires a CO_2 charge, separating the rocket in two parts and freeing the parachute and recovery harness. The cords extend and the parachute begins inflating. The redundant CO_2 charge controlled by the second altimeter fires one second after the first.

Sustainer Descent Under Drogue Parachute

Start: Cords and parachute are fully extended
End: Ignition of main parachute deployment system

The drogue parachute of the sustainer slows down the rocket from a velocity of 175 m/s to 25 m/s during the descent until it reaches 1500 ft.

Booster Main Parachute Deployment

Start: Ignition of deployment system
End: The main parachute has fully deployed and inflated

When an altitude of 1500 ft is detected, the first altimeter fires a pyrotechnic charge, separating the rocket in two parts and freeing the parachute and recovery harness. The cords extend and the parachute begins inflating. The redundant pyrotechnic charge controlled by the second altimeter fires at 1300 ft.

Final Descent Under Main

Start: Cords and parachute are fully extended
End: The first part of the rocket touches the ground

The main parachute of the booster slows down the rocket to a velocity of 5.45 m/s during the descent until touchdown 276 seconds after liftoff.

Sustainer Main Parachute Deployment

Start: Ignition of deployment system
End: The main parachute has fully deployed and inflated

When an altitude of 1500 ft is detected, the first altimeter fires a pyrotechnic charge, separating the rocket in two parts and freeing the parachute and recovery harness. The cords extend and the parachute begins inflating. The redundant pyrotechnic charge controlled by the second altimeter fires at 1300 ft.

Final Descent Under Main

Start: Cords and parachute are fully extended
End: The first part of the rocket touches the ground

The main parachute of the booster slows down the rocket to a velocity of 10 m/s during the descent until touchdown 643 seconds after liftoff.

Touchdown

Start: The first part of both stages has touched the ground
End: The rocket and parachutes have stopped moving

The booster and sustainer land completely and slowly on the ground, waiting to be recovered.

Recovery Operations

Start: Landing of the rocket
End: Recovery team has returned to mission control

The GPS positions of both stages are determined, as well as the planing of a route to recover them. After arriving at the rocket's landing spot, the avionics is disarmed and deployment charges are verified. The rocket is then brought back to mission control for final inspection.

3 Requirements

3.1 Mission requirements

No	Description	Specification	Source	Planned Way to Show Compliance
1	Thrust to weight	Staged flights shall have a minimum thrust-to-weight ratio of 10 on the booster.	DTEG 3.1.2	Motor selection, system weight optimization
2	Thrust to weight	The sustainer shall have a minimum thrust-to-weight ratio of 5.	DTEG 3.1.3	Motor selection, system weight optimization

Table 1: General requirements

No	Description	Specification	Source	Planned Way to Show Compliance
1	Velocity off-rail	Launch vehicles shall have sufficient velocity upon "departing the launch rail" to assure they will follow predictable flight paths.	DTEG 10.2.1	Motor selection and vehicle design
2	Stability	Launch vehicles shall remain "stable" for the entire ascent.	DTEG 10.3.1	Fins design
3	Stability	Two plots of stability (CP/CG) as a function of time shall be provided as part of the team's design review documents. The first shall detail the initial phase of flight up to rail departure, and the second shall cover the full ascent phase.	DTEG 10.3.2	Flight analysis
4	Stability	All launch vehicles should avoid becoming "over-stable" during their ascent.	DTEG 10.4.1	Fins design

Table 2: Flight requirements

3.2 System Requirements

No	Description	Specification	Source	Planned Way to Show Compliance
1	Structure	Launch vehicles shall be designed and constructed to withstand the operating stresses and retain structural integrity under the conditions encountered during handling as well as during the most severe possible conditions experienced during flight.	DTEG 8.3.1	Structure design and analysis
2	Materials	PVC, PCL, PLA, PET, rPLA, rPETG, PE(and other similar low-strength low-temperature polymers) and Public Missiles Ltd. (PML) Quantum Tube shall not be used in any structural (i.e., load bearing) capacity, most notably as launch vehicle airframes.	DTEG 8.4.1	Material selection (composites)
3	Materials	Additively Manufactured Parts shall be designed to exceed loading requirements.	DTEG 8.4.4	Design and load analysis
4	SRAD tubes	SRAD tubes shall not present obvious gross manufacturing defects.	DTEG 8.6.2	Post-fab. inspections
5	Fins resistance	All aerodynamic surfaces shall be designed to avoid flutter and divergence under all anticipated operating conditions	DTEG 8.7.1	Fin flutter analysis
6	Motor attachment	Motors shall be secured to the airframe by mechanical retention.	DTEG 8.7.2	Bulkhead design
7	Motor attachment	Motor mounts shall secure or retain the motor such that no loads are transferred through any motor components that are not specifically designed to be in the load path.	DTEG 8.7.3	Precise positioning of structural components
8	Motor attachment heat	Motor mounts and all materials composing them shall be designed to withstand all temperatures to which they will be exposed.	DTEG 8.7.4	Material selection
10	Couplers	Airframe joints, regardless of the specific implementation (e.g., coupler tube, RADAX or other joint types), shall be sufficiently stiff to limit bending under flight loads.	DTEG 8.7.5	Material selection
11	Couplers	Airframe joints which implement ”coupling tubes” should be designed such that the coupling tube extends no less than one body caliber on either side of the joint – measured from the separation plane.	DTEG 8.7.7	Design
12	Interstage	Multi-stage rocket interstage couplers on vehicles in the Advanced category shall not have a discontinuity extreme enough to render the vehicle unstable during accent.	DTEG 8.7.10	Design of smooth transitions
13	Venting	Launch vehicles shall be adequately vented to prevent unintended internal pressures developed during flight from causing either damage to the airframe or any other unplanned configuration changes.	DTEG 8.8.7	Planning placement of holes
14	Rail guides	Rail guides shall be mounted to reinforced ”hard points” for mechanical attachment to the launch vehicle airframe.	DTEG 8.8.10	Planning of attachment points
15	Rail guides	Both rail guides shall be able to support the launch vehicle’s fully loaded launch weight while vertical.	DTEG 8.8.11	Design of strong holding points

Table 3: Structure requirements

No	Description	Specification	Source	Planned Way to Show Compliance
1	Official apogee measurement	Both stages shall carry a COTS altimeter with onboard data storage capable of recording apogee for official scoring.	R&R 3.4	Post-flight data retrieval and altimeter documentation.
2	High-altitude GPS scoring and tracking	The vehicle shall include a COTS GPS that outputs GPS altitude and position data for scoring	R&R 3.4, DTEG 6.4.1	GPS datasheet review and flight telemetry logs.
3	Controlled upper-stage ignition logic	Upper-stage ignition shall be performed by a COTS flight computer with altitude checks, boost-burnout detection, tilt inhibition, and permanent inhibit if conditions are unmet.	DTEG 3.2.1, 3.2.2, 3.2.3, 3.4.1, 3.4.2	Configuration of appropriate COTS flight computer.
4	Ground and non-nominal ignition prevention	The system shall prevent upper-stage ignition on the ground, during arming, during misfire, or during non-nominal flight, and allow full disarming if launch does not occur.	DTEG 3.1.4, 3.3.1, 3.3.3	Use of shunts and appropriate arming system.
5	Ignition system safety and isolation	Upper-stage ignition electronics shall comply with safety-critical wiring and electrical isolation requirements to prevent unintended ignition.	DTEG 3.1.1, 6.1.3	Wiring review and following of DTEG recommendations.
6	Dual-event recovery concept	Each independently recovered vehicle section shall implement a dual-event recovery architecture suitable for flights above 1,500 ft AGL.	DTEG 4.1.1	Recovery sequence diagrams and flight profile analysis.
7	Redundant recovery systems	The vehicle shall include two fully independent and redundant recovery systems such that failure of one ensures safe recovery by the other.	DTEG 4.8.1, 4.8.2	System architecture diagrams.
8	Proven recovery avionics and validation	At least one recovery system shall use a proven COTS flight computer, and any SRAD recovery avionics shall be thoroughly ground and flight tested.	DTEG 4.8.3, 4.9.1, 4.9.2	Test reports and prior flight data.
9	RF integrity and range compatibility	Antennas and GPS systems shall be located in RF-transparent sections and support rapid frequency changes as required by the launch range.	DTEG 6.3.2, 6.10.1	Communications test with fully assembled avionics bay.
10	Battery life	The entire avionics subsystem shall have a minimum autonomy of a full launch day, being 8 hours.	Team	Battery life computing and testing
11	Operational range	Communication with the rocket should be available up to a distance of 30km with the line-of-sight.	Team	Communication link budget and testing.

Table 4: Avionics requirements

No	Description	Specification	Source	Planned Way to Show Compliance
1	Parachutes	Each independently recovered launch vehicle body anticipated to reach an apogee above 1,500 ft (457 m) above ground level (AGL) shall follow a "dual-event" recovery operations concept (CONOPS).	DTEG 4.1.1	Two parachutes configuration
2	Apogee deployment	The initial deployment event shall occur at or near apogee to stabilize the vehicle's attitude (i.e., prevent or eliminate tumbling) and prevent or eliminate a ballistic landing.	DTEG 4.2.1	Apogee deployment
3	Drogue descent	The drogue chute shall also reduce the vehicle's descent rate enough to permit the main deployment event yet not so much as to exacerbate wind drift.	DTEG 4.2.2	Parachute selection
4	Main descent	The main deployment event shall occur at an altitude no higher than 1,500 ft (457 m) AGL and reduce the vehicle's descent rate sufficiently to prevent excessive damage upon impact with ground (i.e., less than 30 ft/s [9 m/s)]). The touch down velocity should be closer to 20 ft/s to reduce damage upon touchdown when calculated drift allows.	DTEG 4.3.1	Parachute selection
5	Parachute protection	The recovery system shall implement adequate protection (e.g., fire resistant material, pistons, baffles etc.) to prevent hot ejection gases (if implemented) from causing burn damage to retaining chords, parachutes, and other vital components as the specific design demands.	DTEG 4.4.1	Nomex protective sheets
6	Parachute links	The recovery system rigging (e.g., parachute lines, risers, tethers, etc.) shall implement swivel links at connections to relieve torsion as the specific design demands.	DTEG 4.5.1	Harness design
7	Tests	All recovery system mechanisms shall be successfully (without anomalies) tested prior to the LC Challenge, either by flight testing, or through one or more ground tests of key subsystems.	DTEG 4.11.2	Ground tests

Table 5: Recovery system requirements

4 Systems Architecture

The proposed vehicle is a two-stage, minimum-diameter rocket optimized for flight performance and maximum achievable altitude. The design philosophy prioritizes simplicity, mass reduction, and structural efficiency while ensuring reliable integration of propulsion, avionics, recovery, and staging systems.

Both stages are designed around commercially available 98mm solid rocket motors (N1000W from Aerotech for the sustainer and O3400 from CTI for the booster). The airframe diameters and structural interfaces are directly driven by the motor casings, enabling a minimum-diameter configuration for both stages. Figure 2 illustrates the fully integrated vehicle, highlighting the major subsystems and their interfaces.

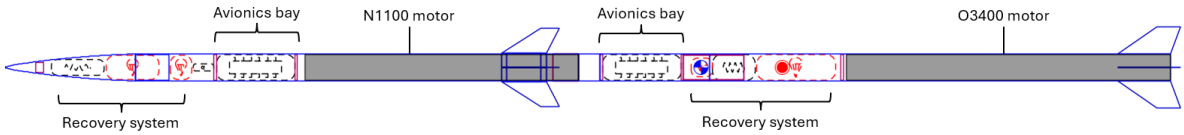


Figure 2: System architecture overview

Sustainer

The sustainer consists of a long, minimum-diameter composite body tube enclosing the propulsion system and avionics bay, terminated by a nose cone at the forward end. The sustainer recovery system is housed primarily within the nose cone and the forward section of the body tube.

Due to the minimum-diameter configuration, the motor is retained at its forward end by the lower bulkhead of the avionics bay. This bulkhead is structurally bonded to the airframe and serves as both the motor retention interface and the primary thrust transfer path from the motor to the vehicle structure. Although direct thrust transfer via the motor casing's aft retaining ring would be preferable, implementing both forward and aft load paths would impose strict position constraints between the motor hardware and internal bulkheads, significantly complicating integration. The chosen configuration therefore represents a balance between structural robustness and manufacturability.

The aft end of the body tube does not extend to the motor end. Instead, a dedicated fin can section in two parts is used. The fin assembly is made entirely out of aluminium with the fins screwed to two centering rings that slide over the motor casing. Then at the aft end there is another small fiberglass tube section. This approach simplifies fin integration while maintaining the minimum-diameter.

The avionics bay is located immediately forward of the motor. The motor igniter

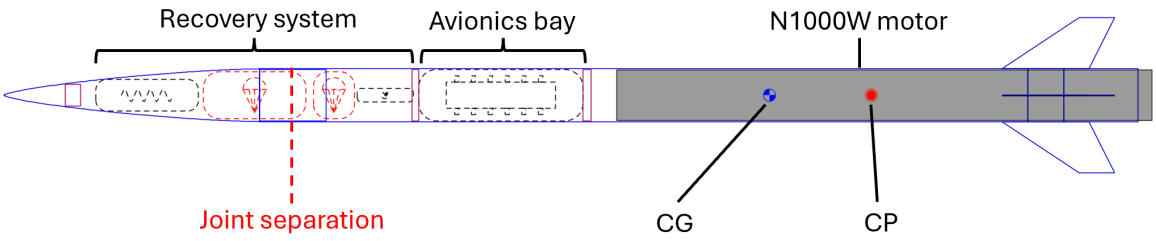


Figure 3: Sustainer overview

is routed externally along the airframe and re-enters at the aft end of the motor.

Above the avionics bay is the recovery bay, which contains both the drogue and main parachute systems, as well as the CO_2-based deployment hardware. The CO_2 system is responsible for apogee separation and drogue parachute deployment. The main parachute is housed within the same compartment but operates independently: during drogue descent, it remains folded and tethered externally, and is released at a predefined altitude using a line cutter.

The recovery harness is attached between the nose cone tip and the upper bulkhead of the avionics bay.

Booster

The booster stage shares several architectural similarities with the sustainer but incorporates key differences driven by staging and recovery requirements.

As with the sustainer, the booster motor is retained at its forward end by a structural bulkhead, which also serves as the thrust transfer interface. Directly above this bulkhead is the booster recovery bay, containing both parachutes and the recovery harness.

Above the recovery bay, a short body tube section is connected via an internal coupler. Immediately above this coupler is the lower avionics bulkhead, which houses the deployment pyrotechnic charges. The avionics bay extends from this bulkhead to near the top of the booster airframe.

The interstage interface consists of an external coupler attached to the forward end of the booster. This coupler slides over the exterior of the sustainer's aft section, providing structural alignment and load transfer during boost. Stage separation is mainly acheived by aerodynamic drag, but also uses a redundant pyrotechnic charge initiated by the booster avionics system.

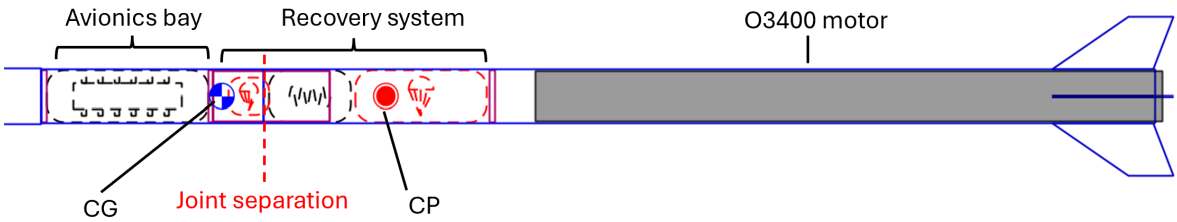


Figure 4: Booster overview

Characteristic	Value
External diameter (mm)	100
Stack length (mm)	4573
Sustainer length (mm)	2438
Propulsion type	Solid COTS
Total impulse (Ns)	35,188
Fin flutter speed sustainer (m/s)	1762
Booster burn duration (s)	6.2
Separation delay (s)	0.3
Sustainer ignition delay (s)	2.0
Sustainer burn duration (s)	13.1
Apogee deployment method	CO_2
Low-altitude deployment method	Pyrotechnic

Table 6: High-level performance metrics and system characteristics

4.1 Mass Budget

With the current assembly state of the rocket, most of its parts can be weighed. However, since some aluminium parts have been harder to source than initially planned, their mass is being estimated using CAD.

The mass budget is detailed in the following two tables:

Part Name	Mass (kg)	Source	margin
Structure			
Nose tip	0.51	CAD	± 0.01
Nose cone	0.294	Weighed	± 0.001
Nose coupler	0.156	Weighed	± 0.001
Body tube	1.362	Weighed	± 0.001
Fin can	1.10	CAD and Weighed	± 0.01
Fins	0.61	CAD	± 0.01
Avionics upper bulkhead	0.09	CAD	± 0.01
Avionics lower bulkhead	0.11	CAD	± 0.01
Total	4.21	-	-
Recovery system			
Main Parachute	0.076	Weighed	± 0.001
Drogue Parachute	0.074	Weighed	± 0.001
Shock cords	0.106	Weighed	± 0.3
Harness hardware	0.245	Weighed	± 0.001
Total	0.501	-	-
Propulsion			
Wet mass	12.71	Vendor	-
Dry mass	4.852	Vendor	-
Avionics			
Total	1.5	Weighed	± 0.1
Total wet mass	17.41	-	-

Table 7: Sustainer Mass budget

Part Name	Mass (kg)	Source	margin
Structure			
Interstage coupler	0.190	Weighed	± 0.001
Avionics body tube	0.508	Weighed	± 0.001
Motor body tube	1.382	Weighed	± 0.001
Body coupler	0.206	Weighed	± 0.001
Fins and tip to tip	0.47	Weighed	± 0.01
Avionics upper bulkhead	0.09	CAD	± 0.01
Avionics lower bulkhead	0.08	CAD	± 0.01
Motor bulkhead	0.09	CAD	± 0.01
Total	3.02	-	-
Recovery system			
Main Parachute	0.652	Weighed	± 0.001
Drogue Parachute	0.092	Weighed	± 0.001
Shock cords	0.112	Weighed	± 0.001
Harness hardware	0.263	Weighed	± 0.001
Total	1.056	-	-
Propulsion			
Wet mass	16.84	Vendor	-
Dry mass	10.93	Vendor	-
Avionics			
Total	1.5	Weighed	± 0.1
Total wet mass	22.42	-	-

Table 8: Booster Mass budget

5 Performance Analysis

5.1 Flight/Demo Performance

Predicted flight performance and stability analyses are currently based on multiple independent simulation tools to increase confidence in the results and reduce reliance on a single modeling approach. The tools used are OpenRocket, RASAero II and RocketPy, which provide predictions for apogee, velocity, acceleration, and stability throughout the full flight profile.

Characteristic	Value
Apogee sustainer (ft)	106 000
Apogee booster (ft)	13 500
Maximum velocity (m/s [mach])	827 [2.74]
Rail exit velocity (m/s)	31.3
Maximum acceleration (G)	11.7
Lift-off thrust-to-weight ratio	> 11
Time to apogee (s)	83

Table 9: Simulation results

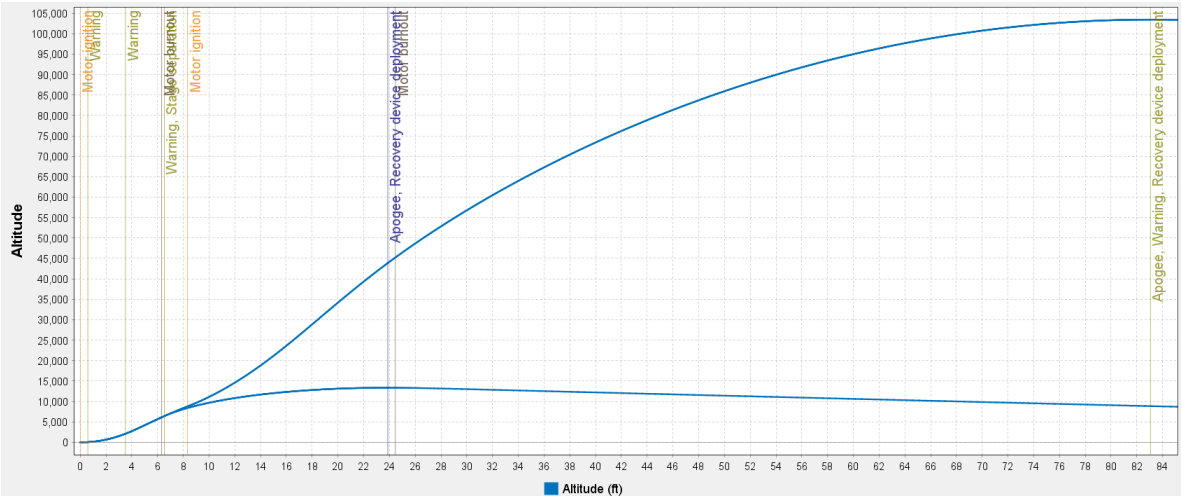


Figure 5: Evolution of altitude in time

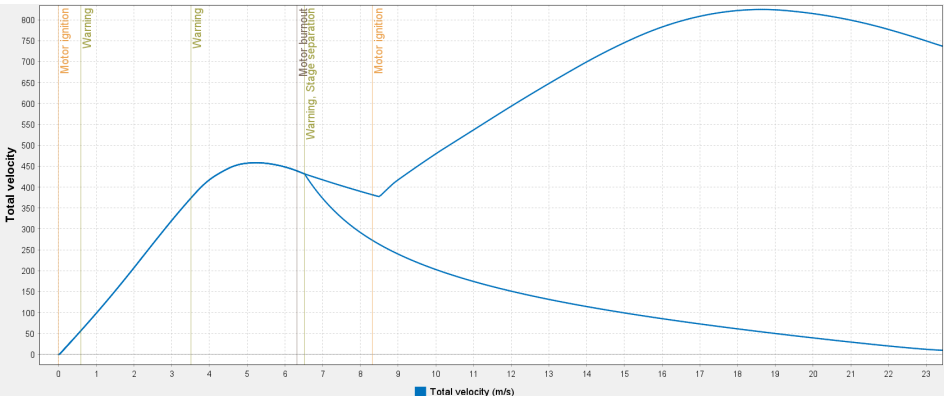


Figure 6: Evolution of velocity in time

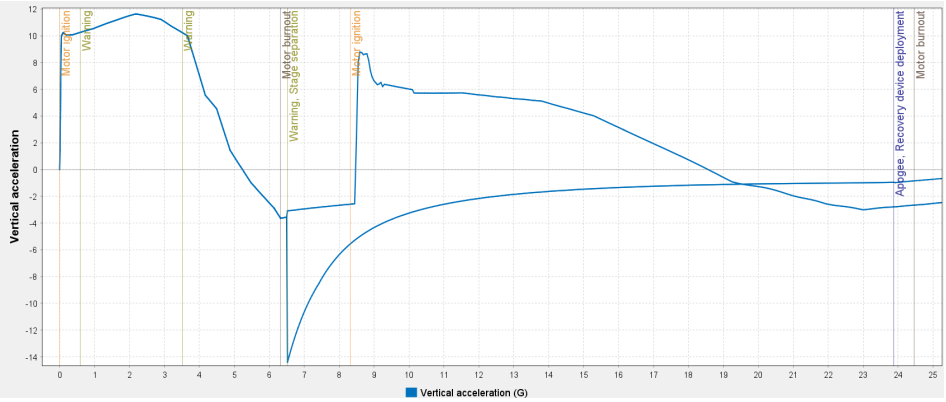


Figure 7: Evolution of acceleration in time

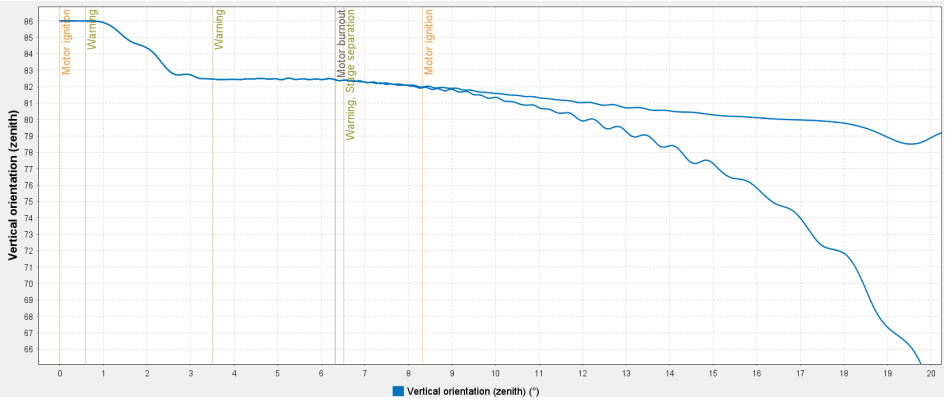


Figure 8: Evolution of the rocket's angle relative to horizontal during ascent

Vehicle stability is evaluated using OpenRocket, RASAero II, and AeroLab, with the latter providing additional insight into the center of pressure (CP) location across varying Mach numbers. Stability is quantified using the static stability margin, defined as

the distance between the center of gravity (CG) and the center of pressure, normalized by the vehicle diameter and expressed in calibers.

For a two-stage configuration, stability is assessed for two distinct and critical flight phases:

Stacked configuration. This phase extends from rail departure through first-stage burnout and up to stage separation. During this phase, a minimum target stability margin of 3 calibers is maintained to ensure robust stability during launch rail exit and early atmospheric flight. Figure 10 illustrates the stability margin evolution during this phase, confirming that the vehicle remains stable throughout.

Sustainer alone. Following stage separation, stability is reassessed for the sustainer alone from separation through burnout and coast to apogee. For this configuration, a target stability margin of 2 calibers is maintained, reflecting reduced aerodynamic disturbances and the absence of booster-induced mass asymmetry. Simulation results confirm that the sustainer remains statically stable throughout this phase of flight.

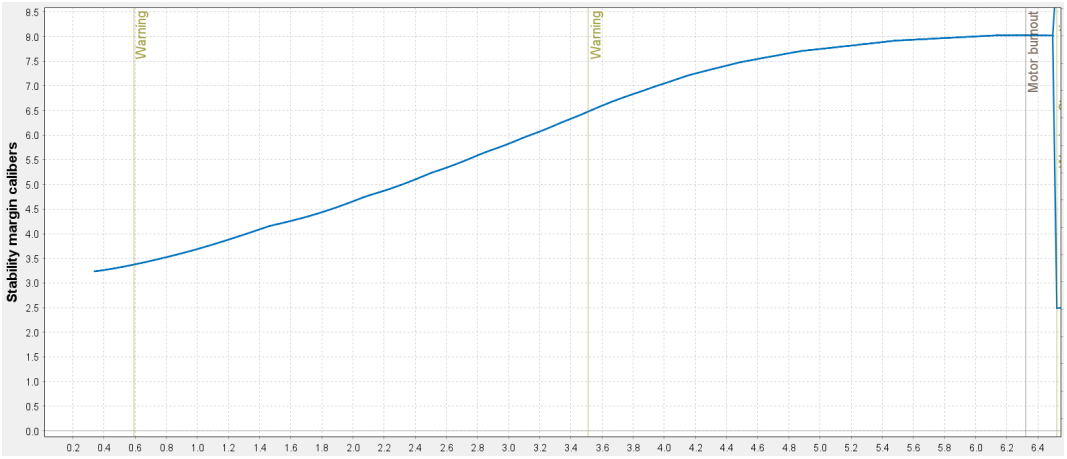


Figure 9: Stack stability margin from OpenRocket

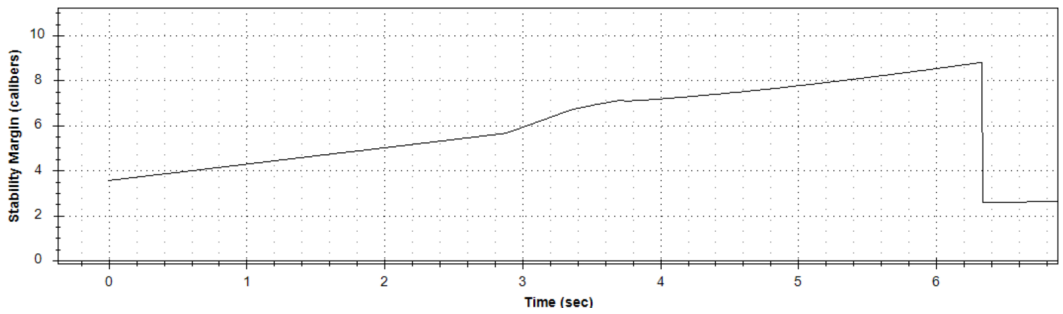


Figure 10: Stack stability margin from RASAero

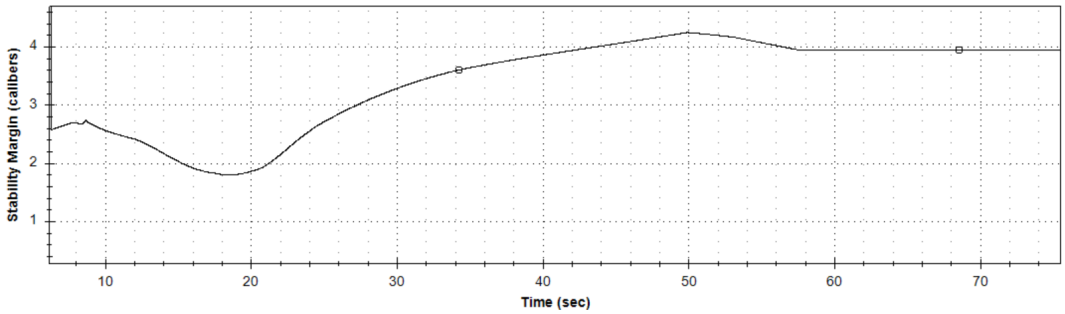


Figure 11: Sustainer stability margin

Aerodynamic performance prediction represents the largest source of uncertainty in the flight dynamics analysis. More specifically, the prediction of the drag coefficient as a function of Mach number has the greatest influence on the expected apogee and therefore on the overall mission performance. For this reason, four independent methods are used to determine the drag coefficient curve. Three are based on COTS software (OpenRocket, RasAero, and AeroLab), while the fourth is the internally developed DATCOM-based method.

The resulting drag coefficient curves are first compared directly as a function of Mach number to identify differences in the prediction of the transonic and supersonic regimes, where aerodynamic uncertainties are generally the highest. Since small differences in drag can accumulate over the entire ascent and significantly affect the final altitude, each drag model is then incorporated into the flight simulation to compare the resulting predicted apogee. This second comparison provides a single performance metric that quantifies the practical impact of the different aerodynamic models and helps identify which flight regimes require further investigation or refinement. Using both curve-to-curve and trajectory-level comparisons provides greater confidence in the selected aerodynamic model while highlighting the remaining sources of uncertainty in the performance predictions.

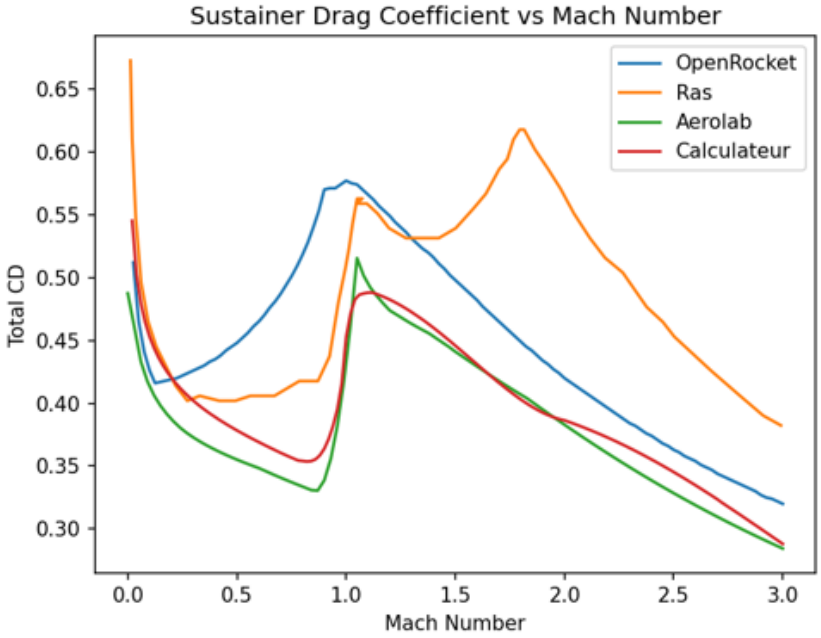


Figure 12: Sustainer Drag coefficient

Software	Drag source	Apogee deviation (ft)
OpenRocket	OpenRocket	0
RASAero	RASAero	+800
RocketPy	OpenRocket	+1500
RocketPy	RASAero	-23 000
RocketPy	Aerolab	+ 18 200
RocketPy	Custom calculator	+ 8600

Table 10: Normalized predicted apogee comparisons

It can be observed that the most effective way to predict the effect of wind on apogee, without performing simulations using a complete atmospheric wind profile, is to consider the wind conditions at an altitude of 1000 m ASL. Wind direction has the greatest influence on the predicted apogee. When the wind is aligned with the launch rail direction, the rocket experiences a larger weathercocking effect during the initial ascent, resulting in a greater inclination from the vertical. Conversely, when the wind blows opposite the launch rail direction, the vehicle tends to correct back toward the vertical, leading to a higher predicted apogee. Wind speed primarily affects the magnitude of this directional effect. Above approximately 6 m/s, increasing wind speed generally amplifies the influence of wind direction and, in most cases, results in a decrease in the predicted apogee.

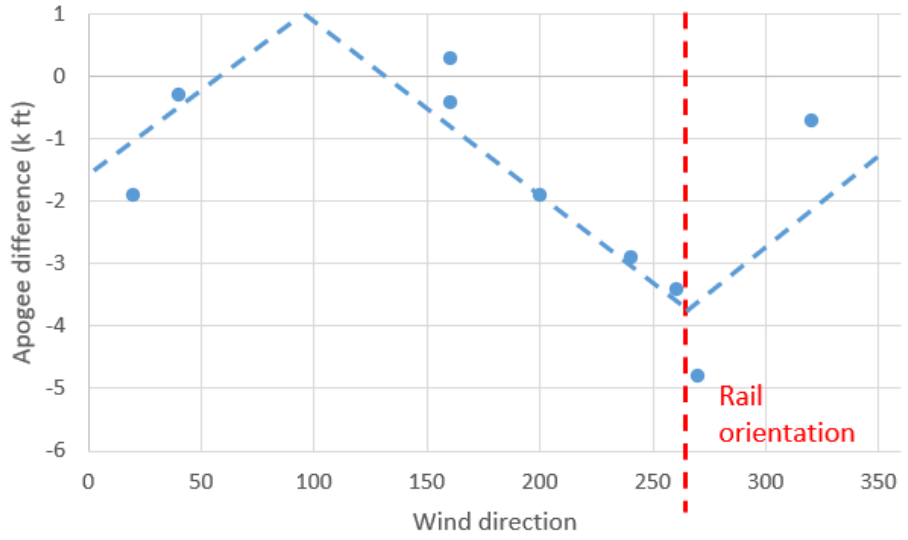


Figure 13: Effect of 1000 m ASL wind orientation on apogee predictions

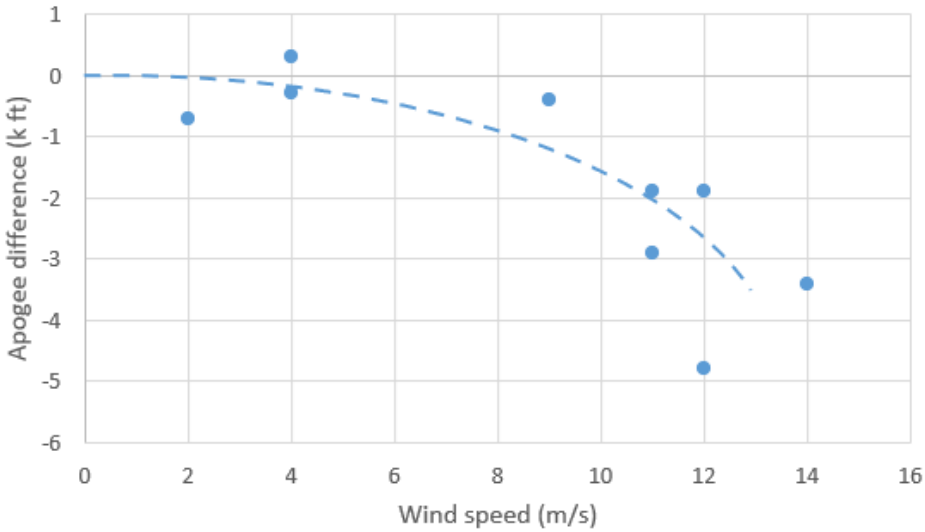


Figure 14: Effect of 1000 m ASL wind speed on apogee predictions

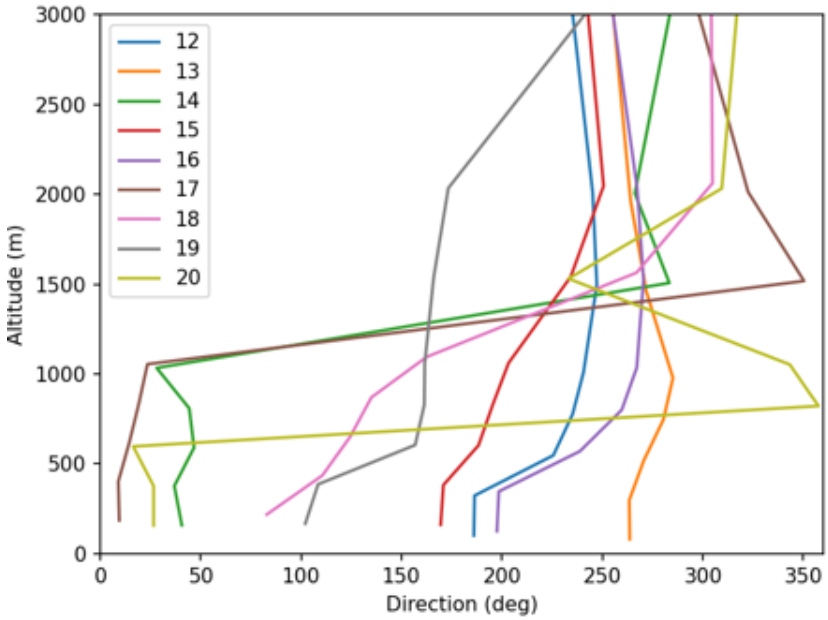


Figure 15: Mid-august 2025 low altitude wind direction

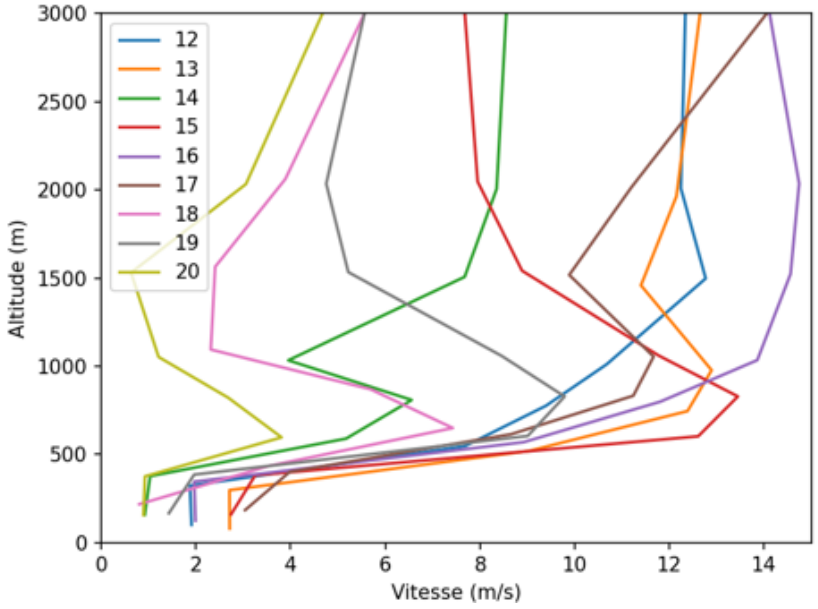


Figure 16: Mid-august 2025 low altitude wind speed

5.2 Recovery Dispersion Analysis

Multiple dispersion analysis were done to evaluate the risk of the rocket falling outside the recovery area. Openrocket as well as our Simuloutil SRAD software were used to estimate landing position after the flight. With both Openrocket and Simuloutil, we

simulated the landing areas for two scenarios: Every valid August 2025 wind levels from NavCanada and high ground winds wind levels from available NavCanada data (17 kt for OpenRocket and 17-18 kt for Simuloutil). From those four analysis, a maximum average wind threshold as well as a maximum wind in one level threshold were determined. The tresholds were then used to filter the historical wind data to verify that they were adequate.

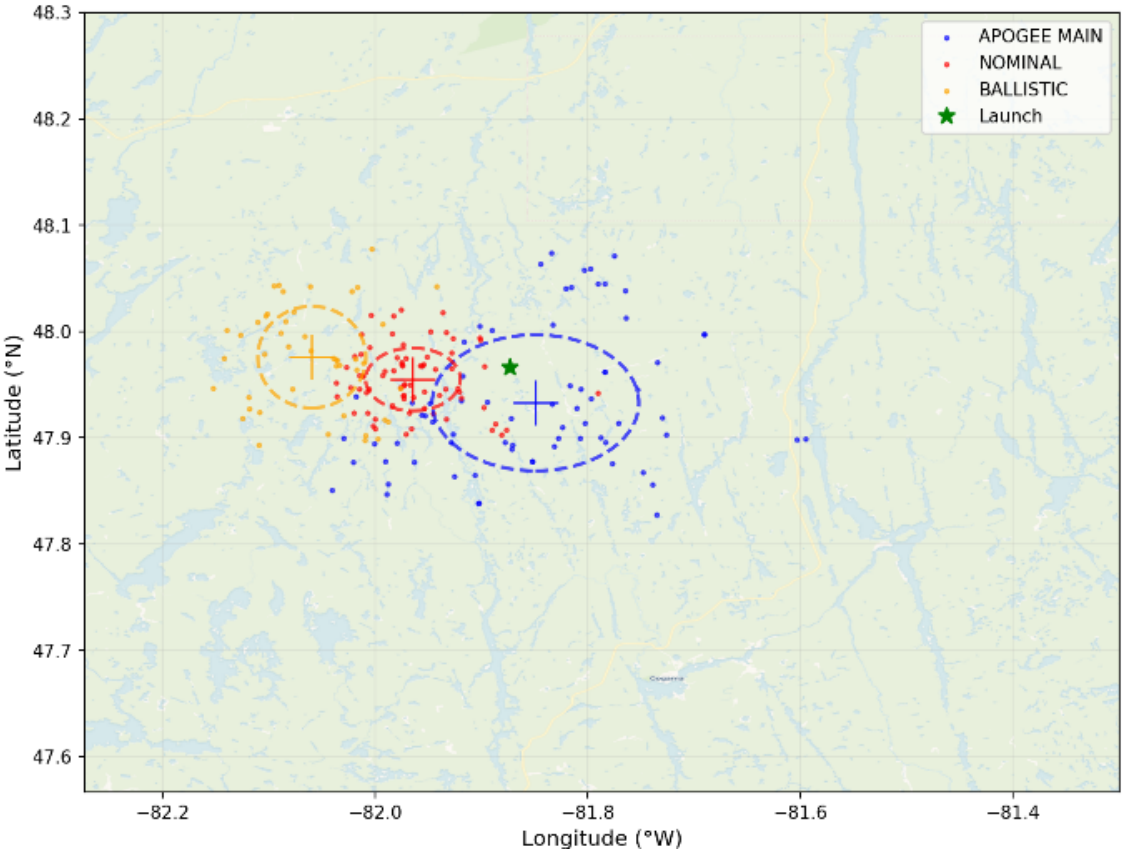


Figure 17: Landing dispersion using Openrocket for valid winds in August 2025

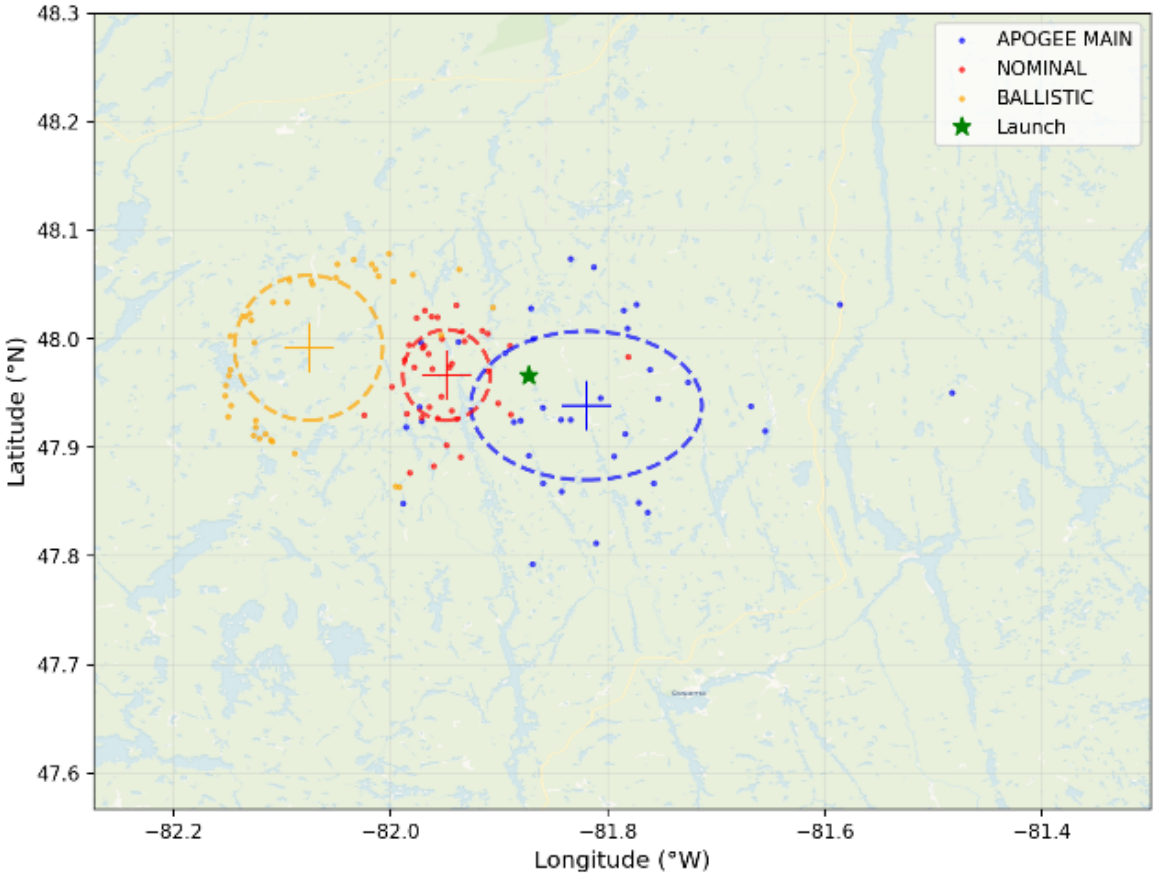


Figure 18: Landing dispersion using Openrocket for 17 kt ground winds

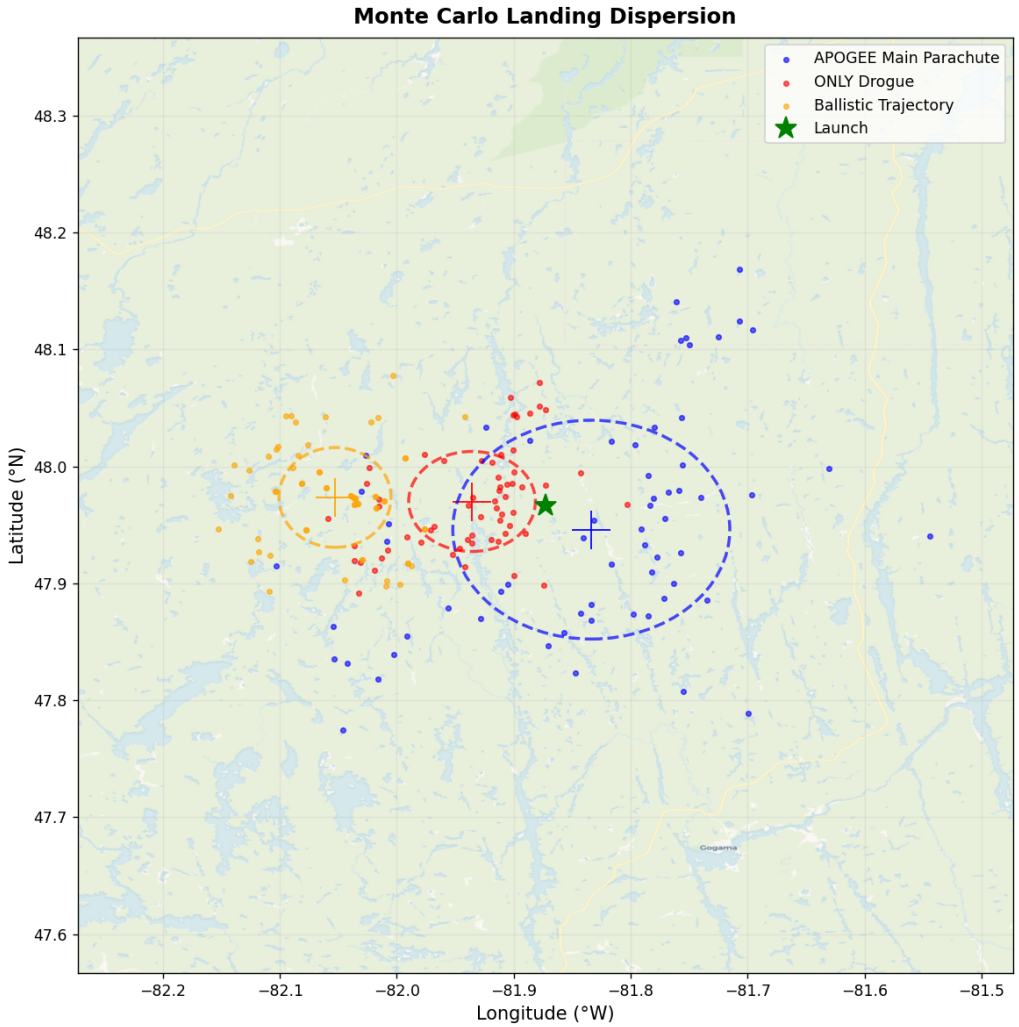


Figure 19: Landing dispersion using Simuloutil for valid winds in August 2025

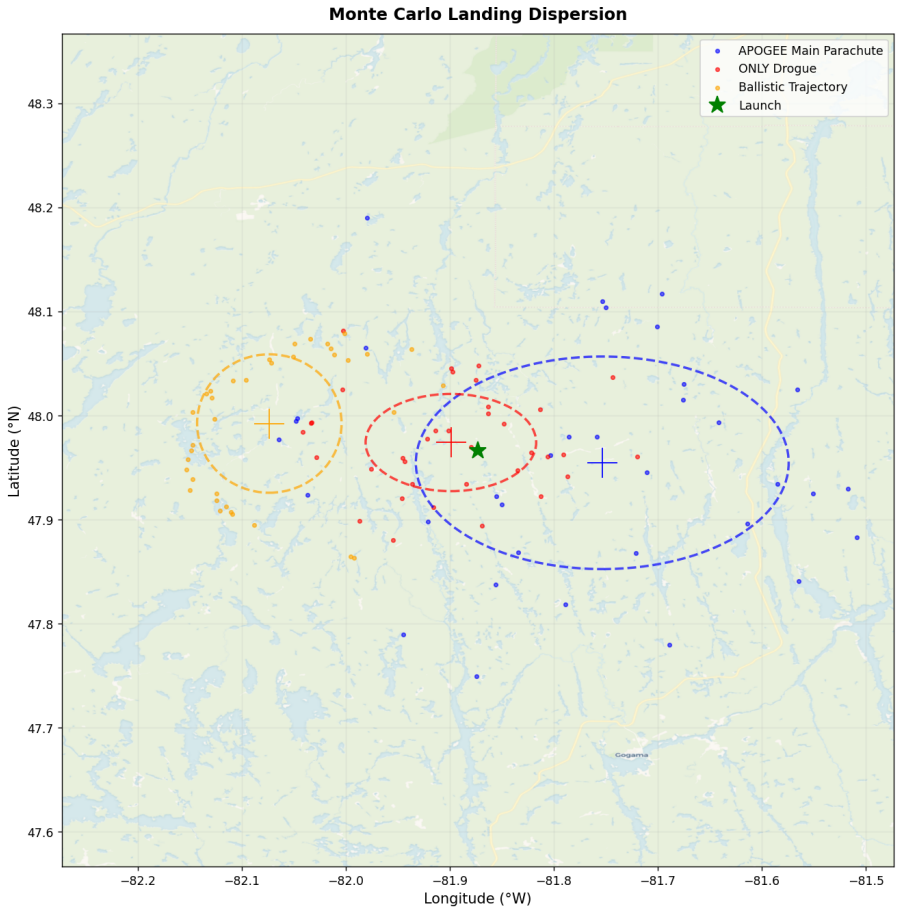


Figure 20: Landing dispersion using Simuloutil for 17-18 kt ground winds

By examining the wind levels that brought the rocket outside the recovery zone, we set the following two thresholds:

Average wind speed	$\leq$ 43 kt
Maximum wind speed in any level	$\leq$ 103 kt

Table 11: Wind speed tresholds set to not exit the landing zone in the event of a main parachute deployed at apogee.

The maximum wind speed is usually at the 39,000 ft level because of the jetstream. The winds were then filtered and analysis reexecuted with Simuloutil giving the following graphs:

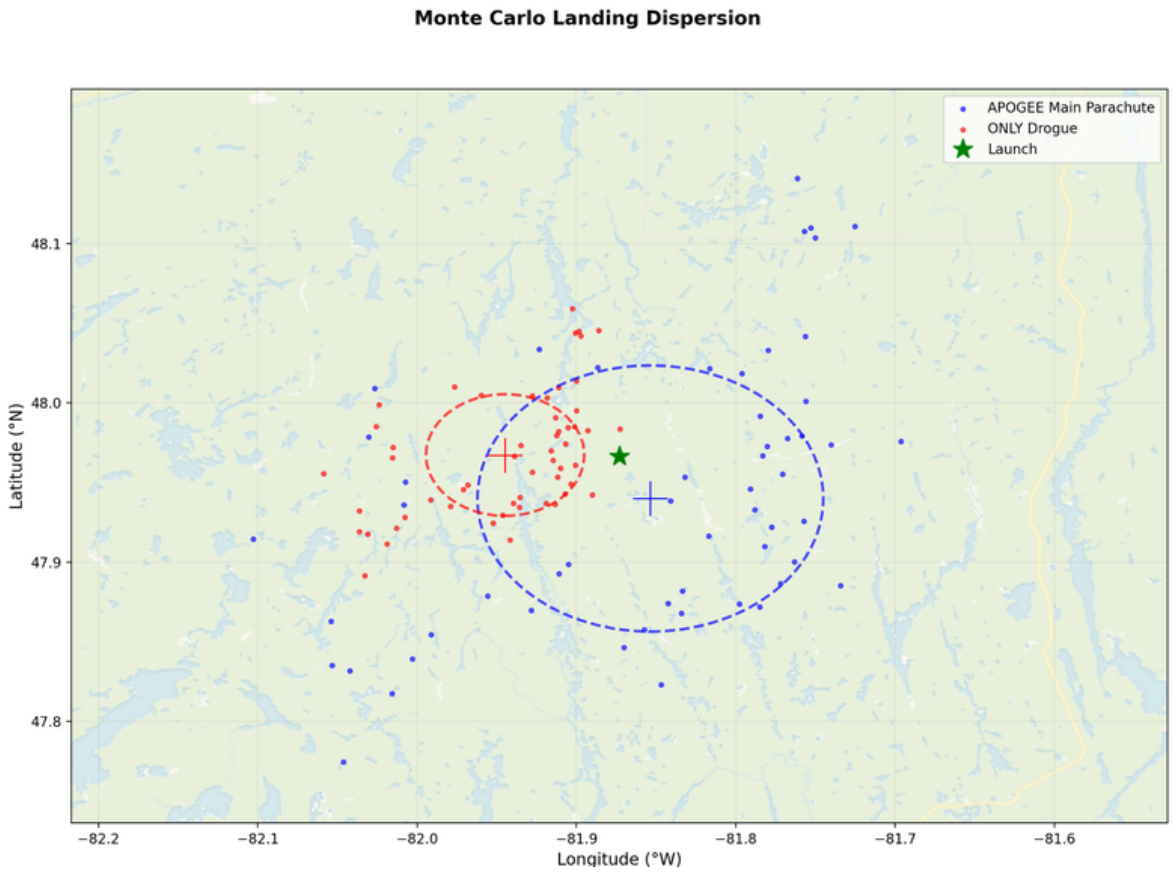


Figure 21: Landing dispersion using Simuloutil for valid winds in August 2025 (filtered)

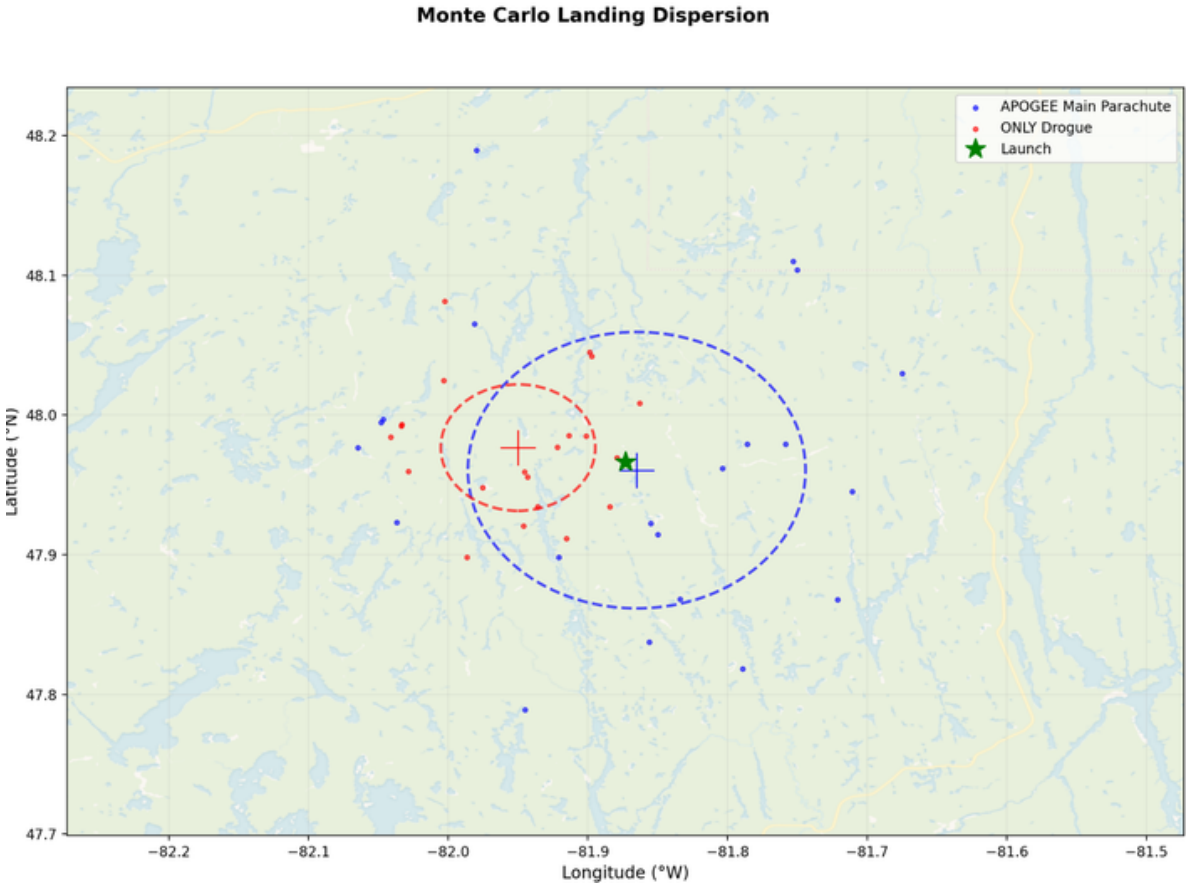


Figure 22: Landing dispersion using Simuloutil for 17-18 kt ground winds (filtered)

6 Aerostructure

6.1 Nose Cone

6.1.1 Detailed Design

The nose cone follows a 5:1 Von Karman profile, a shape selected for its low drag characteristics at supersonic velocities, consistent with the vehicle's flight regime near Mach 2.6. The outer mold line was modeled in Onshape using the Haack series formula.

The vehicle's overall diameter is driven by a minimum-diameter design philosophy aimed at maximizing aerodynamic performance: the airframe inner diameter is set at 99 mm, sized to accommodate a 98 mm diameter motor with sufficient tolerance to account for motor casing manufacturing variation. The nose cone's aft end is sized to mate with this 99 mm sustainer airframe tube.

The shell is fabricated from E-glass fiberglass sleeve, 332 g/m², with a nominal cured ply thickness of 0.25 mm, laid up in five plies with a ±45ř twill braid orientation relative to the tube axis. While the theoretical laminate thickness at five plies is approximately 1.25 mm, the as-built shell thickness is closer to 1.5 mm due to resin uptake and the inherent thickness variation of hand wet layup. Sleeve fabric was chosen over flat plain-weave fiberglass specifically because a braided sleeve conforms far more consistently to a continuously varying diameter, such as that of a nose cone, than a flat ply would; this avoids the wrinkling and darting that flat fabric requires on double-curvature geometry. Fiberglass sleeve was selected over a carbon fiber sleeve of equivalent weave primarily on a cost basis, since the nose cone does not carry significant structural load (see Analysis).

The nose cone tip is machined aluminum, fitted and bonded into the forward end of the composite shell. Aluminum was retained for the tip rather than composite for two reasons: it withstands the aerodynamic heating generated at the stagnation point at the vehicle's peak Mach number, and it is significantly easier to manufacture to the tight geometric tolerances required at the tip apex than a composite layup would allow.

6.1.2 Manufacturing Process

The nose cone is manufactured by hand wet layup of fiberglass sleeve over an internal mold. An external mold produced by Elite Composite used for our IREC rocket was available for resin infusion of von karman nosecones, but is not sized for this vehicle's 99 mm diameter, and producing a new external mold at this diameter was not economically justified given the team's budget and timeline. The nose cone was therefore manufactured using the team's established internal-mold method instead.

The internal mold is 3D-printed in PLA and coated with a release agent prior to layup. Five plies of ±45ř fiberglass sleeve are pulled over the mold and wet out by hand with

West System epoxy resin without a vacuum bag to not add wrinkles in the part. This makes the final part a bit thicker and heavier but saves time on sanding and does not fragilise the outer layer.

Once cured, the part is demolded by applying localized heat to the PLA mold and pushing it out through the tip end; the release agent layer allows the softened mold to separate cleanly from the composite shell without damaging the internal surface. This heat-and-push demolding approach is necessary because, unlike an external mold, an internal mold cannot simply be pulled away from the finished part and must instead be softened and pushed out.

Following demolding, the nose cone is sanded and finished with epoxy filler to achieve a smooth aerodynamic surface prior to installation of the aluminum tip.

6.1.3 Analysis

Aerodynamic loading on the nose cone was characterized through a CFD simulation of the supersonic flow field around the 5:1 Von Karman profile, performed in STAR-CCM+, capturing the shock structure at the vehicle's maximum velocity.

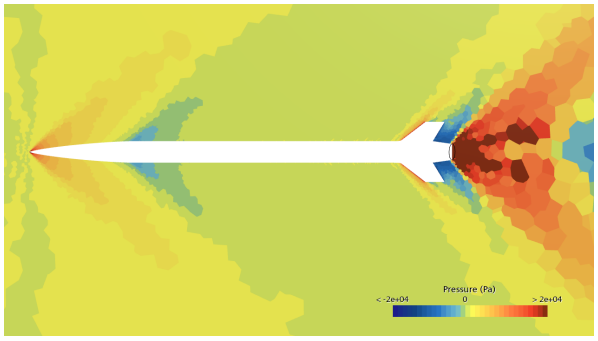


Figure 23: Pressure scene from Starr CCM+ of the Von Karman 5:1 nosecone at supersonic flow

The resulting aerodynamic pressure distribution, combined with axial loads from motor thrust, was used to compute the force and stress state on the nose cone shell in the team's structural analysis tool. Classical laminate theory was used for this calculation: an ABD stiffness matrix was assembled from the ply layup, accounting for the ±45ř sleeve orientation, and Kirchhoff's hypothesis was applied to determine the through-thickness stress distribution in each ply. Material properties used as inputs to this model: Young's modulus, shear modulus, and ultimate strength were obtained from laboratory tensile testing of the team's own composite laminates rather than manufacturer datasheet values.

By propagating this method along the full length of the vehicle, a minimum safety

factor was extracted at each station. The calculation method can be found in Appendix E and the final result in Figure 26.

6.1.4 Testing and Validation

The material properties used in the structural analysis above were validated experimentally through tensile testing of composite coupons, manufactured using the same wet layup process and materials used for the flight nose cone, to ensure the calculated safety factors reflect as-built laminate performance rather than nominal fiber/resin datasheet values.

6.2 Airframe and Bulkheads

6.2.1 Detailed Design

Bulkheads

Bulkheads serve as rigid structural interfaces within the rocket's internal structure, transferring mechanical loads between subsystems. Their main functions are anchoring parachute shock cords during recovery deployment, securing avionics and payload components to the fuselage, and transmitting motor thrust to the airframe. To fulfill these roles, the bulkheads must satisfy the following requirements.

All bulkheads are machined from 6061-T6 aluminium, with an outer diameter of 98 mm for the booster and sustainer to match their respective fuselage internal diameters, and are secured to the fuselage by four radial M5 screws. Parachute cords are anchored through 3/8" U-bolts, except on the booster motor bulkhead which uses a central eyebolt that also connects to the motor casing. The sustainer motor bulkhead uses a 3/8" bolt that connects it to the motor casing. To minimize mass while resisting peak bending stresses, the bulkheads feature I-shaped or X-shaped internal ribs, resulting in masses ranging from 87.5 g for the sustainer upper avionics bay bulkhead to 113 g for the booster motor bulkhead. This mass optimization is critical to reach the 97 000 ft target apogee, as every gram saved on structural components directly contributes to the achievable altitude.

Airframe

The sustainer/booster airframe is composed of three SRAD body tubes, all with an inner diameter of 99 mm. This diameter is driven by the motor: the vehicle uses a 98 mm diameter motor, and 99 mm was selected as the minimum inner diameter that reliably accommodates the motor with sufficient tolerance for casing manufacturing variation. This minimum-diameter approach reflects the vehicle's design philosophy of maximizing aerodynamic performance by keeping frontal area, and therefore drag, as low as possible.

Material selection follows a radio-transparency constraint, consistent with the nose cone: any tube housing avionics must allow unobstructed RF transmission for telemetry, and is therefore fabricated from E-glass fiberglass. Tubes with no avionics content are fabricated from carbon fiber to minimize mass. The three tubes are summarized below.

Tube	**Stage**	**Material**	**Layup**	L **(mm)**	**RF transparent**
Motor	Booster	Texonic CF 3k 2×2 twill, bidirectional	$[0\check{r}_5]_T$	1760	No
Avionics	Booster	Texonic E-glass 320 g plain weave	$[90\check{r}_5]_T$	420	Yes
Motor & Avionics	Sustainer	Texonic E-glass 320 g plain weave	$[0\check{r}_5]_T$	1480	Yes

Table 12: Body tube specifications. All tubes have an inner diameter of 99 mm.

The booster motor tube is built from five plies of Texonic bidirectional carbon fiber, 3k 2×2 twill, oriented at 0° along the tube axis, and carries no avionics, allowing the use of carbon fiber for a lighter, stiffer structure without a radio-transparency penalty.

The booster avionics tube is built from five plies of Texonic E-glass, 320 g/m^2 plain weave, at 90°, and is fabricated in fiberglass specifically to maintain radio transparency for avionics telecommunication.

The sustainer motor tube also houses the sustainer avionics in the same tube, and is therefore fabricated entirely in E-glass (five plies, 0°) to preserve radio transparency along its full length. A split design was considered, using carbon fiber for the lower (motor-only) portion of this tube to save mass while keeping fiberglass only around the avionics section, but this approach was rejected.

6.2.2 Manufacturing Process

Bulkheads

The manufacturing of the bulkhead was sponsored by a company near the university.

Airframe

All tubes are student-designed and student-manufactured (SRAD) using a vacuum-assisted resin infusion process on a reusable aluminum mandrel. A mylar release sheet is applied to the mandrel prior to layup to ensure clean demoulding without surface damage. [1]

Dry fibre layup: Dry fabric plies are roll-wrapped tightly onto the mandrel using 3M spray adhesive to hold each ply in place. This dry application method ensures a

[1]Here is short video of us making the tubes: `https://www.instagram.com/reel/DZVCCwlR2td/?igsh=MWNhbTlkMHhxdzBhdA==`

uniform, wrinkle-free layup and eliminates the veining defects commonly introduced when a wet-wound tube is consolidated under vacuum bag pressure.

Resin infusion: Once the desired ply count is reached, a peel ply and infusion mesh are applied over the stack to promote uniform resin flow. Resin feed lines are distributed along the part, with a resin intake puck at one end and a vacuum puck at the other. The assembly is sealed in a vacuum bag with sealant tape, and resin is drawn through the dry preform by vacuum pressure alone. This simultaneously degasses the resin, reducing porosity and improving fibre-to-resin ratio consistency and strength compared to wet layup. Infusion continues for approximately 30 minutes until resin breakthrough is observed at the vacuum puck.

Resin quantity calculation: The resin mass required to achieve the target 60/40 fibre-to-resin ratio is calculated prior to each infusion, ensuring full wetout without excess matrix mass. For a tube of length L and inner diameter d, with n plies of areal weight A_f:

$$m_{\text{resin}} = \left(\frac{n \cdot A_f}{1.5} + m_{\text{losses}} \right) \cdot L \cdot \pi d \tag{1}$$

where the factor of 1.5 follows from the 60/40 ratio ($m_{\text{resin}} = 0.4/0.6 \cdot m_{\text{fibre}}$), and m_{losses} accounts for resin absorbed by the peel ply and feed lines ($700\,\text{g/m}^2$). For example, for the sustainer tube

$$\begin{aligned} m_{\text{resin}} &= \left(\frac{5 \times 320}{1.5} + 700 \right) \times 1.73 \times \pi \times 0.099 \\ &= 1766.6\,\text{g/m}^2 \times 0.5326\,\text{m}^2 = 940.97\,\text{g} \end{aligned} \tag{2}$$

The West System epoxy used requires a 5:1 epoxy-to-hardener mix ratio by mass, applied to the total resin mass above.

Cure and finishing: The bagged assembly is oven-cured at 75 °C for 90 minutes. After demoulding, the peel ply is removed, the tube is cut to the required length, and the outer surface is sanded to the specified finish.

6.2.3 Analysis

Bulkheads

The mechanical analysis of each bulkhead followed a two-step approach. A first iteration based on the beam theory was used to size the components and identify the dimensioning load cases, followed by a finite element analysis in Siemens NX Nastran to validate the final design under realistic boundary conditions. Section 12.1 in the appendix presents the finite element analysis (FEA) results for each bulkhead, with the corresponding stress distributions shown in Figures (42 43, 44, 45, 46 & 47). The

following figures shows FEA of the motor bulkhead of the sustainer part of the rocket. Note : the safety factor is calculated this way:

$$S.F = \frac{\sigma_{\text{rup,AL6061T6}}}{\sigma_{max}} = \frac{275\ \text{MPa}}{\sigma_{max}} \tag{3}$$

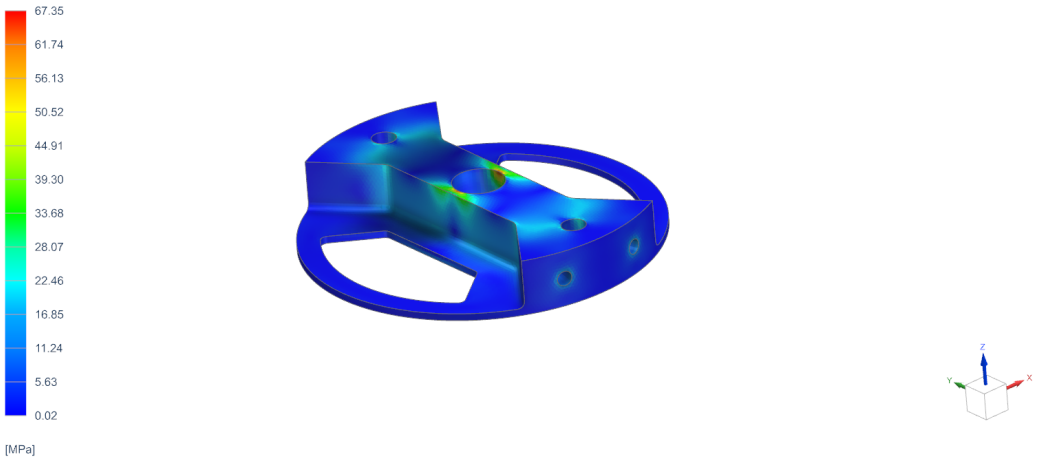


Figure 24: FEA : Bulkhead motor sustainer - isometric view

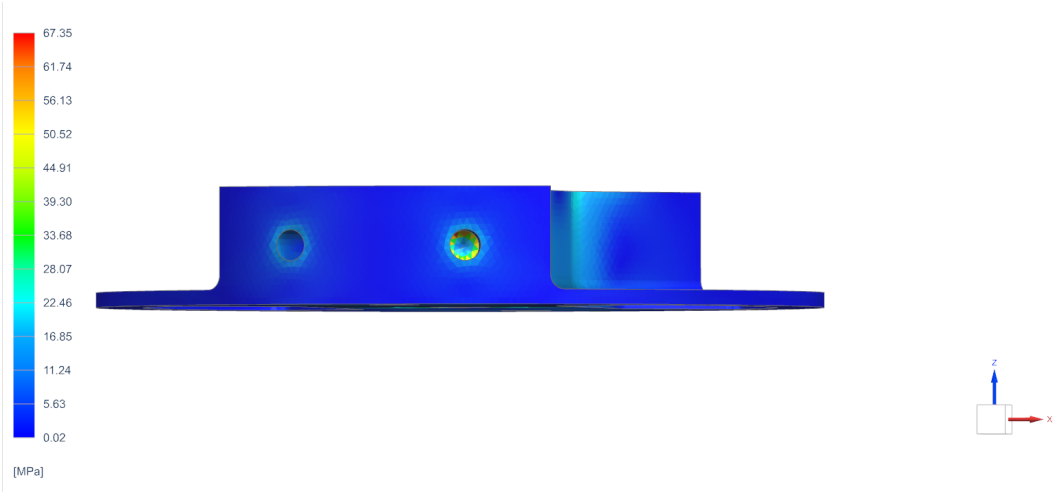


Figure 25: FEA : Bulkhead motor sustainer - side view

The principal results from the simulations are summarized in the next table.

Table 13: Summary of the maximum stresses and minimum factors of safety for each bulkhead.

Sustainer – Upper Avionics Bulkhead		
Minimum Factor of Safety		11.14
I-beam corner	[MPa]	10.12
Above U-bolt hole	[MPa]	7.44
Below U-bolt hole	[MPa]	9.45
Radial bolt hole	[MPa]	24.69
Sustainer – Motor Bulkhead		
Minimum Factor of Safety		4.08
Above Eye-bolt hole	[MPa]	63.86
Below Eye-bolt hole	[MPa]	55.09
Radial bolt hole	[MPa]	67.35
Booster – Upper Avionics Bulkhead		
Minimum Factor of Safety		14.30
Above U-bolt hole	[MPa]	8.48
Below U-bolt hole	[MPa]	7.11
Radial bolt hole	[MPa]	19.23
Booster – Motor Bulkhead		
Minimum Factor of Safety		4.22
Above U-bolt hole	[MPa]	23.13
Below U-bolt hole	[MPa]	24.90
Radial bolt hole	[MPa]	65.10

Airframe

Structural loading on the airframe was determined using the same classical laminate theory framework described for the nose cone. Flight simulation data was used to extract aerodynamic and motor thrust loading along the vehicle, which was then combined with the laminate ABD stiffness matrix and Kirchhoff's hypothesis to compute ply-level stresses at each station along the airframe. Material properties used in this calculation (Young's modulus, shear modulus, ultimate strength) were obtained from laboratory tensile testing of the team's composite laminates.

The full calculation methodology, including the flight loads extraction and the laminate stress derivation, is provided in Appendix E due to its length. By iterating this method along the length of the vehicle, the minimum safety factor at each station was obtained and is shown in Figure 26; this result also constitutes the structural substantiation

referenced in the nose cone analysis (Section 6.1.3).

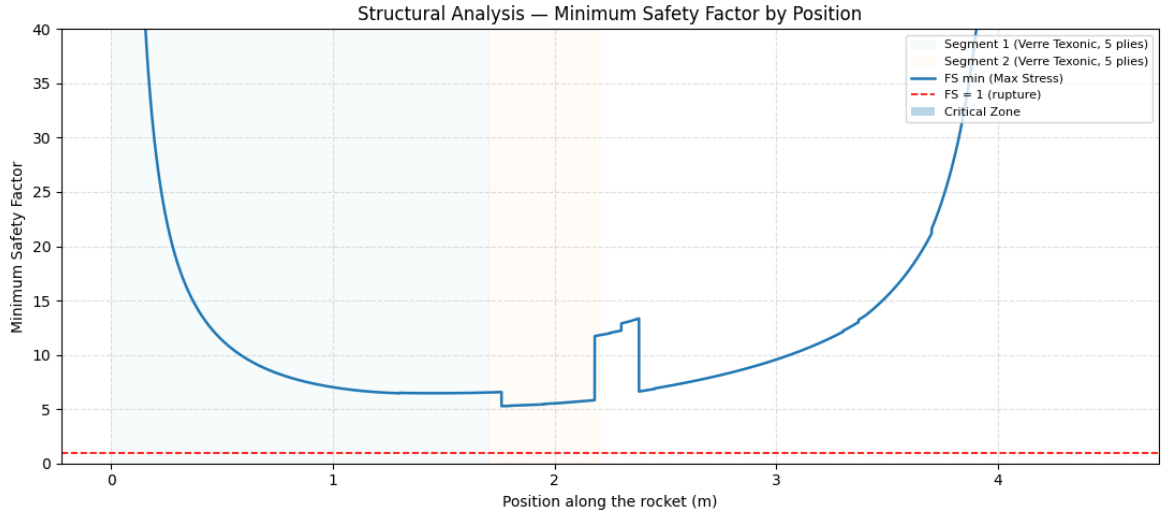


Figure 26: Minimum safety factor along the length of the rocket in stack configuration due to motor thrust and aerodynamic loading

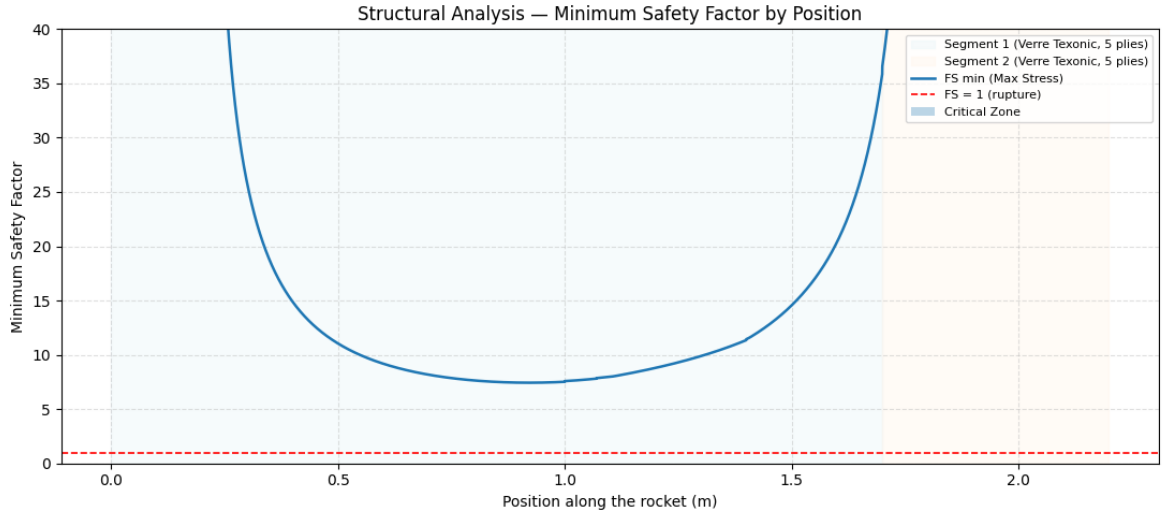


Figure 27: Minimum safety factor along the length of sustainer due to motor thrust and aerodynamic loading

6.2.4 Testing and Validation

Material properties used in the structural analysis were obtained through tensile testing of composite coupons, manufactured using the same resin infusion process and materials as the flight tubes, following ASTM test standards. Coupons were tested in the 0°, 90°,

and 45° fibre directions to extract the Young's modulus in each principal direction as well as the shear modulus and max strength. Four samples were tested per orientation, per material, to allow averaging and provide statistically reliable property values for use in the structural analysis.

6.3 Couplers

6.3.1 Detailed Design

The airframe uses three SRAD couplers, each sized and located according to the joint it serves.

The lowest coupler joins the booster motor tube to the booster avionics tube. It is internal to both tubes, with an outer diameter of 98.5 mm, and is 230 mm long. It is bonded to the avionics tube over a 100 mm length at its upper end, while the remaining 130 mm slides freely inside the motor tube, allowing the motor tube to be removed independently of the avionics section. It is fabricated from five plies of fiberglass at 0°.

The second coupler, located further up the vehicle, joins the interstage. Because the sustainer motor occupies the internal volume at this station, leaving no room for an internal coupler, this joint is made externally: the coupler fits over the outside of the airframe, with an inner diameter of 102 mm and a length of 200 mm, of which 100 mm is bonded to the booster avionics tube. It is fabricated from six plies of carbon fiber at 0°.

The third coupler joins the nose cone to the sustainer airframe. Unlike the other two, it is not a simple cylinder: it transitions from a cylindrical section, which fits inside the sustainer tube, to a profile matching the internal contour of the nose cone base, which is where it is bonded to the nose cone shell. It is fabricated from five plies of sleeve fiberglass at 45°.

6.3.2 Manufacturing Process

All three couplers are manufactured by hand wet layup over 3D-printed molds. Because each coupler differs in diameter, length, and/or geometry, a single-use, 3D-printed mold was produced for each part rather than attempting to share tooling between them; this was simpler and more cost-effective than designing reusable or reconfigurable tooling for only one part per geometry.

6.3.3 Analysis

The couplers are analyzed using the same classical laminate theory framework described for the airframe (Section 6.2.3), with one key difference: because the couplers are designed to slide within their mating tubes rather than being fully constrained, they do

not carry axial compressive loading. Instead, their structural analysis considers only the bending moments transmitted through the joint. The full calculation methodology is provided in Appendix E, and the resulting safety factors along the vehicle, including at each coupler location, are shown in Figure 26.

6.3.4 Testing and Validation

As with the airframe tubes, the material properties used in the coupler analysis were obtained through tensile testing of composite coupons manufactured with the same process and materials, tested in the 0°, 90°, and 45° fibre directions with four samples per orientation per material to obtain reliable Young's modulus and shear modulus values.

6.4 Fins

6.4.1 Detailed Design

The fin design process aims to achieve an optimal balance between aerodynamic efficiency, static stability, and flutter resistance. Design iterations begin with an initial set of fin dimensions that ensure acceptable stability margins in OpenRocket simulations. Based on this baseline geometry, the flutter velocity is analytically evaluated using material properties and aerodynamic loading models. The fin geometry is then iteratively modified to achieve a minimum flutter safety factor of 1.5× the maximum expected flight velocity, after which the updated geometry is re-imported into OpenRocket to verify that the required stability margins are still satisfied. Final refinements are performed to minimize aerodynamic drag while maintaining both stability and flutter constraints.

The sustainer is expected to reach a maximum velocity of approximately 790 m/s, placing stringent requirements on fin flutter resistance and structural integrity. For this reason, aluminum alloy fins were selected over composite alternatives. Aluminum provides significantly higher and more reliably characterized shear modulus and shear strength compared to carbon fiber laminates, enabling improved flutter resistance and reduced uncertainty in high-speed flight regimes.

Although materials such as magnesium, titanium, and other high-performance alloys were evaluated, their higher cost and limited performance gains relative to aluminum led to their exclusion from the final design.

The sustainer fins are integrated into an aluminum fin can, which serves as a structural interface between the fins and the airframe. Due to the minimum-diameter architecture, traditional through-the-wall fin attachment is not feasible. Instead, the fins are mechanically attached to the fin can using integrated brackets, providing a robust load path while maintaining a clean external profile.

The sustainer fins dimensions are shown in Figure 28. Each fin has a uniform thickness of 5 mm, with sharp leading and trailing edges to minimize wave drag in the transonic and supersonic regimes. Attachment details are shown in Figure 29.

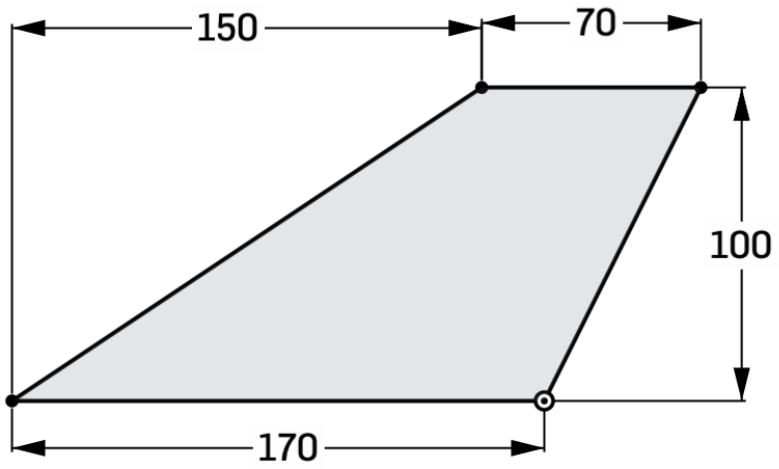


Figure 28: Sustainer fins dimensions

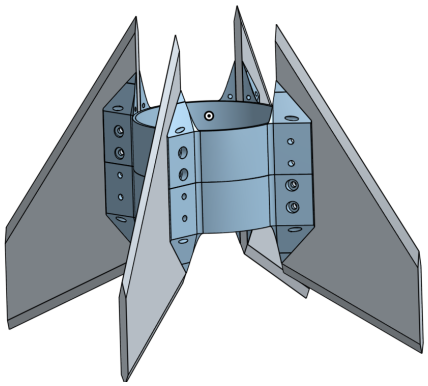


Figure 29: Sustainer fincan assembly

The booster stage experiences significantly lower peak velocities (455 m/s), allowing greater flexibility in material selection and attachment strategy. For this stage, a carbon fiber sandwich construction has been selected to minimize mass while maintaining sufficient stiffness and flutter resistance.

The booster fins are bonded directly to the external surface of the body tube and reinforced using tip-to-tip composite layups, eliminating the need for a dedicated fin can.

The fin geometry is shown in Figure 30, with a total thickness of 6 mm. Each fin consists of a lightweight core approximately 4 mm thick, sandwiched between composite skins. The expected laminate schedule includes 5 plies per side (three primary structural plies and two tip-to-tip reinforcement plies). A divinycell foam core was used for the sandwich. We tested this method at IREC this year in our booster fins and thes held up great even without tip-to-tip.

Unlike the sustainer fins, the booster fin edge geometry is less critical from an aerodynamic standpoint. Square leading and trailing edges are used, as they facilitate the staging drag separation.

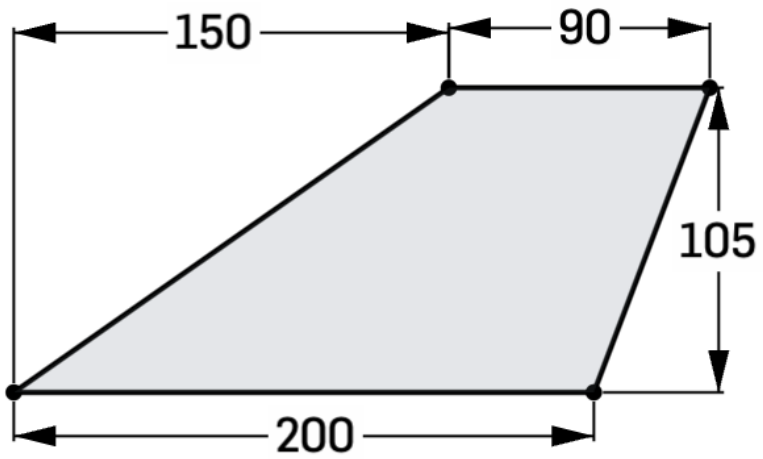


Figure 30: Booster fins dimensions

6.4.2 Manufacturing Process

For the aluminum fins, the fins themselves we cut them from an aluminium sheet and then create the edge. The centering ring were sent to GLC CNC to manufacture since they have a complex geometry.

The booster fins are SRAD and manufactured in-house using a composite sandwich construction with tip-to-tip reinforcement. The manufacturing process consists of three main stages: skin layup under vacuum, sandwich layup, and tip-to-tip integration. We start by doing a wet layup of the fin skins on a sheet of glass. The glass provides a perfectly flat surface which gives a nice surface finish, flat fins and easy to demold. We layup one big panel of 3 plies with a sequence of [45/0/45]. We then let them cure under a perfect vacuum to remove all the excess resin and porosity to leave us with a strong, lightweight and thin sheet of carbon fiber. We then proceed to cut the shape of the skin and the shape of the core. We bond them together, again on the glass panel

using epoxy under a perfect vacuum. We are left with 4 sandwich fin with a sequence of [45/0/45/core/45/0/45] that are strong and light. The layup sequence, fiber orientations, and resin system are selected to maximize bending stiffness and flutter resistance while minimizing mass. The goal is to make a gap between the diverging frequency in flexion and in torsion so we need to increase the stiffness in flexion with 45 degree plies. The fins are bonded to the external surface of the body tube using structural epoxy. A tip-to-tip wet layup is then applied across the fin roots and airframe to reinforce load transfer and improve resistance to aerodynamic and inertial loads.

Quality control is addressed through both process discipline and redundancy. Visual inspection is performed at each manufacturing step to identify voids, dry spots, fiber misalignment, or bonding defects. Dimensional checks are conducted after curing and finishing to ensure geometric conformity. To further mitigate manufacturing risk, a fifth fin is produced in addition to the four flight fins. This extra fin serves either as a contingency replacement in the event of a manufacturing defect or as a destructive test article, allowing evaluation of structural strength and, in particular, the adhesion quality between the printed core and composite skins.



Figure 31: Manufacturing of carbon sandwich fins

6.4.3 Analysis

Fin flutter is assessed using a semi-empirical analytical method based on the formulation presented in Apogee Rockets Newsletter 615, which relates the flutter velocity to

fin geometry, material properties, and atmospheric conditions. The flutter velocity is computed using the following expression:

$$V_f = a \times \sqrt{\frac{G_E}{\frac{39.3 \times A^3}{(\frac{t}{c})^3 \times (A+2)} \times \left(\frac{\lambda+1}{2}\right) \times \left(\frac{p}{p_0}\right)}} \tag{4}$$

This equation has been implemented in a Python-based tool with a dedicated user interface to facilitate parametric studies and reduce the risk of transcription or calculation errors. An image of this interface is shown in Figure 32.

Using the current fin geometries and material selections, the calculated flutter velocities are: Sustainer (1762 m/s) and Booster (955 m/s). These values correspond respectively to flutter safety factors of 2.22 and 2.09, relative to the maximum expected flight velocities. Both values exceed the team's design target of 1.5. In other to get the value for the booster fins we first needed to figure out their sheir modulus since this is what is needed to calcultate the divergence between the oscillation frequency en flexion and in torsion to get flutter speed. We did that using our composite calculator in our program Simuloutil. Using the value for our composite that were tested in lab and entering to ply sequence of the entire fin we can obtain the equivalent shear modulus of the whole composite. For our fin we get a value of 12.185 GPa. The figure 33 show the calculator interface we work with.

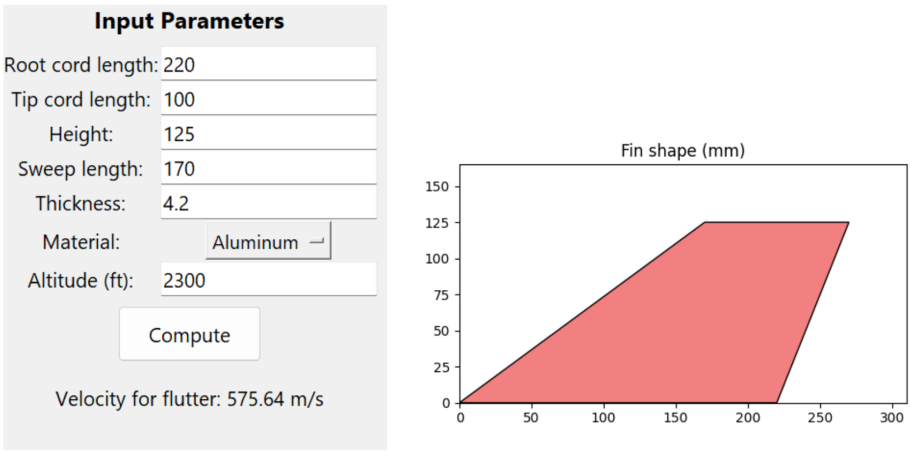


Figure 32: Flutter calculation interface

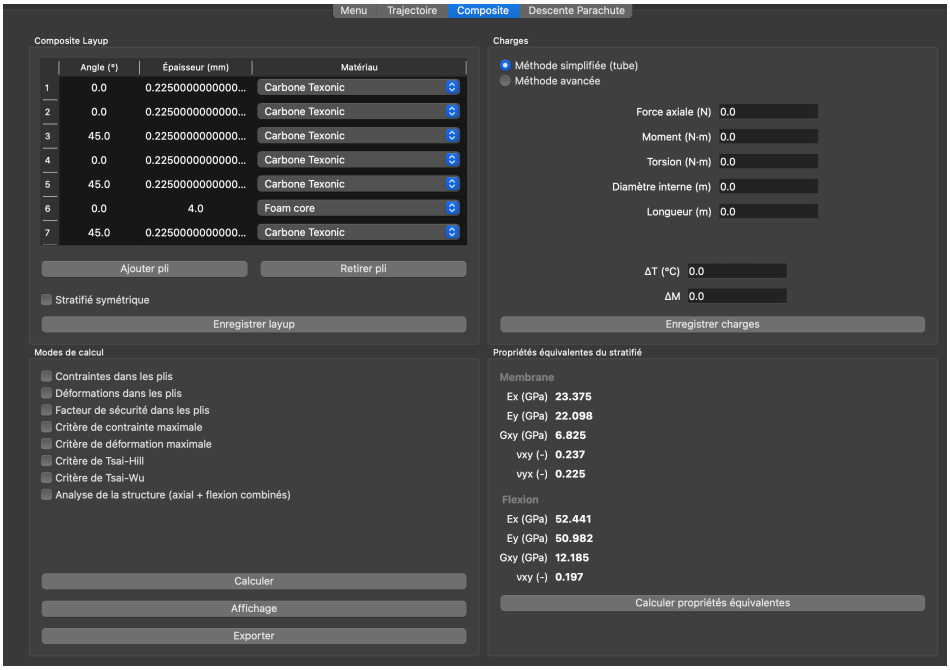


Figure 33: Composite calculator interface

In addition to flutter, the fins are analyzed for static strength under aerodynamic loading caused by vehicle angle of attack. When the rocket experiences an angle of attack, the fins generate lift forces that induce bending stresses both within the fin structure and at the fin–airframe interface. A conservative worst-case angle of attack of 10° is assumed for this analysis. While such a condition is not expected during nominal flight, it is used to ensure sufficient structural robustness under off-nominal scenarios. The analysis is performed at maximum dynamic pressure (max-q), representing the most severe aerodynamic loading environment. For simplicity and conservatism, each fin is modeled as a cantilever beam rigidly embedded in the airframe, with a concentrated lift force applied at approximately three-quarters of the fin span. The fin cross-section is approximated as rectangular with a constant thickness and a chord equal to the mean aerodynamic chord. Under these assumptions, the calculated lift force acting on a single fin is approximately 550 N. Using classical beam theory, the resulting maximum bending stress on the fin is calculated to be 82.5 MPa. For the sustainer fins, this stress level is well below the yield strength of Aluminum 6061-T6 (275 MPa), indicating that the fins remain within the elastic regime even under this hard loading scenario. This result demonstrates adequate structural strength and stiffness for the anticipated flight conditions.

Another analysis on the fins, but not concerning their structural resistance is a CFD analysis of different edge shape to better determine and quantify the effect of this parameter on the general aerodynamics.

6.4.4 Testing and Validation

The same composite tests has the airframe was useful for the booster fins. Without them we could not figure out the properties of the composite sandwich.

7 Propulsion

For this two-stage rocket, both motors are COTS. The first stage is powered by an O3400 motor manufactured by Cesaroni Technology Incorporated (CTI), which provides initial propulsion for the full vehicle stack, while the second stage is sustained by an N1000 motor manufactured by Aerotech. The main characteristics of both motors are summarized in Table 14, followed by their respective thrust curves obtained from thrustcurve.org.

Placement	Designation	Dimensions [mm]	Loaded weight [kg]	Total impulse [Ns]	Burn time [s]	Maximum thrust [N]	Propellant
Booster	O3400	98 x 1239	16.842	21 062	6.2	4750	Imax
Sustainer	N1000	98 x 1046	12.777	14 138	13.4	2120	White lightning

Table 14: Motor specifications

We then obtain from the summation of both motors that the total impulse of this complex rocket is of 35 200 Ns.

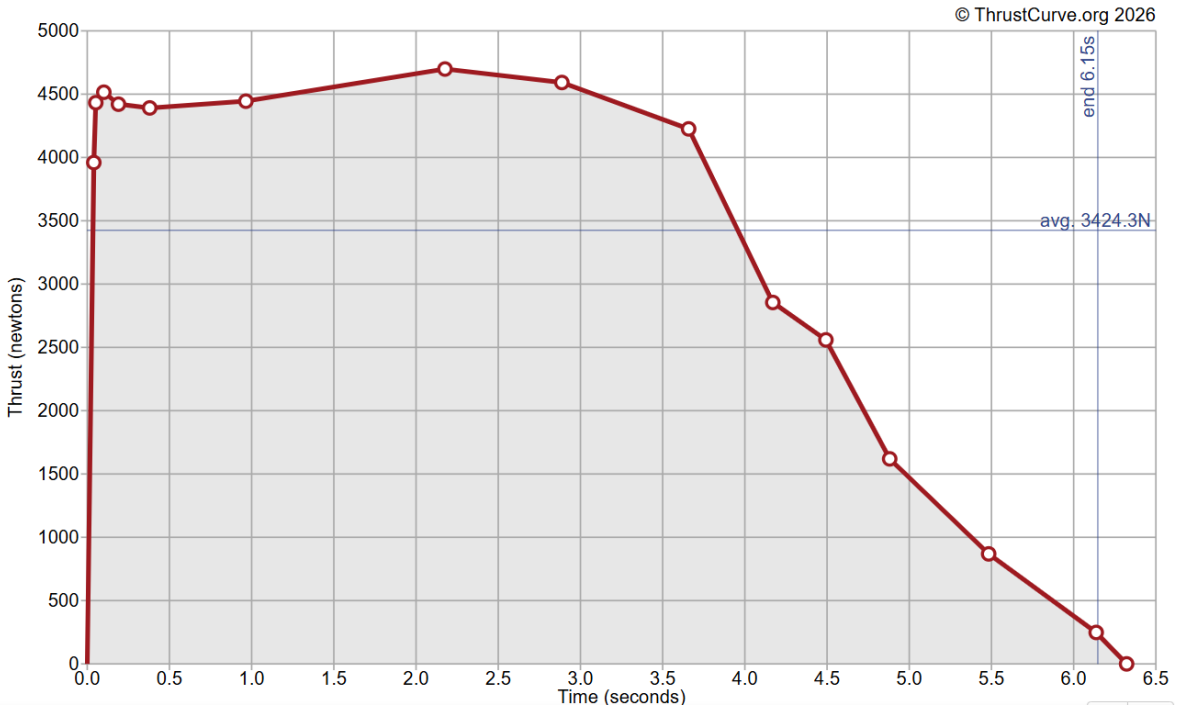


Figure 34: O3400 thrust curve

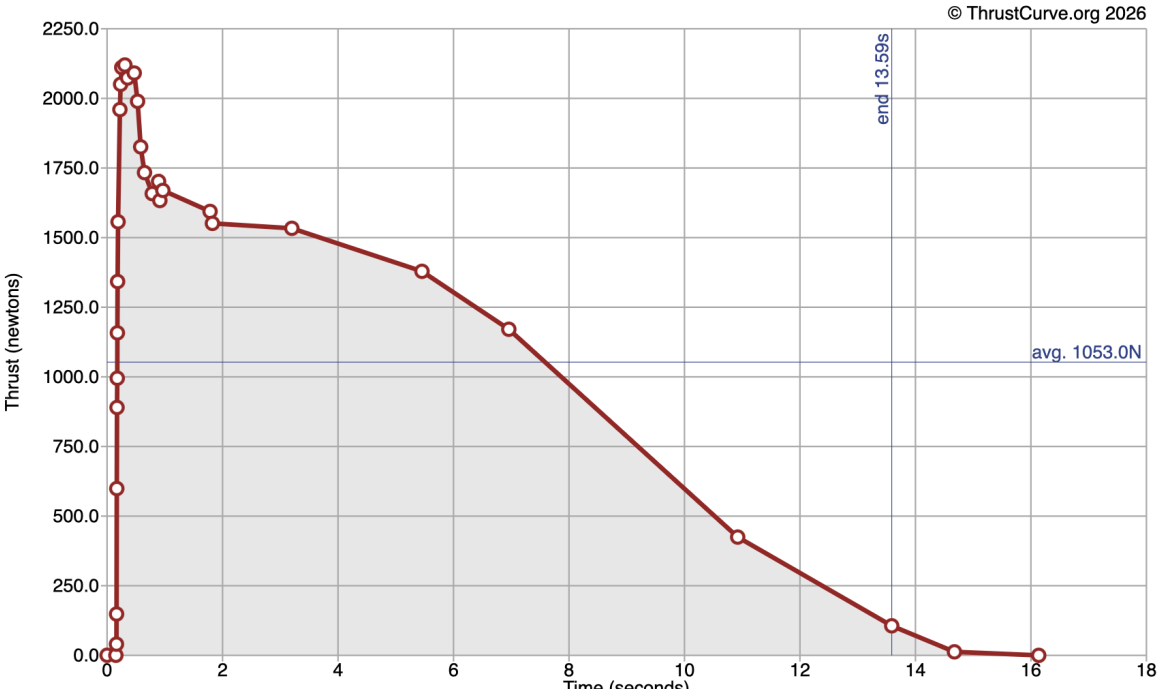


Figure 35: N1000 thrust curve

The selection of these motors was primarily driven by their availability, the booster motor is manufactured by cesaroni but they did not manufacture the initial choice for a sustainer motor. We then decided to change for the N1000 aerotech motor that we had from IREC but did not fire. In addition, preliminary flight performance assessments conducted using OpenRocket simulations indicated that this specific motor combination best met the mission objectives

8 Recovery

8.1 Deployment Systems

Requirement	Description
Electronic triggering	System must be electronically activated
Redundancy	Maximize redundancy to minimize failure risk
Environmental tolerance	Withstand acceleration, pressure, temperature, and vibration
Ignition reliability	Reliable ignition mechanism for deployment sequence

Design

The recovery system relies on three independent deployment mechanisms: a high-pressure carbon dioxide (CO_2) system, a black powder charge well, and a line cutter system.

CO_2 **system:** Now in its fifth design iteration, the SRAD CO_2 system has been progressively refined for improved reliability in rupturing the cartridge. It uses a 25 g high-pressure CO_2 cartridge, of the type used for bicycle tire inflation, threaded onto a custom base mounted to the rocket's bulkhead. A small black powder chamber with an electric igniter is integrated into this base. Unlike earlier versions, the piercing spike is no longer spring-retained; instead, it is held in place by the friction of a tightened o-ring within the tube, eliminating the spring as a failure point. The spike length was also revised to better maintain alignment with the weakened section of the cartridge throughout its stroke. The gas relief port is positioned such that black powder combustion gases are only released once the spike has completed its stroke and pierced the cartridge; the released CO_2 then pushes the spike back down its channel, allowing it to exit through lateral ports and actuate the deployment mechanism. Because this system is used for the high-altitude sustainer separation event, its small black powder charge is fully contained within an isolated chamber.

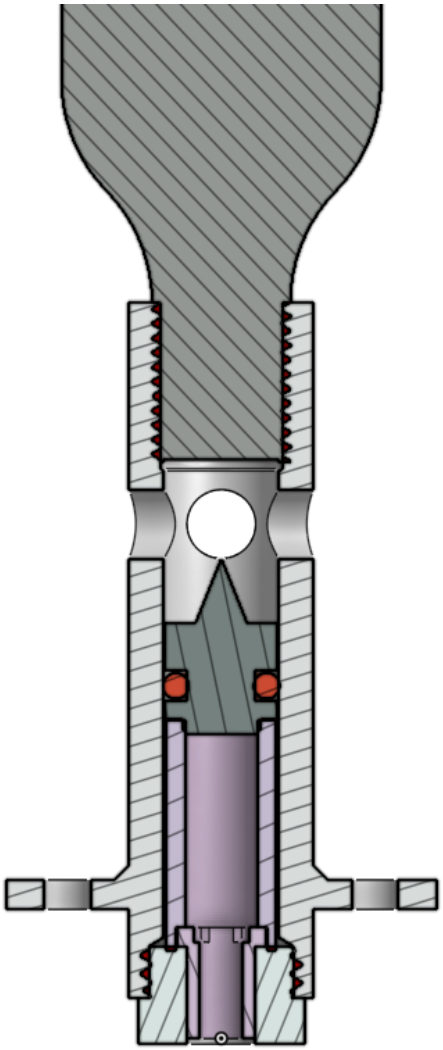


Figure 36: Cross section of the assembly of the SRAD CO2 deployment system

Black powder system: Stage separation between the booster and sustainer is achieved using black powder charge wells, 3D-printed in Onyx. A calculated and empirically tested quantity of black powder is loaded into each well to produce reliable separation at this lower-altitude event. The charge is compressed against a paper-based insulating wadding, and sealed at the top with aluminum tape. Aluminum tape was selected specifically because it is more resistant to rupture than lighter alternatives, which allows pressure to build further before the seal breaks, resulting in more complete combustion of the black powder charge and a higher peak pressure at separation.

Line cutter system: The line cutter system releases the main parachute later in the descent, at 1500 ft. The team previously used Hallfcat Rocketry line cutters for two competitions but experienced reliability issues, including black powder leaking out of the unit prior to activation due to insufficient sealing. In response, the team developed

an in-house line cutter design, based directly on the CO_2 system architecture the team has refined over two years of iteration. Like the CO_2 system, it uses a piercing spike, an o-ring, and a gas relief port positioned to vent only once the spike has completed its stroke. The SRAD line cutter has since been tested and performs reliably.

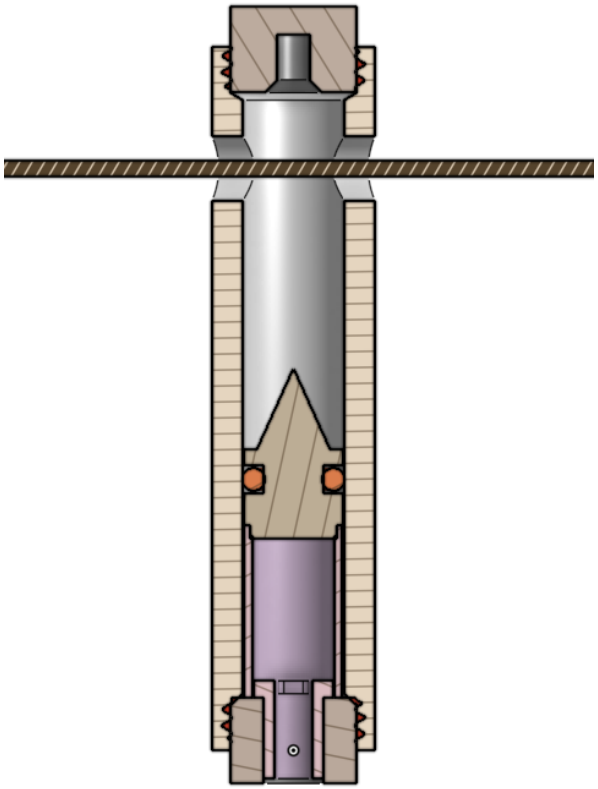


Figure 37: Cross section of the assembly of the SRAD line cutter system

For redundancy, each deployment stage is equipped with two identical systems of the same type, the second triggered a few seconds after the first as a backup in case of failure.

8.2 Parachutes and Harness

Sustainer Deployment Architecture: The sustainer carries a single parachute bay, located in the nose cone, housing both the drogue and main parachutes. Both parachutes are deployed simultaneously at apogee; the main parachute remains closed, restrained by the line cutter system, until it is released at 1500 ft.

Sustainer Parachutes: The drogue is an 18 in Fruity Chutes elliptical parachute ($C_D = 1.55$), giving a descent velocity of 26.5 m/s at the main deployment altitude.

The main is a 36 in Rocketman parachute ($C_D = 2.2$), giving a landing velocity of 10.05 m/s. The main parachute size was selected to keep the sustainer within the required landing speed even under a worst-case main-at-apogee deployment scenario: the sustainer's landing velocity approaches, but stays under, the 12 m/s landing speed limit even with the added descent rate that would result from a heavier effective mass in that contingency. This margin also accounts for the requirement to remain within the designated airspace, since a slower descent increases exposure to wind drift.

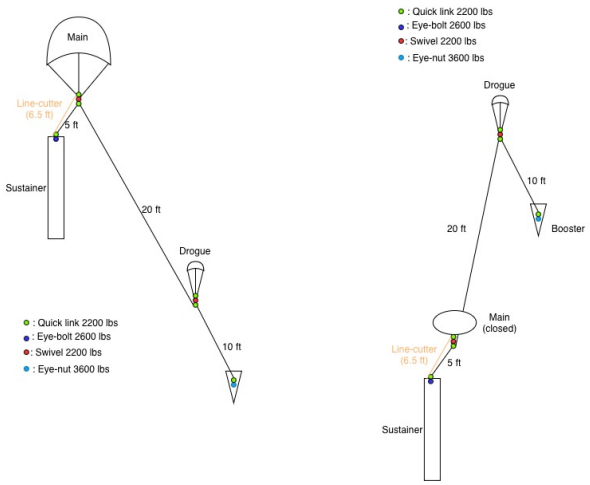


Figure 38: Recovery harness configuration for the sustainer

The recovery harness on the sustainer uses 1/4 in Kevlar shock cord, rated to 4000 lbf.

Booster Deployment Architecture: The booster similarly carries a single parachute bay, with both drogue and main parachutes deployed simultaneously at apogee; the main remains closed until released by the line cutter system.

Booster Parachutes: The drogue is a 24 in Fruity Chutes elliptical parachute ($C_D = 1.55$), giving a descent velocity of 20.11 m/s. The main is an 84 in Fruity Chutes parachute ($C_D = 2.2$), giving a landing velocity of 5.18 m/s. Lower descent velocities were targeted for the booster relative to the sustainer, since the booster does not carry the payload or face the same drift-related airspace constraint; the priority for the booster is instead to minimize impact loads and reduce the risk of structural damage on landing.

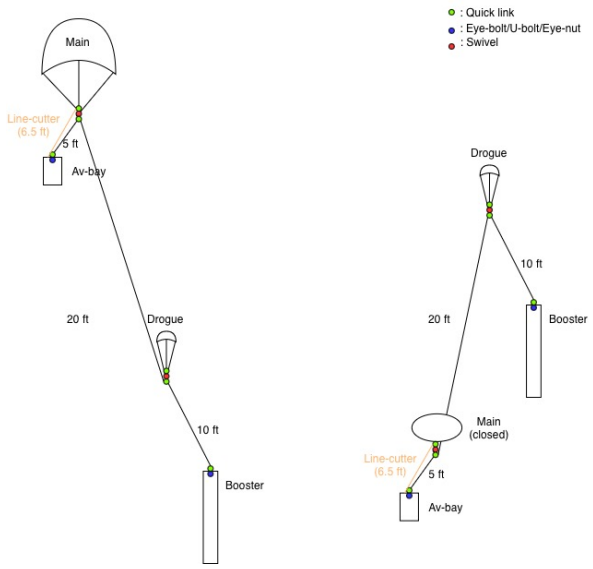


Figure 39: Recovery harness configuration for the booster

The booster recovery harness also uses 1/4 in Kevlar shock cord, rated to 4000 lbf.

Analysis

The load at each bulkhead anchor point is calculated using the two-anchor rope load distribution equation derived in Appendix F, applied to the worst-case deployment scenario for each stage. The resulting worst-case anchor loads are summarized in Table 15.

Anchor point	**Worst-case force (N)**
Sustainer anchor	1000
Booster, avionics side	600
Booster, motor side	2000

Table 15: Worst-case recovery harness anchor loads by deployment event.

Additional recovery-related analysis, specifically wind drift dispersion simulation, is presented separately in Section 5.2.

9 Avionics, Telemetry and Software

9.1 Avionics Overview

The rocket uses two identical avionics bays, one installed in each stage. The hardware configuration is identical for both bays, while the firmware and flight computer configuration are adapted to the specific role of each stage. This standardization simplifies manufacturing, assembly, and pre-flight verification.

The primary recovery system is based on two independent COTS flight computers. A Featherweight Blue Raven performs all nominal deployment and staging events, while a PerfectFlite Stratologger provides redundant parachute deployment.

Tracking capability is provided through multiple independent systems. A Featherweight GPS Tracker provides the primary real-time tracking solution and is used for Launch Canada altitude scoring. An SRAD IdéFIX 433 MHz beacon enables short-range directional tracking during the final recovery phase, while an SRAD ODBlix flight computer transmits real-time telemetry and records flight data for post-flight analysis. A SPOT Trace satellite tracker provides a last resort recovery solution.

Each stage also carries a RunCam Split 4 camera for onboard video recording.

As for extra documentation : Altimeter programming settings are detailed in appendix B, full wiring diagrams are included in appendix C and avionics specific checklists are provided in appendix D. The bay is depicted in figure 40.

Table 16 summarizes the avionics hardware installed in each avionics bay.

Table 16: Avionics hardware summary

Category	**Components**
Deployment altimeters	Featherweight Blue Raven PerfectFlite Stratologger
Tracking & telemetry	Featherweight GPS Tracker SPOT Trace IdéFIX beacon (SRAD) ODBlix flight computer (SRAD)
Camera	RunCam Split 4
Power and arming	SRAD arming PCB (8 switches) Three push-pins SRAD 2S 16.4 Ah Li-ion battery pack SRAD 2S 4.2 Ah Li-ion battery pack 9 V alkaline battery

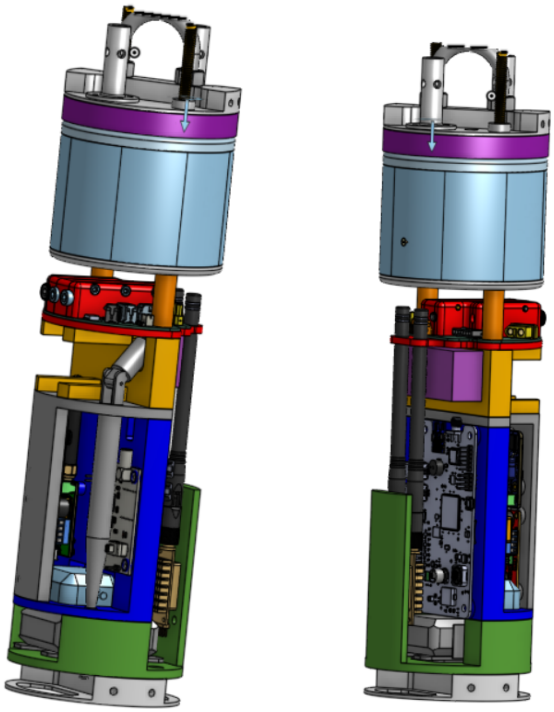


Figure 40: View of the avionics bay

9.2 Deployment Altimeters

Each avionics bay contains two independent commercial flight computers: a Featherweight Blue Raven and a PerfectFlite Stratologger. The hardware configuration is identical in both stages; only the flight configuration differs. The detailled programming setting of all the following altimeters are shown in appendix B.

9.2.1 Featherweight Blue Raven

The Featherweight Blue Raven is the primary flight computer. It combines a barometric altimeter, a 9-axis IMU, onboard flight logging, four programmable pyrotechnic outputs, and Bluetooth configuration.

Assigned functions
Booster

- Primary drogue deployment
- Primary main deployment
- Stage separation charge deployment

Sustainer

- Primary drogue deployment
- Primary main deployment
- Sustainer motor ignition

The Blue Raven was selected because its 9-axis IMU enables IMU-based apogee detection. This provides a deployment method independent of barometric measurements, which can become unreliable at high altitude.

9.2.2 PerfectFlite Stratologger

The PerfectFlite Stratologger is the redundant deployment altimeter. It provides barometric altitude measurement, dual-deployment capability, and onboard flight logging.

Assigned functions
Booster

- Redundant drogue deployment
- Redundant main deployment

Sustainer

- Redundant drogue deployment
- Redundant main deployment

The Stratologger was selected based on its demonstrated reliability over several years of operation within the team. It is also rated for operation up to 100,000 ft, making it suitable for our target altitude.

9.3 Telemetry and Tracking Solutions

Each stage carries multiple independent tracking systems. The primary tracking solution is a commercial GPS tracker used for both recovery and competition scoring. Additional tracking systems provide independent recovery capability and real-time telemetry.

9.3.1 Featherweight GPS Tracker

The Featherweight GPS Tracker is the primary tracking system. It determines the vehicle position using GPS and transmits it to a dedicated ground station through a long-range LoRa radio link. Position data is displayed through the Featherweight mobile application.

The Featherweight GPS Tracker is also used as the official altitude scoring system for Launch Canada, since barometric altimeters cannot provide a valid competition score above 60,000 ft.

9.3.2 SPOT Trace

The SPOT Trace is an independent commercial satellite GPS tracker. It periodically transmits the vehicle position through the Globalstar satellite network and does not rely on the ground telemetry station.

The SPOT Trace serves as the last-resort recovery system in the event of a failure of all primary tracking systems.

9.3.3 IdéFIX Beacon

IdéFIX is a student-developed 433 MHz tracking beacon. It periodically transmits a GFSK-modulated beacon containing the team callsign and is tracked using a handheld directional Yagi antenna.

The beacon is intended for short-range recovery after landing, particularly when the rocket is concealed by dense vegetation during the final few hundred metres of recovery.

9.3.4 ODBlix Flight Computer

ODBlix is a student-developed flight computer currently undergoing flight qualification. It records flight data using onboard inertial, barometric, and GPS sensors while transmitting real-time telemetry through an external RFD900x radio modem. The deployment algorithm implemented on ODBlix is also evaluated during flight.

Table 17: RF transmitters

Device	**Stage**	**Frequency band**	**Bandwidth**	**TX Power**
Featherweight GPS Tracker	Sustainer	915 MHz	800 kHz	100 mW
RFD900x		915 MHz	200 kHZ	1W
IdéFIX Beacon		433 MHz	100 kHZ	100 mW
SPOT Trace		1615 MHz	-	300 mW
Featherweight GPS Tracker	Booster	915 MHz	800 kHz	100 mW
RFD900x		915 MHz	200 kHZ	1W
IdéFIX Beacon		433 MHz	100 kHZ	100 mW
SPOT Trace		1615 MHz	-	300 mW

9.4 Power Distribution and Arming

Power distribution and arming are centralized on a custom arming PCB located in the avionics bay. The PCB distributes power to all subsystems and serves as the electrical interface between the batteries, flight computers, payloads, and pyrotechnic devices. The PCB can be seen in figure 41. They are currently fully assembled and tested (continuity, correct switching, reverse polarity protection).

The arming system consists of three push-pins operating a total of eight switches. Each electrical interface has a dedicated connector on the PCB, eliminating the possibility of wiring errors due to fatigue or stress during the final assembly.

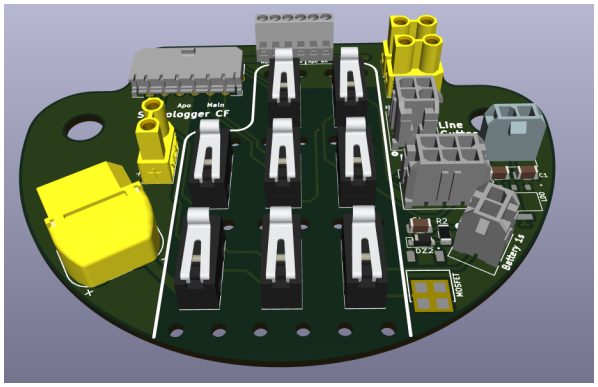


Figure 41: View of the bare arming PCB. A 3D printed assembly covers the switches and acts as the casing for the push-pins (not seen on the picture).

9.4.1 Power Distribution

The avionics are divided into three electrically independent power circuits.

- **2S 16.4 Ah Li-ion battery** — Telemetry and tracking circuit:
  - ODBlix flight computer
  - Featherweight GPS Tracker
  - IdéFIX beacon
  - RunCam Split 4

  The Featherweight GPS Tracker is supplied through a dedicated 4.2 V LDO voltage regulator integrated on the arming PCB.
- **2S 4.2 Ah Li-ion battery** — Featherweight Blue Raven.
- **9 V alkaline battery** — PerfectFlite Stratologger.

The recovery system is electrically independent from the telemetry and tracking system. The biggest flaw of this design is that all tracking solutions are on the same circuit, exposing a single failure point for all tracking systems. It is for this reason that the SPOT Trace tracker is carried as a completely independent last resort GPS tracking system.

9.4.2 Arming Functions

The eight switches perform the following functions. The specific layout of the switches on the arming PCB is detailed in the wiring diagrams in appendix C:

- Shunt primary drogue e-match

- Shunt redundant drogue e-match
- Shunt primary main e-match
- Shunt redundant main e-match
- Shunt sustainer igniter (sustainer) or stage separation charge (booster)
- Power telemetry and tracking circuit
- Power Featherweight Blue Raven
- Switch PerfectFlite Stratologger

Complete wiring diagrams are provided in Appendix C.

Table 18: Power circuits and battery life estimation

Circuit	**Voltage**	**Capacity**	**Average Current**	**Expected Battery Life**
Telemetry and tracking	7V4	16.8 Ah	1.8 A	9h20
Blue Raven	7V4	4.2 Ah	20 mA	210h
Stratologger	9V	450 mAh	1.5 mA	300h

10 Ground Support Equipment

10.1 Ground Station

The ground station was designed to receive, decode, and visualize telemetry data transmitted from the rocket during flight. It also provided local atmospheric measurements for comparison and redundancy. The system was engineered around a modular and scalable architecture to facilitate integration, robustness, and expandability.

- **Main Computing Unit**

  A Raspberry Pi 4 Model B serves as the primary computing platform. It manages the user interface, receives incoming telemetry via USB-connected RFD900x radio, and communicates with the custom ground station board through a GPIO bus.

- **Telemetry Reception**

  The RFD900x radio modem receives live telemetry data streamed from the ODBlix flight computers onboard the rocket. Data packets included altitude, velocity, GPS coordinates, and system status. A custom software interface on the Raspberry Pi processes and displays the incoming data in real time on 2 LCDs and a touch screen.

- **Environmental Sensing**

  The ground station includes local environmental sensors:

  - NEO-8M GPS Module for obtaining relative ground position and time synchronization.
  - BMP280 Barometric Sensor for local pressure and altitude readings.

- **User Interface and Displays**

  Multiple I2C displays were connected to the ground station board to provide secondary visual outputs. This allowed quick access to critical parameters such as altitude, battery voltage, or GPS coordinates without relying solely on the touch screen.

- **Cooling and Thermal Management**

  Cooling fans powered at 5V were integrated to maintain operational temperatures of critical components during long mission durations.

10.2 IdéFIX substation

The IdéFIX substations is a portable device designed to locate the rocket based on transmitting power on the 433 MHz band. It will assist the recovery team during the

final steps of the recovery or when the team is within a 200m range from the rocket. The receiver provides two methods to locate the rocket: either by reading the precise GPS coordinates sent by the beacon, or by triangulating the signal using a directional Yagi antenna. The IdéFIX system has a modular design, allowing for fast development, and future scalability.

- **Radio Transceiver Module** The radio module contains all the essential electronics, including the STM32 microcontroller, power supplies, various interface ports, and the RFM22B radio transceiver. This modules can function both as the beacon and as the receiver: a standalone module is placed in the rocket as the beacon, and another module is used in the substation as the receiver.

- **User Interface** An interface board is connected to the radio module in the substation to make the user interface more convenient. 4 buttons are used to control the substation and navigate in the interface. A buzzer is used to indicate signal strength. Finally, an I2C LCD display is also included in the interface for displaying the RX frequency, signal strength, and received GPS coordinates.

- **Yagi Antenna** The Yagi antenna is a foldable directional antenna with a gain of 7dBi. The recovery team is able to point the antenna in any direction, and read the received signal strength (RSSI) from the LCD screen. By pointing the antenna in various directions and moving frequently, the team is able to triangulate the rocket's position.

11 Range safety

For range safety during the flight, there is a designated place at the competition for the team to be during the entirety of the flight. We will also limit the number of people on the pad during the arming phase to make sure we don't put unnecessary team members at risk.

As for the features on our vehicle, our arming system is composed of push-pins. Those pins are put into the rocket when it is ready to fly and arms all electronic by shutting on limit switches in the rocket. This ensures that the electronics is activated in a secure manner on the pad and not anytime before.

Furthermore, numerous checklists, ranging from the preparation of the separation charges to the assembly of the avionics bay, were produced and are presented in Appendix D and Appendix H.

11.1 Hazard Analysis

The following table enumerates the potential hazards that may be encountered and the measures taken to mitigate the risks.

Hazard	Possible Causes	Risk of Mishap	Mitigation Approach	Residual Risk
Motor explosion during launch	Grain cracks, structural failure, overpressure	Medium – limited testing and student manufacturing	Pressure testing of motor casing, visual inspection of grain, use of non-fragmenting materials, restricted safety perimeter during launch	Low
Stage separation failure	Improper charge sizing, mechanical misalignment, wiring failure	Medium – critical event with multiple failure points	Ground testing of separation system, redundant charges where possible, strict arming procedures	Low
Premature upper stage ignition	Avionics error, wiring fault, accidental arming	High – could occur near personnel	Safe/arm systems, continuity checks, ignition only enabled at final countdown	Low
Failure of upper stage ignition	Igniter failure, battery failure, timing error	Medium – mission failure but limited safety impact	Redundant ignition checks, battery testing, verified sequencing logic	Low
Recovery system failure	Improper packing, charge failure, altimeter malfunction	High – uncontrolled descent	Redundant deployment charges, ground testing, proper packing procedures	Medium
Rocket deviates from flight path	Aerodynamic instability, wind, CG/CP miscalculation	High – risk to personnel	Stability simulations, launch constraints (wind limits), safety exclusion zone	Low
Hang fire (delayed ignition)	Ignition delay or partial ignition	High – personnel may approach rocket	Strict launch protocol, minimum wait time before approach, remote ignition	Low
Pyrotechnic accidental ignition	Static discharge, wiring errors, premature connection	High – handling hazard	Shorting plugs, delayed connection, trained personnel only	Low
Exposure to composite dust	Sanding fiberglass/carbon fiber	Medium – health hazard over time	Use of respirators, designated ventilated workspace	Low
Black powder mishandling	Improper storage or transport	Medium – fire/explosion hazard	Storage in fire-resistant locked container, compliance with transport regulations	Low

11.2 Risk Assessment

The following table enumerates the potential risks that may be encountered and the measures taken to mitigate them.

Failure Mode	Mission Phase	Cause	Likelihood	Impact	Mitigation Approach
Ignition failure (stage 1)	Launch	Igniter failure, wiring issue	Low	High – no liftoff	Pre-launch continuity checks, igniter redundancy, battery verification
Thrust underperformance	Ascent	Improper grain casting, pressure loss	Medium	High – reduced altitude	Static motor testing, grain inspection, design margin
Aerodynamic instability	Ascent	CG/CP miscalculation, fin misalignment	Low	High – trajectory deviation	Stability simulation (OpenRocket), CG verification before launch
Stage separation failure	Stage separation	Charge failure, mechanical issue	Medium	High – upper stage unusable	Ground testing, redundant charges, robust mechanical design
Upper stage ignition failure	Stage separation	Igniter or avionics failure	Medium	High – mission incomplete	Redundant ignition checks, validated timing logic
Avionics malfunction	All phases	Software bug, wiring fault, power loss	Medium	High – loss of control/data	Code verification, secure wiring, battery testing
Altimeter failure	Recovery	Sensor failure, improper configuration	Medium	High – parachute not deployed	Redundant altimeters, pre-flight configuration check
Parachute deployment failure	Recovery	Packing error, charge failure	Medium	High – vehicle loss	Redundant deployment charges, packing verification
Excessive drift during descent	Recovery	High winds, late deployment	Medium	Medium – difficult recovery	Wind constraints, deployment altitude tuning
Structural failure during flight	Ascent	Material defect, excessive load	Low	High – mission loss	Structural analysis, safety factor in design

12 References

[1] Launch Canada, *Launch Canada Rules and Requirements Guide*, Revision R3, 2026.

[2] NASA, 2016. *NASA Systems Engineering Handbook*. [pdf] [online] Available at: <Nasa Systems Engineering Handbook>

[3] Launch Canada, *Launch Canada's Design, Test & Evaluation Guide*, Revision R4, 2025.

Appendix A: Detailed Design Trade Studies and Calculations

12.1 Bulkheads

This section regroups the FEA results of the bulkheads

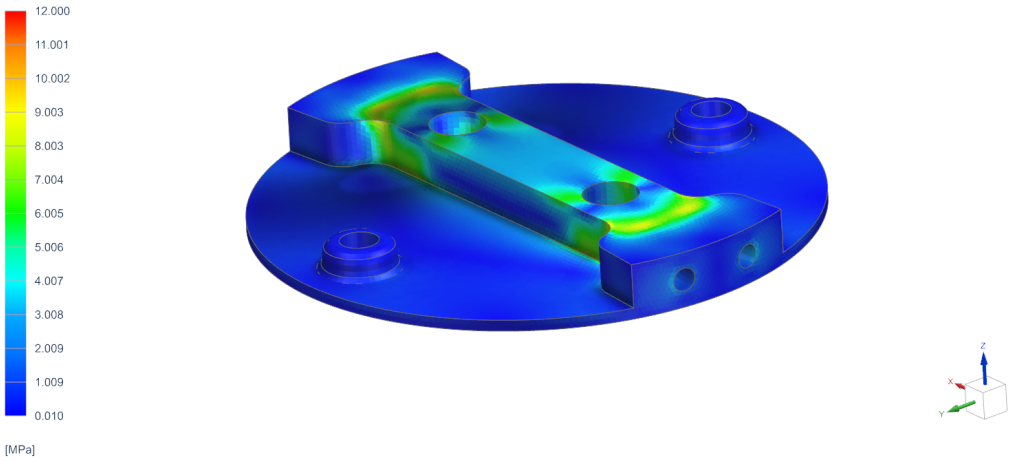


Figure 42: FEA - Bottom bulkhead booster avionic bay - isometric view



Figure 43: FEA - Bottom bulkhead booster avionic bay - side view

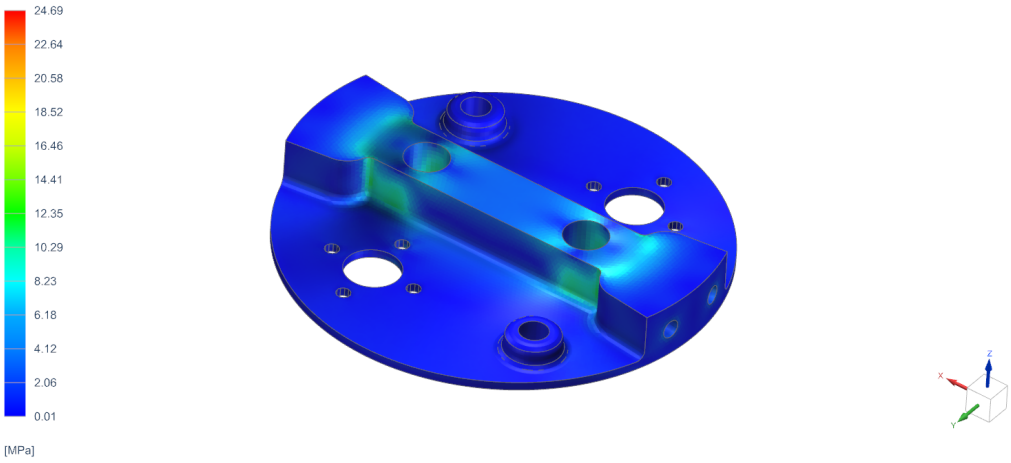


Figure 44: FEA - Top bulkhead sustainer avionic bay - isometric view

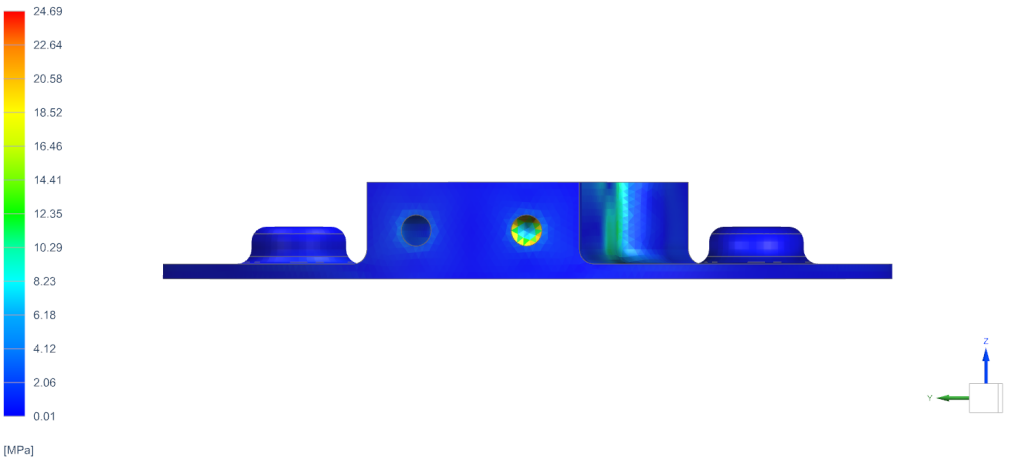


Figure 45: FEA - Top bulkhead sustainer avionic bay - side view

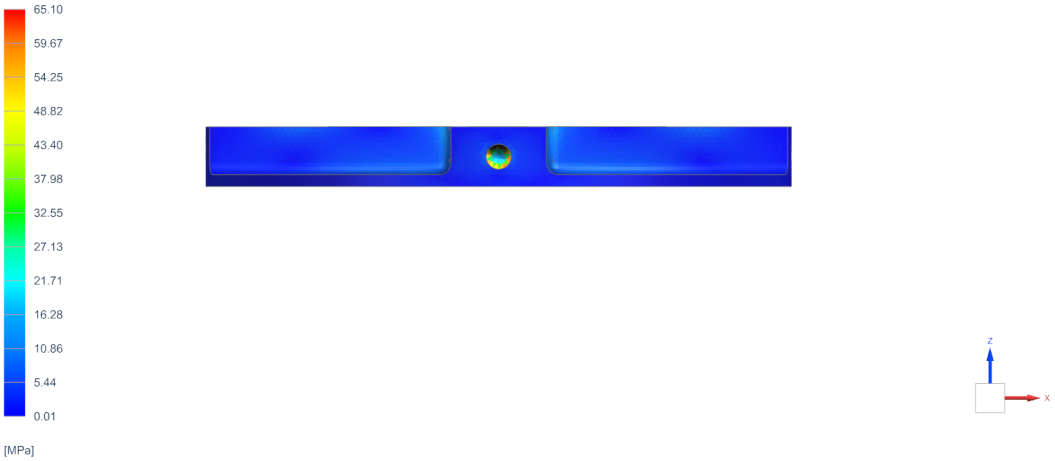


Figure 46: FEA - Bulkhead motor booster - side view

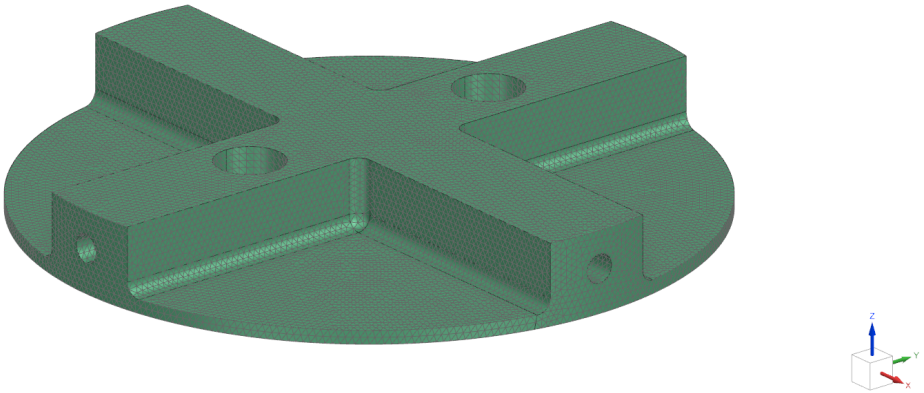


Figure 47: Example of the bulkhead mesh

Appendix B: Altimeter Programming

Blue Raven Booster

The booster's configuration is that of a standard sub-60k-feet dual deployment flight. The 3rd channel is used to deploy the redundant stage separation charge. Since we want the stage to separate no matter what, only a timer after liftoff is used to deploy this charge. There is no altitude, velocity or tilt angle check. This was recommended to us by the staff at IREC 2026 for our last double stage project, Ozyman.

Event Name	**Trigger Type**	**Trigger Setting**	**Output Channel**
Stage separation	Timer	6.5 sec after liftoff	3rd
Apogee Drogue Deployment	Mutli-sensor Fault Tolerant	–	Apo
Main Parachute Deployment	Barometric Altitude Detection	1500 ft	Main

Table 19: Flight Event Configuration - Blue Raven Booster

Parameter	**Value**	**Units and Typical Use Case**
TVAL2 (Enable output after)	> 6.4 s	Allows outputs only after this time delay from liftoff.
TVAL1 (Inhibit output after)	< 6.6 s	Prevents outputs from firing too long after liftoff.
NOMANG (Maximum nominal ascent tilt)	$< 60.0°$	Maximum tilt angle during ascent; becomes false after violation.
BVEL (Barometric downward velocity)	< -200 ft/s	Early deployment detection if the rocket is descending too rapidly.
BURNTCOUNT (Motor burnout counter)	> 1	Burnout counter used as part of the staging logic.

Table 20: 3rd Channel Costum Configuration - Blue Raven Booster

Blue Raven Sustainer

The sustainer is configured for a 100k feet flight. The third channel ignites the sustainer's motor and is configured as follows. In opposition to the stage separation charge, we want the motor to ignite only if the flight is nominal. For this reason, there is an altitude, velocity and tilt angle check for the motor ignition.

Event Name	**Trigger Type**	**Trigger Setting**	**Output Channel**
Sustainer Motor Ignition	Timer	7.1 sec after liftoff	3rd
Apogee Drogue Deployment	Mutli-sensor Fault Tolerant	–	Apo
Main Parachute Deployment	Barometric Altitude Detection	1500 ft	Main

Table 21: Flight Event Configuration - Blue Raven Sustainer

Parameter	**Value**	**Units and Typical Use Case**
AGL2 (Staging ignition altitude)	> 6200 ft	Barometric altitude for altitude-based staging ignition.
FANG1 (Current tilt angle limit)	$< 20.0°$	Maximum allowable current tilt angle during nominal ascent.
NOMANG (Maximum nominal ascent tilt)	$< 60.0°$	Maximum tilt angle during ascent; becomes false after violation.
TVAL2 (Enable output after)	> 7.1 s	Allows outputs only after this time delay from liftoff.
TVAL1 (Inhibit output after)	< 7.3 s	Prevents outputs from firing too long after liftoff.
VEL2 (Minimum stage ignition velocity)	> 300 ft/s	Prevents stage ignition if the rocket velocity is too low.
BVEL (Barometric downward velocity)	< -200 ft/s	Early deployment detection if the rocket is descending too rapidly.
BURNTCOUNT (Motor burnout counter)	> 1	Burnout counter used as part of the staging logic.

Table 22: 3rd Channel Costum Configuration - Blue Raven Sustainer

Stratologger Booster

Event Name	Trigger Type	Trigger setting	Output Channel
Booster Redundant Apogee Deployment	Barometric Apogee Detection	0.5 s delay	Apo
Booster Redundant Main Chute Deployment	Barometric Altitude Detection	1300 ft	Main

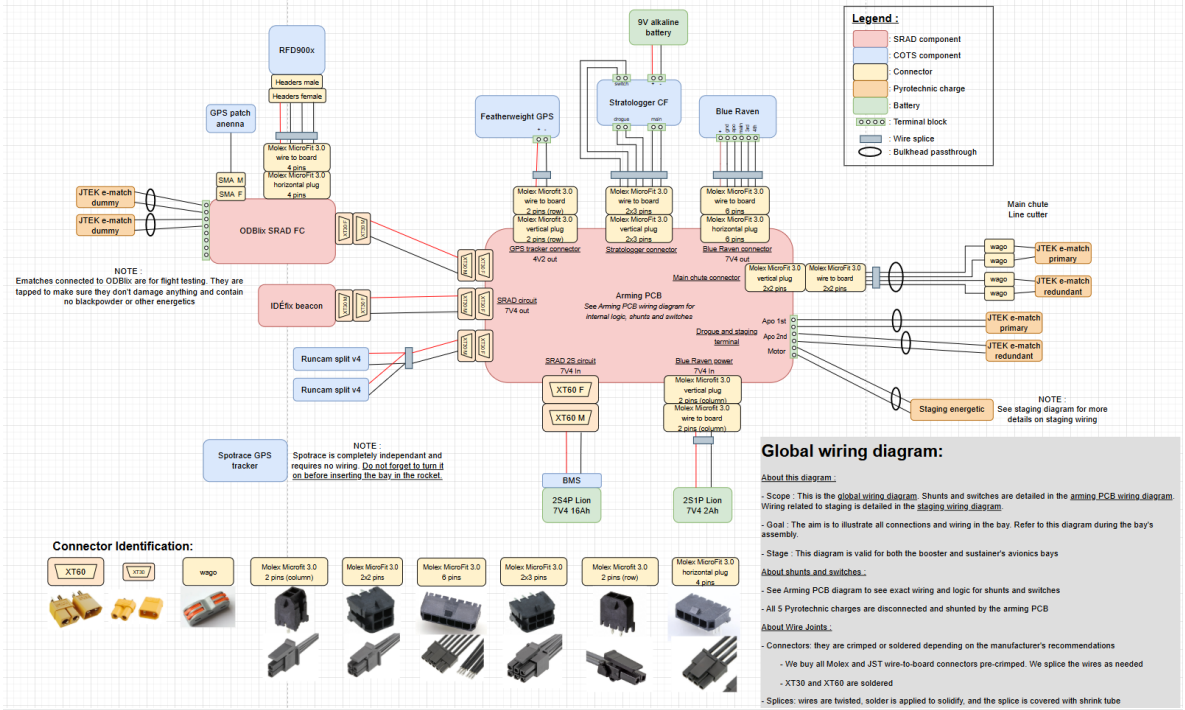
Table 23: Flight Event Configuration - Stratologger Booster

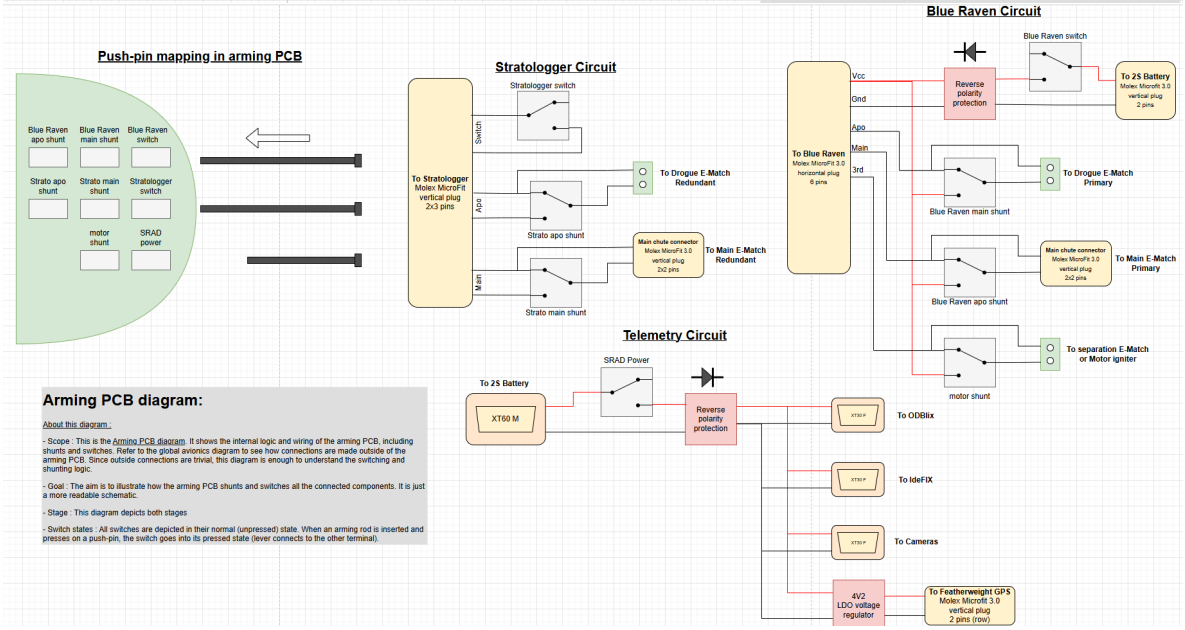
Stratologger Booster

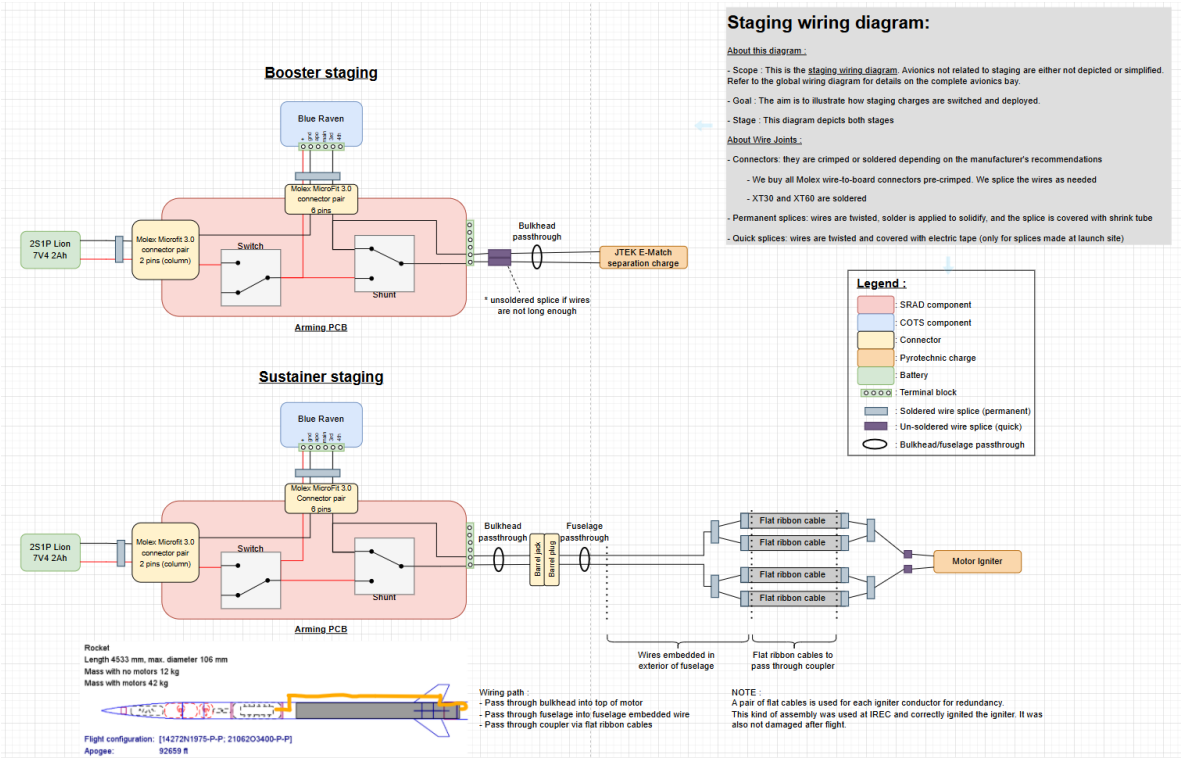
Event Name	Trigger Type	Trigger setting	Output Channel
Booster Redundant Apogee Deployment	Barometric Apogee Detection	0.5 s delay	Apo
Booster Redundant Main Chute Deployment	Barometric Altitude Detection	1300 ft	Main

Table 24: Flight Event Configuration - Stratologger Sustainer

Appendix C: Avionics Wiring Diagrams






Booster staging
Blue Raven
Molex MicroFit 3.0 connector pair 6 pins
2S1P Lion 7V4 2Ah
Molex Microfit 3.0 connector pair 2 pins (column)
Switch
Shunt
Arming PCB
Bulkhead passthrough
JTEK E-Match separation charge
* unsoldered splice if wires are not long enough
Staging wiring diagram:
About this diagram :
- Scope : This is the staging wiring diagram. Avionics not related to staging are either not depicted or simplified. Refer to the global wiring diagram for details on the complete avionics bay.
- Goal : The aim is to illustrate how staging charges are switched and deployed.
- Stage : This diagram depicts both stages
About Wire Joints :
- Connectors: they are crimped or soldered depending on the manufacturer's recommendations
- We buy all Molex wire-to-board connectors pre-crimped. We splice the wires as needed
- XT30 and XT60 are soldered
- Permanent splices: wires are twisted, solder is applied to solidify, and the splice is covered with shrink tube
- Quick splices: wires are twisted and covered with electric tape (only for splices made at launch site)
Legend :
: SRAD component
: COTS component
: Connector
: Pyrotechnic charge
: Battery
: Terminal block
: Soldered wire splice (permanent)
: Un-soldered wire splice (quick)
: Bulkhead/fuselage passthrough
Sustainer staging
Blue Raven
Molex MicroFit 3.0 Connector pair 6 pins
2S1P Lion 7V4 2Ah
Molex Microfit 3.0 connector pair 2 pins (column)
Switch
Shunt
Arming PCB
Bulkhead passthrough
Barrel jack
Barrel plug
Fuselage passthrough
Flat ribbon cable
Flat ribbon cable
Flat ribbon cable
Flat ribbon cable
Motor Igniter
Wires embedded in exterior of fuselage
Flat ribbon cables to pass through coupler
Rocket
Length 4533 mm, max. diameter 106 mm
Mass with no motors 12 kg
Mass with motors 42 kg
Flight configuration: [14272N1975-P-P; 21062O3400-P-P]
Apogee: 92659 ft
Wiring path :
- Pass through bulkhead into top of motor
- Pass through fuselage into fuselage embedded wire
- Pass through coupler via flat ribbon cables
NOTE :
A pair of flat cables is used for each igniter conductor for redundancy.
This kind of assembly was used at IREC and correctly ignited the igniter. It was also not damaged after flight.

Appendix D: Avionics Checklists

Elrina Avionics Checklist

Don't forget to charge battery before launch day and make ALL UP TEST BEFORE BLACK POWDER

- **Verification before leaving base camp :**

Verify igniter wiring : **BOOSTER**	**Verify igniter wiring :** **SUSTAINER**
Follow wiring diagram to verify other connections	**Follow wiring diagram to verify other connections**
Type : [« Igniter name » : « Altimeter » : « Channel »]	Type : [« Igniter name » : « Altimeter » : « Channel »]
- Connection list :	- Connection list :
☐ Primary apogee : Blue Raven : Apogee	☐ Primary apogee : Blue Raven : Apogee
☐ Secondary apogee : Stratologger : Apogee	☐ Secondary apogee : Stratologger : Apogee
☐ Primary main : Blue Raven : Main	☐ Primary main : Blue Raven : Main
☐ Secondary main : Stratologger : Main	☐ Secondary main : Stratologger : Main
☐ Primary separation : Blue Raven : 3rd	☐ Primary 2nd stage ignition : Blue Raven : 3rd

Batteries verifications	**BOOSTER**
Take measure with voltmeter	
☐ 2S Li-ion Batteries 7.4V (BlueRaven) – min 7.4V	V
☐ 9 V alkaline Batterie (Stratologger)	V
☐ 2S Li-ion Battery Pack 7,4 V (ODB, IdéFIX, Cameras, RFD-900X, Featherweight GPS)	V

Batteries verifications	SUSTAINER
Take measure with voltmeter	
☐ 2S Li-ion Batteries 7,4V (BlueRaven) – min 7,4V	V
☐ First 9 V alkaline Batterie (Stratologger)	V
☐ 2S Li-ion Battery Pack 7,4 V (ODB, IdéFIX, Cameras, RFD-900X, Featherweight GPS)	V

Batteries verifications	GROUND STATION & PAD
Take measure with voltmeter	

☐ Check Battery pack “Ambatakum” is full (Ground Station)

☐ Check Battery pack “Lilo” is full (Ground Station)

☐ Check Battery Pad Camera

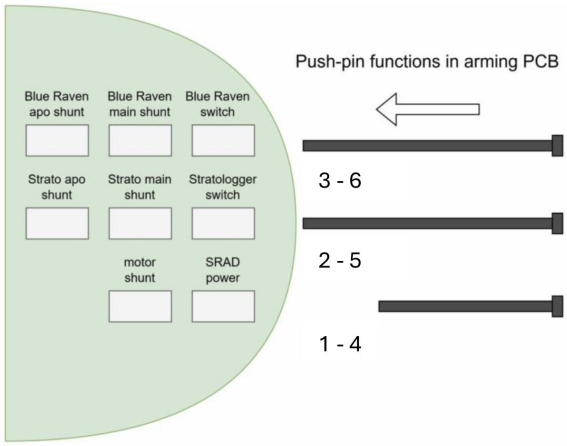
☐ Check Battery Featherweight Ground Station



SUSTAINER



BOOSTER



DON’T Forget material before leaving

All up test **WARNING ! MAKE BEFORE BLACK POWDER MATING**

SUSTAINER

☐ Insert pushpin 5

☐ Listen to Stratologger beeping **REFER BELOW**

☐ Verify programming

☐ Verify continuity

☐ Verify battery voltage (9,6V)

V

☐ Insert pushpin 6 **BEWARE OF MOTOR IGNIT**

☐ Verify BlueRaven startup

☐ Insert pushpin 4 halfway



☐ Verify Featherweight GPS startup

☐ Verify ODB/CAM/IDFX/RFD startup

☐ Insert pushpin 4 all the way in

☐ Verify shunt continuity

GROUND STATION

☐ Turn ground station on

☐ Turn IdéFIX Receiver on

☐ Verify IdéFIX is transmitting to receiver

☐ Verify ODB is transmitting to ground station

☐ Verify ODB data (GPS, altimeter, accelerometer)

☐ Verify ODB SD fix

BOOSTER

☐ Insert pushpin __2_

☐ Listen to Stratologger beeping **REFER BELOW**

☐ Verify programming

☐ Verify continuity

☐ Verify battery voltage (9,6V)

V

☐ Insert pushpin __2 - 3_

☐ Verify BlueRaven startup

☐ Verify Featherweight GPS startup

☐ Verify ODB/CAM/IDFX/RFD startup

CELLPHONE

For booster

☐ Verify Featherweight (Booster).
FIX with Cellphone

☐ Verify continuity Featherweight (Booster).

☐ Verify Featherweight GPS (Booster) battery voltage (7.4 V)

V

For sustainer

☐ Verify Featherweight (Sustainer)
FIX with Cellphone.

☐ Verify continuity Featherweight (Booster).

☐ Verify Featherweight (Sustainer) battery voltage (7.4 V)

V

STARTOLOGGER BEEP SEQUENCE

Digit	*Reported as:*
0	beep-beep-beep-beep-beep-beep-beep-beep-beep-beep
1	beep
2	beep-beep
3	beep-beep-beep
4	beep-beep-beep-beep
5	beep-beep-beep-beep-beep
6	beep-beep-beep-beep-beep-beep
7	beep-beep-beep-beep-beep-beep-beep
8	beep-beep-beep-beep-beep-beep-beep-beep
9	beep-beep-beep-beep-beep-beep-beep-beep-beep

Detected an anormal condition

Ok avec une croix si l'évènement n'arrive pas

Short siren + beep + siren	Total power loss during last flight	
Short siren + beep beep + siren	Momentary power loss during flight	
Short siren + beep beep beep+ siren	Drogue currrent exceeded 6 amps	
Short siren + beep beep beep beep + siren	Main current exceeded 6 amps	
Short siren + beep beep beep beep beep + siren	Problem detected with flash memory	
Short siren + beep beep beep beep beep beep + siren	Problem detected with pressure sensor DO NOT FLY	

Normal informations :

Ok avec croix si l'évènement arrive

Program set (7)	7x beep	
Digit main deploy altitude (1500)	1 x beep 5 x beep 1 x beep 1 x beep	
Ne pas tenir compte		
Check volatge battery (exemple 92 -> 9.2)	Voir bip 8.9 minimum	
Continuité dans tous les systèmes	Beep beep beep every 0,8 sec	

Ok pour stratologger ready for launch

DON'T Forget material before leaving

- **At Launch Site**

Bring laptop for Stratologger, Ground station for ODB, IdeFIX Receiver for IdeFix and cellphone for Featherweight on the launch pad, prepare sufficient battery.

Pre-launch pad (safety inspection)
☐ Wipe SD cards for cameras
☐ Wipe SD ODB
☐ Turn spot trace on
☐ Verify every connection (tirer sur chaque fil / verifier la solidité + wiring diagram)
☐ Check camera in good position
☐ Check Featherweight is in position “on” (switch)
☐ Mark connection with tape when verified
☐ Check wiring during assembly to prevent accident

Cheklist material for pad operation
☐ Ground station
☐ Laptop with Stratologger cable
☐ Security glasses
☐ Security gloves
☐ **Allen key**
☐ IdeFix receiver
☐ Cellphone with Featherweight app
☐ Table
☐ Tuck tape
☐ Pad camera

- **On the launch pad**

On the launch pad

SUSTAINER

☐ Insert pushpin 5

☐ Listen to Stratologger beeping **REFER BELOW**

☐ Verify programming

☐ Verify continuity

☐ Verify battery voltage (9,6V)

V

☐ Insert pushpin 6 **BEWARE OF MOTOR IGNIT**

☐ Verify BlueRaven startup

☐ Insert pushpin 4 halfway

☐ Verify Featherweight GPS startup

☐ Verify ODB/CAM/IDFX/RFD startup

☐ Insert pushpin 4 all the way in

☐ Verify shunt continuity

BOOSTER

☐ Insert pushpin __2_

☐ Listen to Stratologger beeping **REFER BELOW**

☐ Verify programming

☐ Verify continuity

☐ Verify battery voltage (9,6V)

V

☐ Insert pushpin __2 - 3_

☐ Verify BlueRaven startup

☐ Verify Featherweight GPS startup

☐ Verify ODB/CAM/IDFX/RFD startup

GROUND STATION

☐ Turn ground station on

☐ Turn IdéFIX Receiver on

☐ Verify IdéFIX is transmitting to receiver

☐ Verify ODB is transmitting to ground station

☐ Verify ODB data (GPS, altimeter, accelerometer)

☐ Verify ODB SD fix

CELLPHONE

For booster

☐ Verify Featherweight (Booster).
FIX with Cellphone

☐ Verify continuity Featherweight (Booster).

☐ Verify Featherweight GPS (Booster) battery voltage (7.4 V)

V

For sustainer

☐ Verify Featherweight (Sustainer)
FIX with Cellphone.

☐ Verify continuity Featherweight (Booster).

☐ Verify Featherweight (Sustainer) battery voltage (7.4 V)

V

Digit	*Reported as:*
0	beep-beep-beep-beep-beep-beep-beep-beep-beep-beep
1	beep
2	beep-beep
3	beep-beep-beep
4	beep-beep-beep-beep
5	beep-beep-beep-beep-beep
6	beep-beep-beep-beep-beep-beep
7	beep-beep-beep-beep-beep-beep-beep
8	beep-beep-beep-beep-beep-beep-beep-beep
9	beep-beep-beep-beep-beep-beep-beep-beep-beep

Detected an anormal condition

Ok avec une croix si l'évènement n'arrive pas

Short siren + beep + siren	Total power loss during last flight	
Short siren + beep beep + siren	Momentary power loss during flight	
Short siren + beep beep beep+ siren	Drogue currrent exceeded 6 amps	
Short siren + beep beep beep beep + siren	Main current exceeded 6 amps	
Short siren + beep beep beep beep beep + siren	Problem detected with flash memory	
Short siren + beep beep beep beep beep beep + siren	Problem detected with pressure sensor DO NOT FLY	

Normal informations :

Ok avec croix si l'évènement arrive

Program set (7)	7x beep	
Digit main deploy altitude (1500)	1 x beep 5 x beep 1 x beep 1 x beep	
Ne pas tenir compte		
Check volatge battery (exemple 92 -> 9.2)	Voir bip 8.9 minimum	
Continuité dans tous les systèmes	Beep beep beep every 0,8 sec	

Ok pour stratologger ready for launch

On the launch pad

CELLPHONE FINAL VERIF

BOOSTER

Blue Raven

☐ Verify continuity of each channel Battery (7.4) ____ V

☐ Verify flight program

Stratologger

☐ Verify continuity of each channel Battery (9.6) ____ V

☐ Verify flight program

Featherweight GPS

☐ Verify fix Ground station

☐ Verify fix cell phone Battery (7.4) ____ V

☐ Verify frequency

SUSTAINER

Blue Raven

☐ Verify continuity of each channel Battery (7.4) ____ V

☐ Verify flight program

Stratologger

☐ Verify continuity of each channel Battery (9.6) ____ V

☐ Verify flight program

Featherweight GPS

☐ Verify fix Ground station

☐ Verify fix cell phone Battery (7.4) ____ V

☐ Verify frequency

All systems are go for launch!

☐ **Get away from pad**

Appendix E: Airframe calculations

Material Characterization Mechanical properties for both the E-glass and carbon fibre laminates were determined experimentally via uniaxial tensile testing of SRAD coupons conducted in the laboratory. The following orthotropic ply properties were extracted and used as inputs to all subsequent analyses:

Material	E_1 (GPa)	E_2 (GPa)	G_{12} (GPa)	ν_{12}	σ_{1t} (MPa)	σ_{2t} (MPa)	τ_{12f} (MPa)
E-glass 320 g plain weave	28.524	27.056	4.745	0.0845	600.33	420.31	64.58
CF 3k 2×2 twill 205 g	82.885	80.649	4.997	0.0283	934.95	962.24	66.76

Table 25: Experimentally determined ply properties used in laminate analysis.

Load Cases Structural loads are evaluated at the most critical flight instant: immediately before motor burnout, where thrust, velocity, and aerodynamic stability margin are simultaneously near their maximum. Two configurations are analysed independently: the full stack (booster + sustainer) and the sustainer alone after stage separation.

At this instant, an aerodynamic disturbance induces a 10° angle of attack, causing the rocket to resist the off-axis flow with both asymmetric drag and inertial restoring forces. This produces a distributed bending moment along the airframe superimposed on the axial compressive load from motor thrust. The resulting load envelope for each configuration is shown in Figure 48.

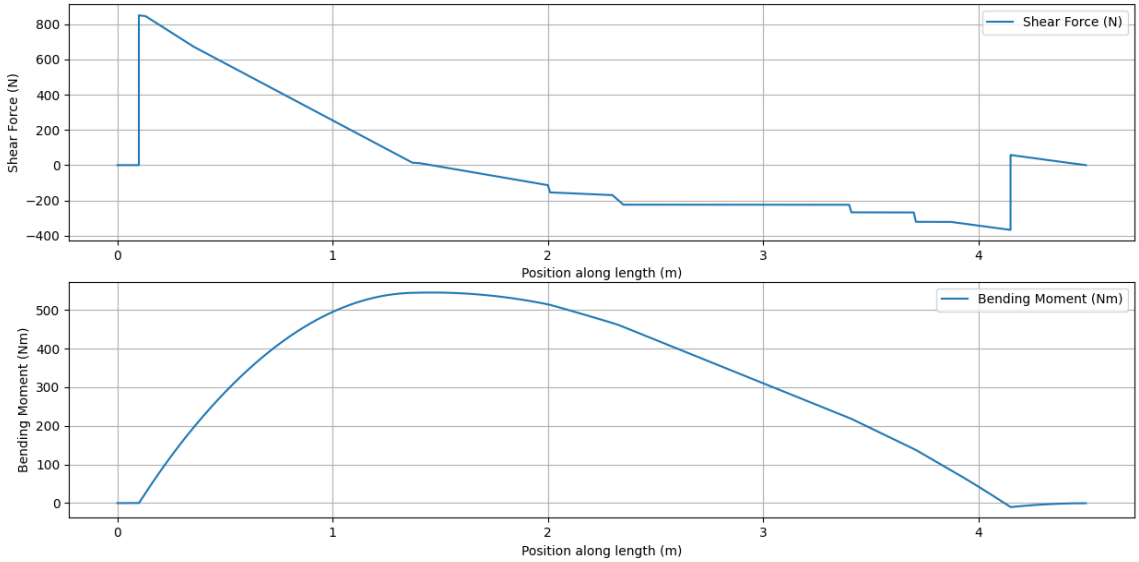


Figure 48: Distributed bending moment along the airframe for the full-stack and sustainer-only load cases at the critical flight instant.

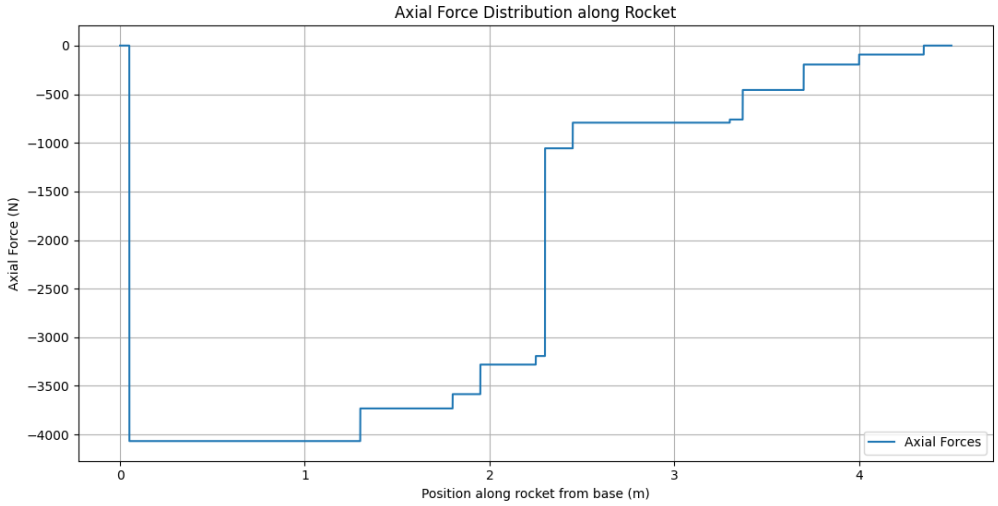


Figure 49: Distributed axial force along the airframe for the full-stack and sustainer-only load cases at the critical flight instant.

Laminate Stress Analysis Stresses are resolved at the ply level using Classical Laminate Theory (CLT) with the Kirchhoff plate hypothesis, which assumes that normals to the mid-plane remain straight and normal after deformation. This yields the through-thickness strain field:

$$\begin{Bmatrix} \varepsilon_x \\ \varepsilon_y \\ \gamma_{xy} \end{Bmatrix} = \begin{Bmatrix} \varepsilon_x^0 \\ \varepsilon_y^0 \\ \gamma_{xy}^0 \end{Bmatrix} + z \begin{Bmatrix} \kappa_x \\ \kappa_y \\ \kappa_{xy} \end{Bmatrix} \tag{5}$$

where ε^0 are the mid-plane strains and κ are the curvatures, both obtained by inverting the ABD constitutive relation:

$$\begin{Bmatrix} \mathbf{N} \\ \mathbf{M} \end{Bmatrix} = \begin{bmatrix} A_{11} & A_{12} & A_{16} & B_{11} & B_{12} & B_{16} \\ A_{12} & A_{22} & A_{26} & B_{12} & B_{22} & B_{26} \\ A_{16} & A_{26} & A_{66} & B_{16} & B_{26} & B_{66} \\ B_{11} & B_{12} & B_{16} & D_{11} & D_{12} & D_{16} \\ B_{12} & B_{22} & B_{26} & D_{12} & D_{22} & D_{26} \\ B_{16} & B_{26} & B_{66} & D_{16} & D_{26} & D_{66} \end{bmatrix} \begin{Bmatrix} \varepsilon_x^0 \\ \varepsilon_y^0 \\ \gamma_{xy}^0 \\ \kappa_x \\ \kappa_y \\ \kappa_{xy} \end{Bmatrix} \tag{6}$$

The ABD matrix is assembled from the individual ply reduced stiffness matrices $\bar{Q}_{ij}$, themselves derived from the experimentally measured ply properties (E_1, E_2, G_{12}, ν_{12}) and the ply orientation angles. The full construction procedure is implemented in Python.

Applied Loads The worst-case loading on a tube section combines axial compression from motor thrust and compression from the bending moment, both acting on the same generator line. The corresponding in-plane running loads are:

$$N_x^{\text{axial}} = \frac{F_{\text{axial}}}{\pi d} \tag{7}$$

$$N_x^{\text{bending}} = \frac{32\, M\, d\, t}{\pi\left[(d+t)^4 - d^4\right]} \tag{8}$$

$$N_x^{\text{total}} = N_x^{\text{axial}} + N_x^{\text{bending}} \tag{9}$$

where d is the inner diameter, t the total laminate thickness, F_{axial} the net axial force, and M the local bending moment at the section of interest.

Ply-Level Stress and Failure Analysis Ply stresses in the fibre coordinate system (σ_1, σ_2, τ_{12}) are recovered by applying the transformation matrix to the laminate strains at each ply mid-plane. Four failure criteria are evaluated simultaneously: Maximum Stress, Maximum Strain, Tsai–Hill, and Tsai–Wu. The stress distribution through the laminate thickness and the resulting failure indices for each ply and criterion are shown in Figures 50 and 51.

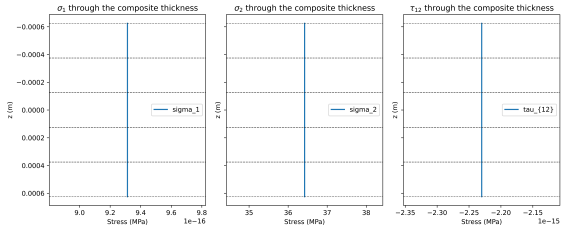


Figure 50: Through-thickness ply stress distribution (σ_1, σ_2, τ_{12}) for the critical tube under worst-case loading.

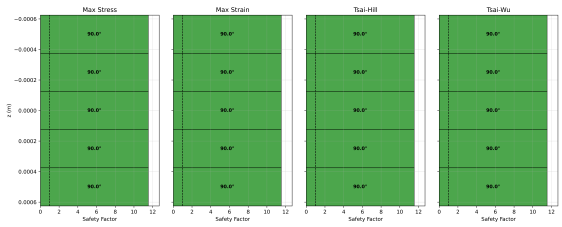


Figure 51: Ply-level safety factors for all four failure criteria (Maximum Stress, Maximum Strain, Tsai–Hill, Tsai–Wu).

Buckling Analysis Local buckling of the tube wall under axial compression is evaluated for the two structurally critical cases: the worst-case carbon fibre tube and the worst-case E-glass tube. The critical buckling load is computed and compared against the applied compressive running load N_x^{total}.

Appendix F: Load distribution for a 2 rope system

Load Distribution in a Two-Anchor Rope System Under Parachute Drag Force

General Derivation for Elastic and Inertial Contributions

System Description

Consider a continuous rope of cross-sectional area A and elastic modulus E, attached at its two ends to structural anchors (bulkheads). A parachute is attached at an intermediate point along the rope, dividing it into two segments:

- **Segment A**: length L_A, connecting the attachment node to Bulkhead A, which carries a concentrated mass m_A.
- **Segment B**: length L_B, connecting the attachment node to Bulkhead B, which carries a concentrated mass m_B.

The parachute exerts a total drag force F at the attachment node, directed along the rope axis (both segments align with the force direction, as a flexible rope cannot sustain transverse loads in equilibrium).

Mechanical Model

Each rope segment is modelled as a linear axial spring with stiffness:

$$k_i = \frac{EA}{L_i}, \qquad i \in \{A,\, B\} \tag{1}$$

The two masses m_A and m_B are rigidly attached at the far ends. Letting u_A, u_B, and u_n denote the axial displacements of mass A, mass B, and the node, respectively, the elongations of the two segments are:

$$\delta_A = u_A - u_n, \qquad \delta_B = u_B - u_n \tag{2}$$

Equations of Motion

Applying Newton's second law to each mass and enforcing equilibrium at the massless node:

$$m_A\,\ddot{u}_A = -k_A\,(u_A - u_n) \tag{3}$$

$$m_B\,\ddot{u}_B = -k_B\,(u_B - u_n) \tag{4}$$

$$k_A\,(u_A - u_n) + k_B\,(u_B - u_n) = F(t) \tag{5}$$

Elastic (Quasi-Static) Contribution

When the load is applied slowly relative to the system's natural period, inertial effects are negligible. The node displacement adjusts such that both segments share the load in proportion to their stiffness. From equation (5) with $\ddot{u}_A = \ddot{u}_B = 0$:

$$T_A^{\text{elastic}} + T_B^{\text{elastic}} = F \tag{6}$$

The compatibility condition (both segments undergo the same node displacement) gives:

$$\frac{T_A^{\text{elastic}}}{k_A} = \frac{T_B^{\text{elastic}}}{k_B} \quad\Longrightarrow\quad \frac{T_A^{\text{elastic}}}{T_B^{\text{elastic}}} = \frac{k_A}{k_B} = \frac{L_B}{L_A} \tag{7}$$

Solving simultaneously:

$$\boxed{T_A^{\text{elastic}} = F\,\frac{k_A}{k_A + k_B} = F\,\frac{L_B}{L_A + L_B}} \tag{8}$$

$$T_B^{\text{elastic}} = F\,\frac{L_A}{L_A + L_B} \tag{9}$$

Note that the *shorter* segment carries the *larger* share of the load, as it is stiffer.

Inertial Contribution

When the load is applied as a brief impulse (duration t_{shock} much shorter than the wave transit time L/c, where $c = \sqrt{E/\rho}$ is the axial wave speed), the rope has no time to deform differentially. The system behaves as a rigid body: both masses experience the same instantaneous acceleration a.

From Newton's second law applied to the total system:

$$F = (m_A + m_B)\,a \quad\Longrightarrow\quad a = \frac{F}{m_A + m_B} \tag{10}$$

The tension in each segment equals the force required to accelerate the mass at its far end:

$$\boxed{T_A^{\text{inertial}} = m_A\,a = F\,\frac{m_A}{m_A + m_B}} \tag{11}$$

$$T_B^{\text{inertial}} = F\,\frac{m_B}{m_A + m_B} \tag{12}$$

General Combined Formula

The physical regime is characterised by the dimensionless coupling parameter α, defined as the ratio of the shock duration to the axial wave transit time:

$$\alpha = \frac{t_{\text{shock}}}{L/c} \in [0,\, 1] \tag{13}$$

The tension in each segment is obtained by linearly interpolating between the two limiting cases:

$$\boxed{T_A = F\left[\alpha\,\frac{m_A}{m_A + m_B} + (1-\alpha)\,\frac{L_B}{L_A + L_B}\right]} \tag{14}$$

$$T_B = F\left[\alpha\,\frac{m_B}{m_A + m_B} + (1-\alpha)\,\frac{L_A}{L_A + L_B}\right] \tag{15}$$

Verification: $T_A + T_B = F\,[\alpha + (1-\alpha)] = F$ ✓

α	Physical regime	Governing mechanism
$\alpha \to 0$	Quasi-static loading	Rope stiffness (segment lengths)
$\alpha \to 1$	Impulsive shock	Mass distribution
$0 < \alpha < 1$	Transient dynamic	Both contributions

Conservative Design Case

For parachute deployment, the shock duration is on the order of milliseconds, while the axial wave transit time in a short nylon or Kevlar rope is of the same order of magnitude. In the absence of measured shock duration data, the conservative design assumption is $\alpha = 1$, which eliminates the geometric contribution entirely:

$$\boxed{T_A^{\text{design}} = F\,\frac{m_A}{m_A + m_B}, \qquad T_B^{\text{design}} = F\,\frac{m_B}{m_A + m_B}} \tag{16}$$

Each bulkhead and its associated U-bolt should be sized to withstand the tension given by equation (16), with appropriate safety factors applied per the applicable structural standard.

Justification of $\alpha = 1$ for Kevlar shock cords

The validity of the $\alpha = 1$ assumption can be verified explicitly by comparing the axial stress wave transit time in the rope to the duration of the parachute opening shock.

Axial wave speed in Kevlar

The speed at which a stress wave propagates along a rope is:

$$c = \sqrt{\frac{E}{\rho}} \tag{17}$$

For Kevlar, the elastic modulus is $E \approx 70$–125 GPa and the density is $\rho \approx 1440$ kg/m^3, giving:

$$c = \sqrt{\frac{70 \times 10^9}{1440}} \approx 7000 \text{ m/s} \tag{18}$$

Wave transit time

The time required for the stress wave to travel the full rope length $L = L_A + L_B$ is:

$$t_{\text{propagation}} = \frac{L}{c} \tag{19}$$

For a representative rope length of $L = 2$ m:

$$t_{\text{propagation}} = \frac{2}{7000} \approx 0.29 \text{ ms} \tag{20}$$

Opening shock duration

The duration of a parachute opening shock depends on deployment speed and system compliance. For small parachutes, values reported in the literature range from $t_{\text{shock}} \approx 5$ to 50 ms.

Resulting value of α

$$\alpha = \frac{t_{\text{shock}}}{t_{\text{propagation}}} = \frac{5 \text{ ms}}{0.29 \text{ ms}} \approx 17 \gg 1 \tag{21}$$

Since α is defined on $[0, 1]$ and the computed ratio far exceeds unity, it is clamped to $\alpha = 1$. This result holds across the full range of expected shock durations: even at $t_{\text{shock}} = 5$ ms (the shortest plausible value), the stress wave completes approximately 17 round trips along the rope before the load peaks. The rope therefore reaches mechanical equilibrium nearly instantaneously relative to the shock timescale.

For a Kevlar shock cord of any length up to several metres, the condition $t_{\text{shock}} \gg t_{\text{propagation}}$ is satisfied for all realistic parachute opening scenarios. The elastic (geometric) contribution in equation (14) is therefore negligible, and the conservative design equations (16) are not merely a safe upper bound, they are physically accurate.

Appendix G: SimulOutil

Simuloutil is a centralized analysis environment developed to address the difficulty of predicting complex rocket flight behavior using only commercial off-the-shelf (COTS) softwares. While COTS tools provide valuable baseline simulations, they often lack flexibility for custom modeling, detailed subsystem analysis, and mission-specific scenarios. To overcome this limitation, additional in-house analysis tools have been developed and integrated into a single Python-based framework. The goal of Simuloutil is to combine these modules into one coherent environment that can be used alongside COTS softwares. The seven currently developed modules are explained in detail below.

1 - Flight and Trajectory

The trajectory tool is an internally developed flight simulation tool whose purpose is to provide an apogee prediction to compare result with other methods like OpenRocket, RasAero and RocketPy.

Calculations:

The simulation is based on Newton's second law applied to the rocket at each time step:

$$\frac{T - D - W}{m} = a \tag{10}$$

Velocity and altitude are then obtained by integration of the acceleration:

$$v = \int a \, dt, \qquad h = \int v \, dt \tag{11}$$

The aerodynamics drag force is calculated as:

$$D = \frac{1}{2} \rho \, A_{\text{ref}} \, C_D(v) \, v^2 \tag{12}$$

The drag coefficient C_D is directly interpolated from a C_D vs mach number curve. The velocity and air density calculated with altitude are computed at each time step and used to find the next values in the iteration, thus needing adequate time step selection. The Motor thrust is also interpolated from the thrust-curve passed as input. Mass is estimated as linearly decreasing during the burn. For the two-stage simulations, the booster and sustainer are solved sequentially, with the final condition of the booster used as the initial condition for the sustainer.

The two-stage calculations is extended to three dimensions. The vector associated with a time step includes the position, velocity and acceleration all in the up, north and east directions, allowing to simulate drift due to wind throughout the flight.

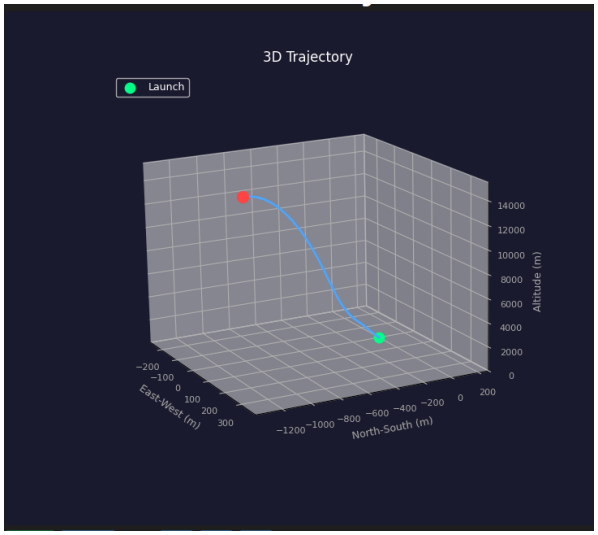


Figure 52: An example of the 3D trajectory calculated with SimulOutil

2 - Aerodynamic (C_D Calculation)

The aerodynamic module computes the drag coefficient as a function of Mach number. These results can then be compared with values obtained from computational fluid dynamics (CFD) simulations, OpenRocket, RasAero and Aerolab to validate the accuracy and reliability of the methodology.
The aerodynamic model considers parasite drag as the dominant contribution, assuming negligible lift effects due to the rocket's near-axisymmetric geometry and small angle of attack. Parasite drag is decomposed into the following components:

1. Friction drag, caused by viscous shear stresses along the rocket surface
2. Pressure drag, resulting from flow separation and shape effects
3. Base drag, generated by the low-pressure wake
4. Wave drag, which becomes significant in transonic and supersonic regimes due to shock formation
5. Interference drag, arising from interactions between components

Each of these contributions is estimated using empirical equations derived from experimental data and established aerodynamic correlations. These equations relate the drag coefficient to parameters such as Mach number, Reynolds number, and geometric properties of the vehicle. The total drag coefficient is obtained by summing all individual contributions, providing a practical and computationally efficient approximation of aerodynamic behavior across different flight regimes.
The tool supports both single-stage and two-stage rocket configurations, allowing users to analyze and compare different flight architectures. Detailed plots are

provided for each individual component of drag (friction, pressure, base, wave, and interference), enabling a deeper understanding of their respective contributions across the flight profile. Furthermore, a continuously updated table displays key parameters throughout the simulation, including a Mach number indicator bar, providing a clear and intuitive representation of the flight regime.
The following graphic illustrates a typical graph that can be obtained with the Aerodynamic module.

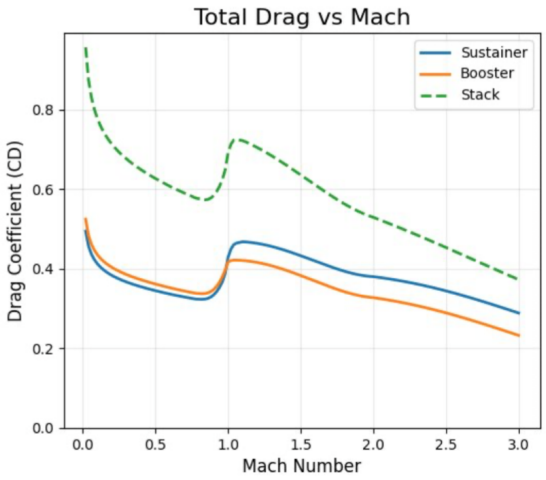


Figure 53: Drag coefficient in function of the Mach number

3 - Aerodynamic analysis (comparison)

The aerodynamic analysis module provides a flexible framework for evaluating drag using multiple modeling approaches. It allows users to import external drag coefficient (C_D) data for validation and comparison, while offering customizable plotting tools to visualize selected curves. This enables efficient analysis and a deeper understanding of aerodynamic performance across different conditions. The following image illustrates the module's interface.

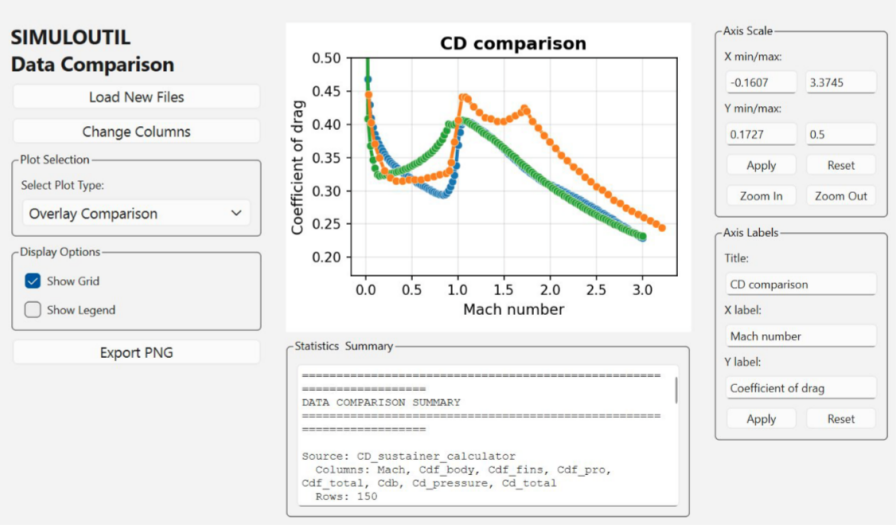


Figure 54: Aerodynamic Analysis User Interface

4 - Landing Dispersion

The landing dispersion module predicts the rocket's landing position using simplified physics-based modeling of the descent phase after apogee. It accounts for parachute and vehicle surface areas, aerodynamic drag on both the rocket and deployed parachute, and the relative airflow velocity vector. Starting from apogee conditions, the equations of motion are integrated over time to simulate the descent under gravity and drag forces. By repeating this simulation multiple times with historical wind data and varying atmospheric conditions, the model generates a statistical dispersion pattern, providing a realistic estimate of potential landing zones and touchdown variability.

The following graphic illustrates a resulting landing scatter on a map.

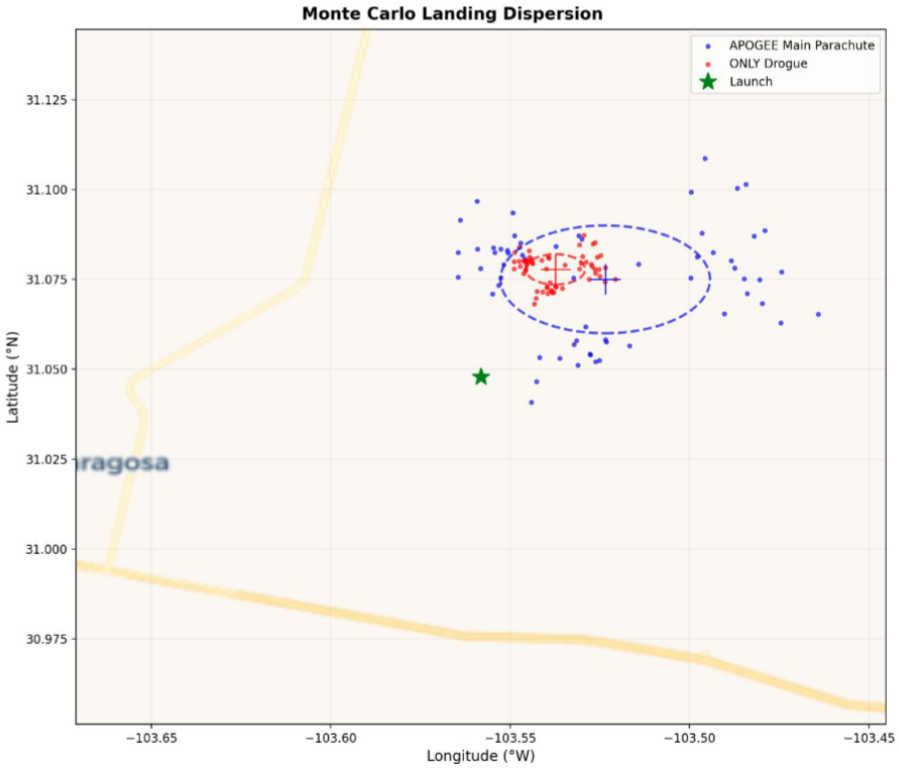


Figure 55: Landing Scatter on a Map

5 - Flight Structural Load

The load analysis module evaluates the structural loads and stresses acting on the airframe throughout flight by considering the most critical aerodynamic and inertial conditions. Two main cases are studied. We suppose that a wind gust orients the rocket at a 10 degrees angle of attack while at maximum dynamic pressure (Max Q), generating bending loads driven by lift forces, aerodynamic side loads, and inertial reactions along the structure. In this scenario, bending results from the combined effect of aerodynamic lift and the vehicle's resistance to motion changes.

In the second case, axial loading during powered ascent is analyzed, where the airframe experiences compression due to thrust, opposed by aerodynamic drag and modified by inertial reactions from acceleration.

Together, these two cases allow the module to identify peak bending moments and maximum axial stresses, ensuring the structural integrity of the rocket under both aerodynamic and propulsion-driven loading conditions. Examples of graphics produced by this module are illustrated in the following appendix I.

6 - Composite materials design

The composite module uses classical laminate theory to calculate safety factors from different criterion (such as Tsai-Hill, Tsai-Wu, max stress, and max strain), deformations, and maximum stress in different sandwich and ply layups. It also calculates such characteristics in individual plies throughout the composite and indicates the most likely point(s) of failure. An example of the equations and uses of the module are demonstrated in the following appendix I with the calculations made directly for our composites.

7 - Fin Flutter

The module determines the critical velocity at which aeroelastic flutter is likely to occur. This is achieved using the following equation:

$$V_f = a \times \sqrt{\frac{G_E}{\left(\frac{39.3 \times A^3}{\left(\frac{t}{c}\right)^3 \times (A+2)}\right) \times \left(\frac{\lambda+1}{2}\right) \times \left(\frac{p}{p_0}\right)}} \tag{13}$$

It features an intuitive and user-friendly interface where the geometrical parameters, material properties and flight conditions are entered. When updated, a geometrical representation of the fin appears to the side and helps to mitigate mistakes in the geometrical parameters often confusing.

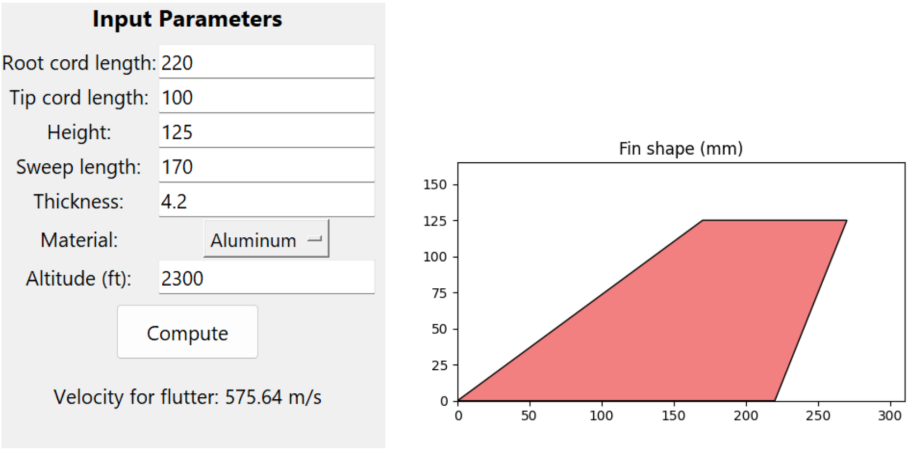


Figure 56: Fin Flutter Module User Interface

Appendix H: Recovery checklists

Parachutes Sustainer

Step	Done	Initials
Ensure proper folding of the booster main parachute in its Nomex sleeve and verify it is indeed the 36 in blue and white Rocketman unit	☐	
Ensure proper folding of the booster drogue parachute in its Nomex sleeve and verify it is indeed the 18 in red and white Fruity Chutes unit	☐	
Form S-folds in the shock cord using masking tape wrapped in electrical tape	☐	
Attach and tighten the quicklink of the booster main parachute to the shock cord at the appropriate location using pliers	☐	
Attach and tighten the quicklink of the booster drogue parachute to the shock cord at the appropriate location using pliers	☐	
Attach and tighten the quicklink to the nosecone eyenut using pliers	☐	
Attach and tighten the quicklink to the top booster avionics bulkhead using pliers	☐	
Verify that the line cutter cable fits properly into the avionics bay and is installed such that pulling on it does not apply tension to the electronics inside the bay	☐	
Attach the line cutter wire to the shock cord using tie-wraps, leaving more slack in the wire than in the shock cord so that tension on the shock cord does not transfer tension to the electrical wire	☐	
Verify that the electrical wire is properly connected to the line cutters via its Wago connectors (lightly pull on the wire to confirm it is securely anchored)	☐	
Wrap the wires and connectors with electrical tape	☐	
Attach the main parachute to the shock cord above the knot using 2 tie-wraps in series, routing the tie-wraps through the line cutters	☐	
Check the line cutter seals by moving the parachute and verifying that no black powder falls out	☐	
Verify that there is a connection point, other than the e-match, between the line cutters and the shock cord to prevent losing them after deployment	☐	
Remove the tape wrapped around the booster parachutes	☐	
Insert the shock cord and sustainer parachutes into the coupler in the nosecone, ensuring they do not get tangled	☐	
Assemble the nosecone and tube together using 3 shear pins	☐	

Parachutes Booster

Step	Done	Initials
Ensure proper folding of the booster main parachute in its Nomex sleeve and verify it is indeed the 84 in red and black Fruity Chutes unit	☐	
Ensure proper folding of the booster drogue parachute in its Nomex sleeve and verify it is indeed the 24 in red and white Fruity Chutes unit	☐	
Form S-folds in the shock cord using masking tape wrapped in electrical tape	☐	
Attach and tighten the quicklink of the booster main parachute to the shock cord at the appropriate location using pliers	☐	
Attach and tighten the quicklink of the booster drogue parachute to the shock cord at the appropriate location using pliers	☐	
Attach and tighten the quicklink to the booster motor bulkhead using pliers	☐	
Attach and tighten the quicklink to the lower booster avionics bulkhead using pliers	☐	
Verify that the line cutter cable fits properly into the avionics bay and is installed such that pulling on it does not apply tension to the electronics inside the bay	☐	
Attach the line cutter wire to the shock cord using tie-wraps, leaving more slack in the wire than in the shock cord so that tension on the shock cord does not transfer tension to the electrical wire	☐	
Verify that the electrical wire is properly connected to the line cutters via its Wago connectors (lightly pull on the wire to confirm it is securely anchored)	☐	
Wrap the wires and connectors with electrical tape	☐	
Attach the main parachute to the shock cord above the knot using 2 tie-wraps in series, routing the tie-wraps through the line cutters	☐	
Check the line cutter seals by moving the parachute and verifying that no black powder falls out	☐	
Verify that there is a connection point, other than the e-match, between the line cutters and the shock cord to prevent losing them after deployment	☐	
Remove the tape wrapped around the booster parachutes	☐	
Insert the shock cord and booster parachutes into the coupler on the avionics bay side, ensuring they do not get tangled	☐	

Assemble the avionics tube and motor tube together using 3 shear pins	☐	

Recovery

Step	Done	Initials
Before leaving, record the final GPS coordinates from the ODB, Featherweight, and Spot Trace trackers and enter them into a map application such as Google Maps, Apple Maps, a Garmin GPS, or a GPS watch	☐	
Plan a route to reach the desired section of the rocket	☐	
Bring water, snacks, a first aid kit, insect repellent, duct tape, gloves, a small saw, a flashlight, a power bank, the Idefix receiver, and the equipment required to safe the rocket (Allen key set, wire cutters)	☐	
Report to the recovery management tent and check in	☐	
Proceed to the rocket using the GPS and Idefix receiver, while following all Launch Control (LC) instructions regarding check-ins and status updates	☐	
Upon arrival, inform recovery management that the rocket has been located and record its GPS position	☐	
Take a video of the rocket's condition on the ground before touching anything, documenting as much as possible the condition of its sections and the deployment of its parachutes	☐	
Safe the rocket by removing all push pins before touching anything else	☐	
Verify that all ejection charges have fired before beginning any work on the rocket	☐	
If the sustainer motor did not ignite, carefully remove the igniter after removing the push pins	☐	
If any charges did not fire, make them safe by carefully removing the black powder	☐	
Pack everything up and return. Do not forget to notify recovery management that you are on your way back	☐	
Report to the recovery tent and check out	☐	